Local Field Theory

— *EYNTKA* Series —

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0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7$\star$; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Introduction

This book is about doing arithmetic one prime at a time. Completing a number field at a single prime concentrates the information carried by that prime into one analytically tractable object, a local field. Questions about ramification, about whether an equation has a solution, about how a Galois group acts near the prime, all sharpen and often become decidable once localized. The technology for that localization, and the dictionary translating between a global field and the family of its completions, is what these pages develop.

The reader is assumed to have met algebraic number theory: rings of integers, Dedekind domains, the splitting of primes, and the ideal class group. From there the treatment is self-contained. It builds the local fields from the ground up, first as completions and as abstract complete discrete valuation rings, then through the fine structure of their extensions, and finally reassembles the local pictures into a global one in the language of places.

The journey runs through five chapters and an appendix. Chapter 1 constructs the objects: the p-adic numbers, discrete valuation rings and their completions, the definition of a local field, the structure of its multiplicative group, and Witt vectors as the canonical source of unramified extensions. Chapter 2 develops the two instruments for locating roots over a complete field, Hensel's lemma and the Newton polygon. Chapter 3 is the technical heart, the extension theory of local fields and the full anatomy of ramification, from the fundamental identity through the higher ramification filtration, the different, and the Hilbert symbol. Chapter 4 completes the development with the local-global correspondence and its consequences, the Kronecker-Weber theorem among them. Chapter 5 recasts everything in the language of places, the uniform treatment of finite and infinite primes that opens onto adelic number theory, and appendix A proves the Hasse-Arf theorem.

The book stops deliberately short of local class field theory itself, the reciprocity map and the classification of abelian extensions by norm groups. That is the subject of the sequel, and the first place to go next.

1

p-adic Numbers, Valuations, and Local Fields

In this chapter, we shall explore special fields which focus into a single prime of a number field. For a prime $p \in \mathbb{Z}$, the fundamental object corresponding to this prime is the p-adic integers $\mathbb{Z}_p$. Intuitively, solving an equation over $\mathbb{Z}_p$ corresponds to solving an equation over $\mathbb{Z}$ mod p^n for each n (which can be thought of as solving the equation over all multiplicities). Each $\mathbb{Z}_p$ shall correspond to a field $\mathbb{Q}_p$, and finite extensions of $\mathbb{Q}_p$, $K_\mathfrak{p}/\mathbb{Q}_p$, shall only contain primes that are over p. The galois group of these extensions shall then be isomorphic to the decomposition group. The field extensions $K_\mathfrak{p}$ shall be called *local number fields*. Finite extensions $K/\mathbb{Q}$ will be called *global number fields*. A large part of the early sections is dedicated to building up the theory of local fields. This work culminates in the local-global correspondence (section 4.2), which readily translates many results between global fields and local fields.

In the process of developing the generalization, we shall see that the local fields $\mathbb{Q}_p$ correspond very naturally to all possible distinct absolute values that can be defined on $\mathbb{Q}$. In fact, the collection of all possible absolute absolute values that can be defined on $\mathbb{Q}$ shall create the following completions:

$$\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7, \mathbb{Q}_{11}, ..., \mathbb{Q}_\infty = \mathbb{R}$$

where there is one distinct completion for each prime, as well as one completion that can in fact be naturally interpreted as the completion at the "prime at infinity". We shall make this idea precise in chapter 5. The idea of places being an extension of primes shall become important when developing class field theory, as well as linking it to the development of extensions of $\mathbb{F}_p(t)$.

Later, in chapter 4, we shall prove the *Kronecker-Weber Theorem*, which shows that every abelian extension is a sub-extension of a cyclotomic extension. This result reveals a deep connection between galois groups and primes, and shall to great effect be generalized: the sequel *Class Field Theory* develops this completely, showing that for every field K there exists a field K_H whose galois group $\mathrm{Gal}_K(K_H)$ is isomorphic to a class group.

1.1 p-adic Integers and p-adic Numbers

Let us start with giving a detailed description of the "new integers" where we only focus on a single prime. Let R be a commutative ring, $\mathfrak{p} \subseteq R$ a prime. Then $R_{\mathfrak{p}}$ will represent the localization of R at $\mathfrak{p}$ (i.e. at the set $R \setminus \mathfrak{p}$). Recall that the completion of R at a prime P is:

$$\varprojlim_n R/\mathfrak{p}^n R = \widehat{R}$$

where the $\mathfrak{p}$ is usually understood from context on the right hand side. For number theory, we are interested in the case where $R = \mathbb{Z}$. This gives us the localizations $\mathbb{Z}_{(p)}$ and the completions $\widehat{\mathbb{Z}}_p$ which we shall denote $\mathbb{Z}_p$. Elements in the completion can be represented as the formal power series:

$$\sum_i^{\infty} a_i p^i = a_0 + a_1 p + a_2 p^2 + \cdots$$

where $a_i \in \{0, ..., p-1\}$. More succidnetly, as $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$, as a sequence

$$(a_0 \bmod p, a_1 \bmod p^2, ...)$$

It is not immediately clear that these two are the same: in the first we have multiplication via convolution, while in the second we have point-wise multiplication. The equivalence in these two representation can be given by the following: any element can be found by seeing that when we are given:

$$s = \sum_i^{\infty} a_i p^i$$

it has to satisfy:

$$s_n = \sum_i^{n-1} a_i p^i$$

hence, for any $[r] \in \mathbb{Z}/p^n\mathbb{Z}$, there exists a representative of the form:

$$[r] = [a_0 + a_1 p + \cdots + a_{n-1} p^{n-1}]$$

Such an expansion can be written down for any $r \in \mathbb{Z}_{(p)}$, that is for any a/b where $p \nmid b$. From this, if we consider $\mathbb{Z}_p$ to be the formal power series with addition/multiplication extending naturally. From this:

Proposition 1.1.1: Representation Of p-addic

Let $\mathbb{Z}_p$ represent the formal p-adic integers and $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ the completion definition. Then:

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

***Proof*:**
Put together all the above assertions.

A motivation for defining the p-adic integers is in the following "universal property": the solutions to Diophantine equations (elements of $\mathbb{Z}[x_1, ..., x_n]$) mod p^n for all $n \in \mathbb{N}_{>0}$ equivalent to finding solution to Diophantine equations over the p-adics. Note that if we ask for solutions over some collection $\bmod m$, by the Chinese Remainder Theorem (CRT) we may reduce the case to solution over powers of primes.

Theorem 1.1.2: Diophantine Equations in p-adic

Let $f \in \mathbb{Z}[x_1, ..., x_n]$. Then the congruence:

$$f \equiv 0 \text{ mod } p^n$$

for all $n \geq 1$ is solvable if and only if f is has solutions in the p-adics.

***Proof*:**
Let's first say f has a solution in the p-adics, meaning we make a choice of elements $(x_1, .., x_n)$:

$$(x_1, ..., x_n) = (x_1^{(\nu)}, ..., x_n^{(\nu)})_{\nu \in \mathbb{N}} \in \mathbb{Z}_p^n$$

such that $f(x_1, ..., x_n) = 0$. Then by construction we get:

$$f(x_1^{(\nu)}, ..., x_n^{(\nu)}) = 0 \text{ mod } p^\nu$$

Conversely, assume for each $\nu \in \mathbb{N}$ there is a solution

$$f(x_1^{(\nu)}, ..., x_n^{(\nu)}) = 0 \text{ mod } p^\nu$$

If by chance we have that the collection $(x_i^{(\nu)})_{\nu \in \mathbb{N}} \in \prod_\nu^\infty \mathbb{Z}/p^\nu\mathbb{Z}$ is already in $\varprojlim_\nu \mathbb{Z}/p^\nu\mathbb{Z}$, we're done. However, having solutions to moduli equaiton doesn't immediately guarantee the regularity needed for the limit. Instead, a p-adic number will be constructed using these numbers.

We shall construct the construction for the case of $n = 1$, the general case follows. Let $x_\nu = x_1^{(\nu)}$. Then this defines a sequence (x_ν) in $\mathbb{Z}$. As $\mathbb{Z}/p\mathbb{Z}$ has finitely many elements, there are infinitely many elements of the sequence that are congruent to the same element $y_1 \mod p$. Then if $(x_\nu^{(1)})$ is the subsequence:

$$(x_\nu^{(1)}) \equiv y_1 \text{ mod } p \qquad \text{hence} \qquad F(x_\nu^{(1)}) \equiv 0 \text{ mod } p$$

From this, another subsequence $(x_\nu^{(2)}) \subseteq (x_\nu^{(1)})$ can be extracted st

$$(x_\nu^{(2)}) \equiv y_2 \text{ mod } p \qquad \text{hence} \qquad F(x_\nu^{(2)}) \equiv 0 \text{ mod } p^2$$

Continuing on, we get y_i where

$$y_k \equiv y_{k-1} \text{ mod } p^{k-1}$$

Hence, we can define $y = (y_k)_{k \in \mathbb{N}} \in \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$, which is our desired p-addic integer:

$$F(y_k) = 0 \text{ mod } p^k \qquad \text{if and only if} \qquad F(y) = 0$$

as we sought to show.

Example 1.1.3: *p*-adic Integers

1. Let $n = 10$. Then $10 = 20_5 = 1010_2 = 101_3 = 10_n$ for $n \geq 10$. For non-prime p, we can still do the n-adic completion $\mathbb{Z}_n$, however we will get zero divisors. For example:

$$\mathbb{Z}_{10} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_5$$

which comes from the direct commuting with completion (see [10]).

2. Let $p = 3$ and considier -1_3. I claim that -1_3 exists. We need to find $[\cdots a_3 a_2 a_1]_3$ such that

$$0001_3 + -1_3 = 0$$

Then $\cdots 222_3$ will give the answer. since:

$$\cdots 222_3 + \cdots 0001_3 = \cdots 000_3$$

Repeat this for any negative number

3. Let $p = 3$ and consider $1/2$. I claim that $(1/2)_3$ exists. We need to find $[\cdots a_3 a_2 a_1]_3$

$$(1/2)_3 \cdot 2_3 = \cdots 0001_3$$

To construct such a number, first:

$$a_1 \cdot 2 \equiv 1 \bmod 3$$

hence $a_1 = 2$. We then need:

$$2 \cdot a_2 + 1 \equiv 0 \bmod 3$$

which gives $a_2 = 1$. Continuing we get:

$$(\cdots 1112_3) \cdot (\cdots 0002)_3 = (\cdots 0001)_3$$

This can be done for any prime $p \neq 2$.

4. Let $a/b \in \mathbb{Q}$. Then if $p \big| b$ there is no p-adic representation. For example, $1/5$ does not have a 5-adic representaiton (you would need to mulitply $1/5$ by 5, but $0005_5 = 0_5$).

5. Using the second example, find the p-adic representation of $\frac{1}{1-p}$.

By the above example, we see that the elements of $\mathbb{Z}_{(p)} \subseteq \mathbb{Q}$ can be represented as by the p-adic's $\mathbb{Z}_p$ and hence there is a natural map:

$$\mathbb{Z}_{(p)} \to \mathbb{Z}_p$$

Note that this map is not surjective (the right hand side is uncountable, or more concretely the right hand side will have $\sqrt{2}$ when $p \neq 2$). This also comes form the more general result that if R is a ring and $\widehat{R}$ is the completion at $\mathfrak{p}$, then if $u \in R/\mathfrak{p}^n$ is a unit, $u \in R/\mathfrak{p}^{2n}$ is a unit. In particular, if $\mathfrak{p} = \mathfrak{m}$ is a maximal ideal, $\widehat{R}_{\mathfrak{m}}$ is a local ring, and hence $\mathbb{Z}_p$ is a local ring (see [10])

Then as $\mathbb{Z}_p$ has a unique maximal ideal, label it $\mathfrak{m}_{\mathfrak{p}}$, let us now consider the localization of $\mathbb{Z}_p$ at $\mathfrak{m}_{\mathfrak{p}}$. Let's denote it $\mathbb{Q}_p$. The completion will consist of all "Laurent series", that is all elements of the form:

$$\sum_{-m}^{\infty} a_i p^i$$

for all $m \in \mathbb{Z}$, $0 \leq a_i < p$. This allows us to represent *any* rational number, for example if $a/b \in \mathbb{Q}$, then represent it as:

$$\frac{a}{b} = \frac{g}{h}p^{-m} \qquad g, h \in \mathbb{Z} \qquad \gcd(gh, p) = 1$$

We can then do the p-adic expansion of $g/h = p^m f$:

$$a_0 + a_1 p + a_2 p^2 + \cdots$$

we then shift everything down by m to get the p-adic expansion of f:

$$a_0 p^{-m} + a_1 p^{-m+1} \cdots + a_{m-1} p^{-1} + a_m + a_{m+1} p + a_{m+2} p^2 + \cdots$$

this is called a p-adic number.

1.1.4 Relation to Complex Analysis Notice the similarity to meromorphic functions in complex analysis (this was where the theory of p-adic integers and numbers where inspired). This will be a general trend, where many of the result for number field have exact parallels in function fields; this shall be made more precise in section 1.3.

Hence, we have the more general embedding:

$$\mathbb{Q} \to \mathbb{Q}_p = (\mathbb{Z}_p)_{\mathfrak{m}_\mathfrak{p}}$$

where $\mathbb{Z}$ injects into $\mathbb{Z}_p$ (if a, b have the same p-addic expansion, $a - b$ is divisible by p^n for every n, hence $a = b$ in $\mathbb{Z}$). Given this, idnetify $\mathbb{Q} \subseteq \mathbb{Q}_p$ and $\mathbb{Z} \subseteq \mathbb{Z}_p$.

Let us now take another perspective on $\mathbb{Q}_p$ that is more analytic. Recall that in [10, chatper 19] we classified all absolute values on $\mathbb{Q}$, namely the p-valuations and the usual absolute value. Recall the p-valuation is defined on $\mathbb{Q}$ as:

$$|a|_p = \left|\frac{b}{c}\right|_p = \left|p^m \frac{b'}{c'}\right|_p = p^{-m} \qquad \gcd(b'c', p) = 1$$

Where in the p-adic representation of a, we may think of ν_p as counting how many 0's there are before hitting the first non-zero digit, for example in $625 = 1000_5$ we have:

$$|625|_5 = \frac{1}{5^4}$$

Or more algebraically, we take the maximal n where $a \equiv 0 \mod p^n$, $a \not\equiv 0 \mod p^{n+1}$. Hence, the larger the order of the prime p in a/b, the smaller the valuation (intuitively, the "less" detectable the number is when taking $\mod p^n$). This is perhaps intuitive if we compare p-adic expansion to decimal expansion, in which we have:

$$a_0 + a_1\left(\frac{1}{10}\right) + a_2\left(\frac{1}{10}\right)^2 + \cdots$$

with $1 \leq a_0 < 10$. Then the more a_i's that are zero, the smaller the valuation. The above representation of a number in decimal-expansion is good for visualizing a valuation called the ∞-valuation, $|-|_\infty$ which is the unique Archimedean valuation on $\mathbb{Q}$ up to equivalence (as all p-valuations induce an ultra-metric, they are all non-Archimedean). In this construction, what is important is the exponent, and not the p. The intuition from this comes from the induced p-adic topology, namely from $|-|_p$ we can define the [ultra]-metric

$$d(x, y) = |x - y|_p$$

from which we can construct $\mathbb{Q}_p$ from $\mathbb{Q}$ by taking it's completion using the above metric. Then choosing another constant c will produce equivalent topologies; what matter's for the definition is the factoring of the p and not the "concrete representation".

To continue creating intuition's, let us motivate the $|-|_\infty$ symbol. This comes from considering $k(t)$ over a finite field k, where the analog of $\mathbb{Z}$ is $k[x]$. In $k[x]$, there are monic irreducible polynomials which are in one-to-one correspondence with the primes $\mathfrak{p}$ of $k[t]$ (recall $k[t]$ is a PID). Hence, define $|-|_\mathfrak{p} : k(t) \to \mathbb{R}$ as:

$$f(t) = \frac{g(t)}{h(t)} = p(t)^m \frac{g'(t)}{h'(t)} \qquad (g'h', p) = 1$$

Then define $\nu_\mathfrak{p}(f) = m$. Then topologically the choice of what to exponentiate doesn't matter, however an informed choice will give us more information, namely let $q_\mathfrak{p} = q^{d_\mathfrak{p}}$ where q is any real number > 1 and d_p is the degree of the residue class field of $\mathfrak{p}$ over k (i.e. of $\kappa(f)$). Then:

$$\nu_\mathfrak{p}(f) = m \qquad |f|_p = q_\mathfrak{p}^{-\nu_\mathfrak{p}(f)}$$

Notice that $\nu_\mathfrak{p}(0) = \infty$ and $|0|_\mathfrak{p} = 0$, giving us that these are indeed the discrete valuations. When $\mathfrak{p} = (t - a)$, the valuation $\nu_\mathfrak{p}(f)$ gives the order of the zero/poles of the function $f(t)$ at $t = a$. Then for the infinite case, we have:

$$\nu_\infty : k(t) \to \mathbb{Z} \cup \{\infty\} \qquad \nu_\infty(f) = \deg(h) - \deg(g)$$

for $f = g/h \neq 0$ and $g, h \in k[t]$. This can be thought of describing the zero and poles at the point at infinity, i.e. the zeros and pole of the function $f(1/t)$ at the point 0. It is in fact associated to a prime ideal, namely $\mathfrak{p} = (1/t)$ in $k[1/t] \subseteq k(t)$. Hence, when considering the absolute value $|-|$ on $\mathbb{Q}$, we may think of this as being the evaluation at the point at infinity, and hence the $|-|_\infty$ symbol. Notice too how this motivates that the size of $|-|_\infty$ diminishes the more zeros' there are in the decimal expansion and taking the reciprocal of integers. As the concrete value of "t" doesn't matter, we can consider any integer and get an expansion that mimics the properties of $|-|_\infty \to 0$, for example

$$a_0 + a_1\left(\frac{1}{2}\right) + a_2\left(\frac{1}{2}\right)^2 + \cdots$$

Finally, for $k(t)$, there is also a 0-valuation given by:

$$|f/g|_0 = \operatorname{ord} g - \operatorname{ord} f$$

where the order represents the first degree of the smallest non-trivial monomial, for example $\operatorname{ord}(ax^3 + bx) = 1$. Then in $\mathbb{Q}$, $|-|_0$ is the trivial valuation.

Coming back to absolute values over $\mathbb{Q}$, as what is important is the exponent and not the choice of constant to be exponentiated, it is often better to define a new functional, namely a discrete valuation, which only captures the exponent information. The reader should recall from [10, chatper 20] the following the function $\nu_p : \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ where $x + \infty = \infty + x = \infty + \infty = \infty$:

1. $\nu_p(a) = \infty$ if and only if $a = 0$ (this mirror's the idea that p divides 0 infinitely many times
2. $\nu_p(ab) = \nu_p(a) + \nu_p(b)$
3. $\nu_p(a+b) \geq \min(\nu_p(a), \nu_p(b))$

The last condition can be thought of as the ultra-metric condition[1]. From this, we may define p-valuations as

$$|-|_p : \mathbb{Q} \to \mathbb{R} \qquad a \mapsto |a|_p = p^{\nu_p(a)}$$

and we may choose to change p to any other constant $c > 1$.

Now, with the p-adic metric properly defined on $\mathbb{Q}$, we may take the completion of $\mathbb{Q}$ under this metric by taking all Cauchy-sequences of $\mathbb{Q}$ and taking the quotient of all null-sequences (sequences that approach 0). Note that null-sequences form a maximal ideal in the ring of Cauchy-sequences. If R is the ring of Cauchy-sequences and $\mathfrak{m}$, then:

$$\mathbb{Q}_p \cong R/\mathfrak{m}$$

and hence, is an equivalent definition. The reader may see that $\mathbb{Q}_p$ is complete, and that $\mathbb{Q}_\infty = \mathbb{R}$. $\mathbb{Q}$ is naturally embedded in $\mathbb{Q}$ via the constant sequence, and as $\mathbb{Q}$ is dense in $\mathbb{Q}_p$ the p-adic valuation naturally extends, namely:

$$|x|_p := \lim_{n\to\infty} |x_n|_p \in \mathbb{R}$$

where x_n are the elements in the sequence[2]. To show that the $\nu_p(x)$ still represents the exponent (i.e. is an integer), if $x \in \mathbb{Q}_p$ is taken as the class of Cauchy-sequences (where $x_n \neq 0$) then:

$$\nu_p(x_n) = -\log_p |x|_p$$

either diverges or converges is a Cauchy-sequence in $\mathbb{Z}$ which eventually must be constant for large enough n. Then:

$$\nu_p(x) = \lim_{n\to\infty} \nu_p(x_n) = \nu_p(x_n) \qquad \text{for } n \geq n_0$$

Thus, we come full circle and have:

$$|x|_p - p^{-\nu_p(x)}$$

We now have a collection of complete fields with different absolute values:

$$\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7, ..., \mathbb{Q}_\infty = \mathbb{R}$$

From each of these, we may extract $\mathbb{Z}_p$ in the following way:

Proposition 1.1.5: Characterizing p-adic Integers

Considering $\mathbb{Q}_p$ as the completion as given above, show that:

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}$$

is a well-defined ring. Then $\mathbb{Z}_p$ is the closure of $\mathbb{Z}$ with respect to $|-|_p$

---

[1]And indeed, we can define discrete valuations to be a generalization of non-Archimedean valuations, see [10, chapter 19]

[2]Technically, this is mod $\mathfrak{m}$; the reader may want to do some practice by checking this is indeed well-defined

***Proof*:**
exercise.

Using the p-adic valuation, we may characterize the units of $\mathbb{Z}_p$ as:

$$\mathbb{Z}_p^* = \{x \in \mathbb{Z}_p \mid |x|_p = 1\}$$

Using this, we may represent every $x \in \mathbb{Q}_p^*$ as :

$$x = p^m u \qquad m \in \mathbb{Z},\ u \in \mathbb{Z}_p^*$$

that is, every element of $\mathbb{Q}_p^*$ is some unit times some power of prime. For if:

$$\nu_p(x) = m \in \mathbb{Z} \quad \Rightarrow \quad \nu_p(xp^m) = 0 \quad \Rightarrow \quad |xp^{-m}|_p = 1$$

thus $u = xp^{-m} \in \mathbb{Z}_p^*$. Going back to the localization definition of $\mathbb{Z}_p$, this can be seen by remembering that all other primes are localized, and hence any prime factorization $\mathbb{Z}_p$ will now only have a single prime left (and the above argument shows this holds in the completion). From this, we can characterize the ideals of $\mathbb{Z}_p$:

Proposition 1.1.6: Classifying Ideals Of p-adic Integers

Let $\mathbb{Z}_p$ be the p-adic integers. Then set set of nonzero ideals of $\mathbb{Z}_p$ are the principal ideals:

$$p^n\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid \nu_p(x) \geq n\}$$

with $n \geq 0$. Furthermore:

$$\mathbb{Z}_p/p^n\mathbb{Z}_p \cong \mathbb{Z}/p^n\mathbb{Z}$$

***Proof*:**
The first part is a result of the theory of valuations, and is left as an exercise (or check [4, Characterizing DVR])

To show the isomorphism, we'll show that the map $\mathbb{Z} \to \mathbb{Z}_p/p^n\mathbb{Z}_p$ where $n \mapsto n \bmod p^n\mathbb{Z}_p$ is surjective. To see this, as $\mathbb{Z}_p$ is the completion of $\mathbb{Z}$ with respect to $|-|_p$, for any $x \in \mathbb{Z}_p$ there exists an $a \in \mathbb{Z}$ such that

$$|x - a|_p \leq \frac{1}{p^n}$$

Thus, $x - a \in p^n\mathbb{Z}_p$ (since $n \in \mathbb{Q}$ is in $\mathbb{Z}_p$ if and only if $|n|_p \leq 1$), i.e. $x \equiv a \bmod p^n\mathbb{Z}_p$. Hence, the map is surjective, the kernel consists of all elements where $a \in \mathbb{Z}$ where $a \in p^n\mathbb{Z}_p$, which is exactly the elements of the form $a = p^n x$ for $x \in \mathbb{Z}$, that is the elements $p^n\mathbb{Z}$, hence:

$$\mathbb{Z}_p/p^n\mathbb{Z}_p \cong \mathbb{Z}/p^n\mathbb{Z}$$

as we sought to show.

Using this, we have that:

Proposition 1.1.7: Next Equivalence Of p-adic Integers

Show that with $\mathbb{Z}_p$ as defined from above we have:

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

***Proof*:**
As $\mathbb{Z}_p/p^n\mathbb{Z}_p \cong \mathbb{Z}/p^n\mathbb{Z}$, we get a surjective homomorphism:

$$\mathbb{Z}_p \to \mathbb{Z}/p^n\mathbb{Z}$$

This gives a family of homomorphisms which can be used to construct a homomorphism:

$$\mathbb{Z}_p \to \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

show this is an isomorphism

Finally, let us characterize p-adic integers in one more way and quickly address a possible miss-conception of the fields $\mathbb{Q}_p$ with respect to $\mathbb{R}$. For the new characterization:

Proposition 1.1.8: Equivalence Of p-adic Via Power-series Quotient

$$\mathbb{Z}_p \cong \mathbb{Z}[[x]]/(x-p)$$

***Proof*:**
exercise.

A common mistake a reader may do is think that $\mathbb{Q}_p \cong \mathbb{R}$ algebraically. We proved this is not the case in [4], however a concrete example should help.

Example 1.1.9: Cautions for p-adic Numbers

1. Recall that $1/p \notin \mathbb{Z}_p$ (review why if this is fuzzy). As $1/p \in \mathbb{Q}_p$, we cannot use it to construct our counter-example. Instead, we shall show that $\sqrt{p} \notin \mathbb{Q}_p$. To see this, notice that $\sqrt{p}$ by definition satisfies the equation $x^2 - p$. Then certainly it would have to satisfy the equation modulo p^n for every n. Then if $\sqrt{p} \in \mathbb{Q}_p$, it must be the case that $\sqrt{p} \in \mathbb{Z}_p$ by theorem 1.1.2. However, $x^2 \equiv p \bmod p^2$ has no solutions, and hence $\sqrt{p} \notin \mathbb{Q}_p$.

   On the other hand, $\sqrt{1-p^3} \in \mathbb{Q}_p$, as the equations $x^2 - 1 + p^3$ have a solution mod each p^n (use quadratic reciprocity). However, $\sqrt{1-p^3} \notin \mathbb{R}$ as this would require a negative square-root, a number we know that $\mathbb{R}$ does not contain.

2. By field theory, we know that $\mathbb{Q}_p$ embeds into $\mathbb{C}$. This is an example where we really require the axiom of choice. The embeddings is far away from preserving the topological structure of $\mathbb{Q}_p$ or being represented in $\mathbb{C}$. To see this, first notice that $\pm\sqrt{-1} = \pm i \in \mathbb{Q}_5$. It may seem like we should map i to either $\pm i$. However, note that i is closer to 2 while $-i$ is closer to 3. In fact, i is even closer to 57 while the other is much closer to 68. In fact, without

the axiom of choice, this embedding would not be possible. Algebraically, you may also try to check that $\mathbb{Q}_5$ has no automorphisms that swap i and $-i$ (in fact it has *no* non-trivial automorphisms, mirroring how $\mathbb{R}$ has no non-trivial automorphisms)[a].

3. Another interesting distinction between $\mathbb{Q}_p$ and $\mathbb{R}$ the degree of $\overline{\mathbb{Q}_p}/\mathbb{Q}_p$ vs $\overline{\mathbb{R}}/\mathbb{R}$. In the latter, we have $\overline{\mathbb{R}} \cong \mathbb{C}$, and we have a degree two extension. After covering Hensel's lemma, we shall see that $x^n - p$ is irreducible for each n in $\mathbb{Q}_p$. It can be shown that there are finitely many extensions for each degree n, and so $\overline{\mathbb{Q}_p}/\mathbb{Q}_p$ is a countable extension.

[a]see exercise ref:HERE in [4]

double-check intuition here Finally, we introduce the following generalization of prime decompositions to $\mathbb{Q}$[3]:

Proposition 1.1.10: Artin-Whaples Product Formula

Let $0 \neq r \in \mathbb{Q}$. Then

$$\prod_{p,\infty} |r|_p = 1$$

where p varies over all primes number and ∞

***Proof*:**
Any $r \in \mathbb{Q}$ can be factored as :

$$r = \pm \prod_{p \neq \infty} p^{\nu_p(r)}$$

and the $\pm$ is equal to $a/|a|_\infty$, giving us the result.

Note that this result work in for the elements of $k(t)$, namely

$$\prod_{\mathfrak{p},\infty} |f|_{\mathfrak{p}} = 1$$

According to nlab (https://ncatlab.org/nlab/show/global+field) global fields are characterized by this product.

Also, I just had another intuition. Recall that the infinite prime feels like it is generically close to all primes. The infinite product shows that we can define:

$$|r|_\infty := \prod_p |r|_p^{-1}$$

showing that it is define as the product of all other primes, explaining this generic feeling.

Exercise 1.1.1

1. Let $V \subseteq \overline{k}[x_1, ..., x_n]$ be a variety so that $\mathcal{I}(V) = (f_1, ..., f_n)$. Let $V(\mathbb{Q})$ be the rational points of the variety. Show that if $V(\mathbb{Q})$ is non-empty, then $V(\mathbb{Q}_p)$ is non-empty for each prime p, and that $V(\mathbb{R})$ is nonempty.

[3]this result shall be generalized to more general fields in section 5.3

Note the converse need not be true: it was shown that the equation $3x^3 + 4y^3 + 5z^5 = 0$ has a solution over all $\mathbb{Q}_p$ and $\mathbb{R}$, but no real solution. This is an example of the *Hasse Principle* or *local-to-global principle* not being an iff.

1.2 Discrete Valuation Rings and Completions

This construction started with the integers, i.e. the "fundamental" ring of integers. We now need to have a way to generalize this notion for rings of integers more generally. Recall that Dedekind domains are locally DVRs (see [10, chapter 22.2]). The intuition is that the localization of a Dedekind domain to the DVR gives information of the multiplicity of a prime. Topologically, we may take advantage of the fact that ideals in $\mathbb{Z}_p$ form a chain:

$$\mathbb{Z}_p \supseteq (p) \supseteq (p)^2 \supseteq \ldots$$

More generally, as any discrete valuation ring $\mathcal{O}$ has a chain of ideals, we have:

$$\mathcal{O} \supseteq \mathfrak{p} \supseteq \mathfrak{p}^2 \supseteq \ldots$$

Then these ideals form a neighborhood-basis:

Definition 1.2.1: Complete DVR

let $\mathcal{O}$ be a DVR with discrete valuation ν. Then $\hat{\mathcal{O}}$ is its completion with respect to ν.

If ν is a normalized exponential valuation (see [10, section 20.7]) and $|-| = q^{-\nu}$ for $q > 1$, we have:

$$\mathfrak{p}^n = \left\{ x \in K \;\middle|\; |x| < \frac{1}{q^{n-1}} \right\}$$

and we can define a topology on K^* through the sets:

$$U^{(n)} = 1 + \mathfrak{p}^n = \left\{ x \in K^* \;\middle|\; |1 - x| < \frac{1}{q^{n-1}} \right\} \qquad n > 0$$

so that we have the descending chain:

$$\mathcal{O}^* = U^{(0)} \supseteq U^{(1)} \supseteq U^{(2)} \supseteq \cdots$$

Notice these are all subgroups under multiplication (use that the valuation is a homomorphism and that $1 + \mathfrak{p}^n = \left\{ x \in K^* \;\middle|\; |1 - x|_p < \frac{1}{q^{n-1}} \right\}$ for some choice of q to define the p-absolute value). These subgroups are called the *higher unit groups* and $U^{(1)}$ is called the group of *principal units*. These groups have the following property:

Proposition 1.2.2: Higher Unit Group

For $n \geq 1$,

$$\mathcal{O}^*/U^{(n)} \cong (\mathcal{O}/\mathfrak{p}^n)^* \qquad \text{and} \qquad U^{(n)}/U^{(n+1)} \cong \mathcal{O}/\mathfrak{p}$$

***Proof*:**
Take the surjection $\mathcal{O}^* \to (\mathcal{O}/\mathfrak{p}^n)^*$. The kernel then is certainly $U^{(n)}$. Similarly for the 2nd isomorphism, where $U^{(n)} = 1 + \pi^n \mathcal{O}$ which maps $1 + \pi^n a \mapsto a \bmod \mathfrak{p}$.

Naturally, the completion of $\mathcal{O}$ is best characterized as a colimit of $\mathcal{O}/\mathfrak{p}^n$:

Proposition 1.2.3: Characterizing Complete DVR Via Colimit

Let $\mathcal{O}$ be a DVR. Then:

$$\mathcal{O} \cong \varinjlim_n \mathcal{O}/\mathfrak{p}^n \qquad \mathcal{O}^* \cong \varinjlim_n \mathcal{O}^*/U^{(n)}$$

showing that each complete valuation ring is naturally characterized as a colimit given by its unique maximal ideal.

***Proof*:**
exercise (Neukirch p.128 if you're stuck)

Completion of Discrete Valuation Ring

Let k be a field with a discrete valuation defined on it. This discrete valuation can be extended to $\widehat{k}$ and denoted $\widehat{\nu}$, the completion of k by setting:

$$\widehat{\nu}(a) = \lim_{n\to\infty} \nu(a_n)$$

Now, for any arbitrarily small ϵ, there exists an a_n such that $|a - a_n| < \epsilon$. Letting $\epsilon = p^{-N}$ for some large n ,we see that at some point it must be that:

$$\widehat{\nu}(a - a_n) \geq \widehat{\nu}(a)$$

and so

$$\nu(a_n) = \widehat{\nu}(a_n - a + a) = \min\{\widehat{\nu}(a_n - a), \widehat{\nu}(a)\} = \widehat{\nu}(a)$$

showing that the valuation extends to the completion:

$$\nu(k^*) = \widehat{\nu}(\widehat{k}^*)$$

Proposition 1.2.4: Valuation Ring Quotient And Completion

Let K be a field with valuation ν. Let $\mathcal{O} \subseteq K$ be the valuation ring of K and $\widehat{\mathcal{O}} \subseteq \widehat{K}$ with completed valuation $\widehat{\nu}$. Then:

$$\widehat{\mathcal{O}}/\widehat{\mathfrak{p}} \cong \mathcal{O}/\mathfrak{p}$$

If ν is discrete, then:

$$\widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n \cong \mathcal{O}/\mathfrak{p}^n$$

In particular, the size of the residue field, $\#\widehat{\mathcal{O}}/\widehat{\mathfrak{p}}$, is finite.

***Proof*:**
generalize proposition 1.1.6

Using this, the p-adic expansion extends to the completion of any discrete discrete valuation of K (see Neukirch p.127 for more details). The main advantage of this generalization is to show that the power-series expansion also works for function fields and we get fields of formal power series $K((x))$ with the infinite expansion representation.

Exercise 1.2.1

1. Show that $\sqrt{2} \in \mathbb{Q}_p$ if and only if $x^2 - 2$ factor into linear factors $\mod p$.

1.3 Local and Global Fields

An overarching theme in this entire book is to understand the relation between a field (such as a number field) and its localizations. It was already noted in 1.1.4 that function field and number fields share many of the same characteristics when localizing. Let us box these notions to study this further:

Definition 1.3.1: Global And Local Fields

1. A *global field* if a finite extension of either $\mathbb{Q}$ or $\mathbb{F}_p(t)$.
2. A *local field* is a field which is complete with respect to a discrete valuation ν and its residue field is finite.

Note that the valuation on the completion of global fields is discrete and has a finite residue class field. If we don't specify the finite residue field, we shall get some constructions we are not looking for. We should immediately re-characterize local fields to make them easier to work with:

Proposition 1.3.2: Characterizing Local Fields

Let K be a local field. Then either:

1. K is a finite extension of $\mathbb{Q}_p$
2. K is a finite extension of $\mathbb{F}_p((t))$

Some authors will define local fields to be completion of fields with respect to an absolute value to allow for $\mathbb{R}$ to be a local field, but this will not be done here In particular, Varma does this.

***Proof*:**
Neukirch p.135

I will for now take this as granted, for in my head local fields can be seen as being these by definition. The point of the abstraction is to unify them and not need to keep taking cases, but I will be okay with that for now.

Hence, the local fields of characteristic $p \neq 0$ are of the form $\mathbb{F}_q((t))$ where $q = p^n$, and local fields of characteristic 0 are finite extensions of $K/\mathbb{Q}_p$. The former will be called *power series fields*, and the latter will be called *p-adic number fields*.

Proposition 1.3.3: Compactness Of Local Fields

Let K be a local field. Then K is locally compact, and its valuation ring $\mathcal{O}$ is compact

***Proof*:**
By 1.2.3, $\mathcal{O} \cong \varprojlim \mathcal{O}/\mathfrak{p}^n$ algebraically and topologically. By 1.2.2, $\mathfrak{p}^\nu/\mathfrak{p}^{\nu+1} \cong \mathcal{O}/\mathfrak{p}$, and since $\mathcal{O}/\mathfrak{p}^n$ are finite, they are finite. As $\mathcal{O}$ is a closed subset of the compact product $\prod_{n=1}^{\infty} \mathcal{O}/\mathfrak{p}^n$, we have that the limit is compact, and hence $\mathcal{O}$ is. Next, for every $a \in K$, $a + \mathcal{O}$ is open and a compact neighborhood, so K is locally compact.

For the topologically minded, here is an interesting result: if K is a locally compact field with respect to a nondiscrete topology, it is isomorphic to euclidean $\mathbb{R}$ or $\mathbb{C}$, or to a local field with the valuation topology. Hence, we can say that we are studying locally compact fields.

Definition 1.3.4: Local Number Field

Let $K_\mathfrak{p}/\mathbb{Q}_p$ be a finite extension of $\mathbb{Q}_p$. Then $K_\mathfrak{p}$ is called a *local number field*

Notice that the extension has a sub-script $\mathfrak{p}$ attached to it. Each local number field shall be associated to a prime, as theorem 3.1.2 suggests (but we shall prove it in the proceeding sections).

Proposition 1.3.5: Units Of Local Field

Let K be a local field. Then:

$$K^* = (\pi) \times \mu_{q-1} \times U^{(1)}$$

where π is a prime element, $q = \#(\kappa(\mathfrak{p}) = \#(\mathcal{O}/\mathfrak{p})$, and $U^{(1)} = 1 + \mathfrak{p}$ is the group of principal units

***Proof*:**
For any $x \in K^*$, we have that $k = \pi^n u$ for a appropriate prime π, $n \in \mathbb{Z}$, and $u \in \mathcal{O}^*$. As $\mathcal{O}$ contains the $q-1$ roots of unity μ_{q-1} (by Hensel's lemma, $x^{q-1} - 1$ splits in $\mathcal{O}$). The homomorphism $\mathcal{O}^* \to \kappa^*$ mapping $u \mapsto u \bmod \mathfrak{p}$ has $U^{(1)}$ as a kernel and the group of unity maps bijectively onto κ^*, hence $\mathcal{O}^* = \mu_{q-1} \times U^{(1)}$, as we sought to show.

This decomposition is only the coarse structure. The principal units $U^{(1)}$ carry a much finer description through the p-adic logarithm and exponential, which turn the multiplicative filtration into an additive one and pin down $K^\times$ as a topological group; this is developed in full in section 1.4.

1.3.6 Henselian Fields If you are following Neukirch, this point he takes some time to develop Henselian fields, as it turns out that most of the theory only requires Hensel's lemma (what is covered in the next section), which applies to fields that are more general than Complete fields and local fields. I will omit this for now, as it is not necessary at this stage

The gist is: If R is a DVR, it's henselization is $\widehat{R} \cap K^{\text{sep}}$, where $K = \text{Frac}(R)$. The idea is that Cauchy sequences that converge (in the completion) to the root of a polynomial are *required* to converge, but not every Cauchy sequence needs to converge!

These also show up in algebraic geometry when taking completion at a point of $\operatorname{Spec} R$; we technically add more elements in the global sections than is needed algebraically (the completion will add power series that are not algebraic, ex. possibly e^x), thus we could instead take the *Henselization* instead of the *completion* at a point

1.4 The Multiplicative Group of a Local Field

The additive group of a local field is easy: it is a finite-dimensional vector space over $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$. The multiplicative group carries the arithmetic, and it is the object local class field theory attaches to abelian extensions. This section pins it down completely. The coarse decomposition $K^\times = \pi^{\mathbb{Z}} \times \mu_{q-1} \times U^{(1)}$ of proposition 1.3.5 splits off a copy of $\mathbb{Z}$ (the valuation) and the prime-to-p roots of unity; the work is to understand the principal units $U^{(1)}$, and the tool is the p-adic logarithm, which trades the multiplicative filtration for an additive one. Throughout, K is a p-adic field of residue field $\mathbb{F}_q$ and absolute ramification index $e = \nu(p)$, with ν normalized so that $\nu(K^\times) = \mathbb{Z}$.

The prime-to-p roots of unity have canonical representatives, lifting the residue field multiplicatively.

Proposition 1.4.1: Teichmüller Representatives

There is a unique multiplicative section $\omega\colon \mathbb{F}_q \to \mathcal{O}$ of the reduction map $\mathcal{O} \to \mathbb{F}_q$; that is, $\omega(\bar{a}) \equiv \bar{a} \bmod \mathfrak{p}$, $\omega(\bar{a}\bar{b}) = \omega(\bar{a})\omega(\bar{b})$, and $\omega(\bar{a})$ is characterized as the unique root of $X^q = X$ reducing to $\bar{a}$. Its nonzero values are exactly the group μ_{q-1}, and $\omega\colon \mathbb{F}_q^\times \xrightarrow{\sim} \mu_{q-1}$.

***Proof*:**
The polynomial $X^q - X$ reduces modulo $\mathfrak{p}$ to $\prod_{\bar{a}\in\mathbb{F}_q}(X - \bar{a})$, which is separable (its derivative $qX^{q-1} - 1 \equiv -1$ is a nonzero constant). By Hensel's lemma (theorem 2.1.1) each root $\bar{a} \in \mathbb{F}_q$ lifts to a unique root $\omega(\bar{a}) \in \mathcal{O}$ of $X^q - X$ with $\omega(\bar{a}) \equiv \bar{a}$. Uniqueness of the lift makes ω well defined and forces $\omega(\bar{a}\bar{b}) = \omega(\bar{a})\omega(\bar{b})$, since $\omega(\bar{a})\omega(\bar{b})$ is a root of $X^q = X$ (the roots form a multiplicative set) reducing to $\bar{a}\bar{b}$. For $\bar{a} \neq 0$, $\omega(\bar{a})^{q-1} = 1$, so the nonzero values lie in μ_{q-1}; conversely μ_{q-1} injects into $\mathbb{F}_q^\times$ by reduction and has order $q - 1 = |\mathbb{F}_q^\times|$, so $\omega\colon \mathbb{F}_q^\times \to \mu_{q-1}$ is an isomorphism, as we sought to show.

With the roots of unity understood, only the principal units remain. Their structure is governed by two power series. Recall that the higher unit groups $U^{(n)} = 1 + \mathfrak{p}^n$ filter $U^{(1)}$, with $U^{(n)}/U^{(n+1)} \cong \mathbb{F}_q$ (proposition 1.2.2).

Theorem 1.4.2: The p-adic Logarithm and Exponential

The series

$$\log(1+x) = \sum_{n\geq 1} \frac{(-1)^{n-1}}{n} x^n, \qquad \exp(x) = \sum_{n \geq 0} \frac{x^n}{n!}$$

converge for $\nu(x) > 0$ and for $\nu(x) > \frac{e}{p-1}$ respectively. The logarithm defines a homomorphism $\log\colon U^{(1)} \to K$, and for every integer $n > \frac{e}{p-1}$ the maps

$$\log\colon U^{(n)} \to \mathfrak{p}^n, \qquad \exp\colon \mathfrak{p}^n \to U^{(n)}$$

are mutually inverse isomorphisms of topological groups, the multiplicative $U^{(n)}$ with the additive $\mathfrak{p}^n$.

***Proof*:**
Convergence. For log, $\nu(x^n/n) = n\nu(x) - e\, v_p(n) \geq n\nu(x) - e \log_p n \to \infty$ when $\nu(x) > 0$, so the series converges for $x \in \mathfrak{p}$. For exp, Legendre's formula gives $v_p(n!) = \frac{n - s_p(n)}{p-1} < \frac{n}{p-1}$, so $\nu(x^n/n!) > n\big(\nu(x) - \frac{e}{p-1}\big) \to \infty$ exactly when $\nu(x) > \frac{e}{p-1}$.

Homomorphism. The identity $\log\big((1+x)(1+y)\big) = \log(1+x) + \log(1+y)$ holds as an identity of formal power series over $\mathbb{Q}$, and both sides converge for $x, y \in \mathfrak{p}$, so it holds in K; thus log is a homomorphism $U^{(1)} \to K$.

Isomorphism. Fix $n > \frac{e}{p-1}$ and x with $\nu(x) \geq n$. For $k \geq 2$,

$$\nu\left(\frac{x^k}{k}\right) - \nu(x) = (k-1)\nu(x) - e\, v_p(k) \geq (k-1)\tfrac{e}{p-1} - e \log_p k > 0$$

the last inequality because $(k-1)/(p-1) \geq \log_p k$ with equality only at $k = p$, where $\nu(x) > \frac{e}{p-1}$ makes it strict. Hence the leading term dominates and $\nu(\log(1+x)) = \nu(x)$, so log carries $U^{(n)}$ bijectively onto $\mathfrak{p}^n$ preserving valuation; the same estimate for exp gives $\nu(\exp(x) - 1) = \nu(x)$, carrying $\mathfrak{p}^n$ into $U^{(n)}$. The formal identities $\exp(\log(1+x)) = 1 + x$ and $\log(\exp(x)) = x$ converge in this range, so the two maps are mutually inverse, completing the proof.

The logarithm converts the finer structure of $U^{(1)}$ into a question about a $\mathbb{Z}_p$-lattice, which the next result resolves.

Theorem 1.4.3: Structure of the Principal Units

Let $K/\mathbb{Q}_p$ have degree $d = [K : \mathbb{Q}_p]$. Then

$$U^{(1)} \cong \mu_{p^\infty}(K) \times \mathbb{Z}_p^d$$

where $\mu_{p^\infty}(K)$ is the finite group of p-power roots of unity in K. Consequently

$$K^\times \cong \mathbb{Z} \times \mu(K) \times \mathbb{Z}_p^d$$

where $\mu(K) = \mu_{q-1} \times \mu_{p^\infty}(K)$ is the finite group of all roots of unity in K.

***Proof*:**
Choose $n > \frac{e}{p-1}$. By theorem 1.4.2, log identifies $U^{(n)}$ with the additive group $\mathfrak{p}^n$, which is a free $\mathbb{Z}_p$-module of rank d (it is an open $\mathbb{Z}_p$-lattice in the d-dimensional $\mathbb{Q}_p$-vector space K). Thus $U^{(n)} \cong \mathbb{Z}_p^d$, in particular torsion-free. The quotient $U^{(1)}/U^{(n)}$ is finite (each $U^{(k)}/U^{(k+1)} \cong \mathbb{F}_q$), so $U^{(1)}$ is a finitely generated $\mathbb{Z}_p$-module with free part of rank d; its torsion subgroup is exactly the p-power roots of unity $\mu_{p^\infty}(K)$, finite since $K^\times$ has only finitely many roots of unity of p-power order. The structure theorem for modules over the principal ideal domain $\mathbb{Z}_p$ ([5, Invariant Factor Form of Modules over PID's]) then splits $U^{(1)} \cong \mu_{p^\infty}(K) \times \mathbb{Z}_p^d$. Feeding this into $K^\times = \pi^{\mathbb{Z}} \times \mu_{q-1} \times U^{(1)}$ (proposition 1.3.5) and combining the prime-to-p and p-power roots of unity into $\mu(K)$ gives the second isomorphism, completing the proof.

Example 1.4.4: The Multiplicative Group of $\mathbb{Q}_p$
For $K = \mathbb{Q}_p$ with p odd, $e = 1$ and $\frac{e}{p-1} < 1$, so already $\log\colon U^{(1)} \xrightarrow{\sim} \mathfrak{p} = p\mathbb{Z}_p \cong \mathbb{Z}_p$; there are no nontrivial p-power roots of unity, and

$$\mathbb{Q}_p^\times \cong \mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p$$

the factors being the valuation, the Teichmüller units μ_{p-1}, and the principal units. For $p = 2$, $\frac{e}{p-1} = 1$ so the logarithm is an isomorphism only from $U^{(2)}$ onward; the extra torsion is $\mu_{2^\infty}(\mathbb{Q}_2) = \{\pm 1\}$, and $\mathbb{Q}_2^\times \cong \mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}_2$.

Exercise 1.4.1

1. Compute the Teichmüller representative $\omega(2) \in \mathbb{Z}_5$ of $2 \in \mathbb{F}_5^\times$ modulo 25, using $\omega(2)^4 = 1$.
2. Show that each unit $x \in \mathcal{O}^\times$ factors uniquely as $x = \omega(\bar{x})\,\langle x\rangle$ with $\langle x\rangle \in U^{(1)}$ a principal unit.
3. Using theorem 1.4.2, show that for p odd the group $U^{(1)}$ of $\mathbb{Q}_p$ has no element of order p, and hence is torsion-free.
4. Let $K/\mathbb{Q}_p$ be totally ramified of degree e. Show that $[U^{(1)} : (U^{(1)})^p] = p^d \cdot |\mu_p(K)|$ where $d = e$, and interpret the factor $|\mu_p(K)|$.

1.5 Witt Vectors and the Structure of Complete DVRs

The local fields of characteristic 0 are finite extensions of $\mathbb{Q}_p$, those of characteristic p are the $\mathbb{F}_q((t))$. This raises a structural question: given a residue field k of characteristic p, is there a canonical complete discrete valuation ring with that residue field, and how many are there? In equal characteristic the answer is $k[[t]]$. In mixed characteristic $(0, p)$ the canonical answer is the ring of *Witt vectors* $W(k)$, an arithmetic machine that manufactures $\mathbb{Z}_p$ and the unramified extensions directly from the residue field. This section builds $W(k)$, shows it is the unique absolutely unramified complete DVR over k, and identifies $W(\overline{\mathbb{F}}_p)$ with the ring of integers of the completed maximal unramified extension. This material is optional for the rest of the book, and used later only as background for the arithmetic of the residue field.

Fix a prime p. For $n \geq 0$ the *Witt polynomial* is

$$w_n(X_0, \ldots, X_n) = \sum_{i=0}^{n} p^i X_i^{p^{n-i}} = X_0^{p^n} + pX_1^{p^{n-1}} + \cdots + p^n X_n$$

Given a commutative ring A, the *ghost map* $w \colon A^{\mathbb{N}} \to A^{\mathbb{N}}$ sends $(x_0, x_1, \ldots)$ to its *ghost components* $(w_0(x), w_1(x), \ldots)$. The idea is to make $A^{\mathbb{N}}$ a ring, the ring $W(A)$ of Witt vectors, so that the ghost map becomes a ring homomorphism to the product ring $A^{\mathbb{N}}$; over a $\mathbb{Z}[1/p]$-algebra this forces the operations, and the theorem is that they have integer coefficients.

Lemma 1.5.1: Dwork's Congruence Lemma

Let A be a ring in which p is a nonzerodivisor, equipped with a ring endomorphism φ satisfying $\varphi(a) \equiv a^p \pmod{pA}$ for all $a \in A$. A sequence $(w_0, w_1, \ldots) \in A^{\mathbb{N}}$ is the sequence of ghost components of some (necessarily unique) $x \in A^{\mathbb{N}}$ if and only if

$$w_n \equiv \varphi(w_{n-1}) \pmod{p^n A} \qquad (n \geq 1)$$

***Proof*:**
First note the elementary fact that $a \equiv b \pmod{p^m}$ implies $a^p \equiv b^p \pmod{p^{m+1}}$: writing $a = b + p^m c$, the binomial expansion $a^p = b^p + p\, b^{p-1}(p^m c) + \cdots$ has every term after b^p divisible by p^{m+1}. Iterating, $a \equiv b \pmod{p}$ gives $a^{p^k} \equiv b^{p^k} \pmod{p^{k+1}}$.

($\Rightarrow$) If $w_n = w_n(x)$ then, since $w_n(x) = \sum_{i=0}^{n-1} p^i x_i^{p^{n-i}} + p^n x_n$ and $\varphi(w_{n-1}(x)) = \sum_{i=0}^{n-1} p^i \varphi(x_i)^{p^{n-1-i}}$, the difference is $w_n - \varphi(w_{n-1}) = p^n x_n + \sum_{i=0}^{n-1} p^i \big(x_i^{p^{n-i}} - \varphi(x_i)^{p^{n-1-i}}\big)$. Each bracket vanishes modulo p^{n-i} by the elementary fact (as $\varphi(x_i) \equiv x_i^p \bmod p$ raised to p^{n-1-i}), so each term is divisible by p^n; hence $w_n \equiv \varphi(w_{n-1}) \pmod{p^n}$.

($\Leftarrow$) Conversely, construct x_n recursively so that $w_n(x) = w_n$. Take $x_0 = w_0$. Given $x_0, \ldots, x_{n-1}$ with $w_m(x) = w_m$ for $m < n$, solving $w_n(x) = w_n$ requires $p^n x_n = w_n - \sum_{i<n} p^i x_i^{p^{n-i}}$. By the computation above $\sum_{i<n} p^i x_i^{p^{n-i}} \equiv \varphi(w_{n-1}(x)) = \varphi(w_{n-1}) \pmod{p^n}$, and the hypothesis gives $w_n \equiv \varphi(w_{n-1}) \pmod{p^n}$, so the right-hand side lies in $p^n A$ and $x_n \in A$ exists, uniquely since p is a nonzerodivisor. This determines x, completing the proof.

Theorem 1.5.2: The Ring of Witt Vectors

There are unique polynomials $S_n, P_n \in \mathbb{Z}[X_0, \ldots, X_n, Y_0, \ldots, Y_n]$ with $w_n(S) = w_n(X) + w_n(Y)$ and $w_n(P) = w_n(X)\, w_n(Y)$. For every commutative ring A, the operations $(x) + (y) = (S_n(x, y))_n$ and $(x) \cdot (y) = (P_n(x, y))_n$ make $A^{\mathbb{N}}$ a commutative ring $W(A)$, functorial in A, whose zero and one are $(0, 0, \ldots)$ and $(1, 0, 0, \ldots)$, and for which the ghost map $w\colon W(A) \to A^{\mathbb{N}}$ is a ring homomorphism.

***Proof*:**
Work in the universal p-torsion-free ring $A_0 = \mathbb{Z}[X_i, Y_i]$ with the endomorphism φ raising every variable to its p-th power, so $\varphi(a) \equiv a^p \bmod p$. The sequences $\big(w_n(X) + w_n(Y)\big)_n$ and $\big(w_n(X)\, w_n(Y)\big)_n$ satisfy Dwork's congruence: for the sum, $w_n(X) + w_n(Y) \equiv \varphi(w_{n-1}(X)) + \varphi(w_{n-1}(Y)) = \varphi\big(w_{n-1}(X) + w_{n-1}(Y)\big) \pmod{p^n}$ since each w_n satisfies the congruence and φ is a ring map; likewise for the product. By lemma 1.5.1 each is the ghost sequence of unique $S = (S_n)$, $P = (P_n)$ with $S_n, P_n \in A_0$, giving the polynomials with integer coefficients. For general A, specializing the variables transports the ring axioms: they are polynomial identities that hold in $W(A_0)$, because the ghost map there is injective (as A_0 is p-torsion-free) and the ghost side is the honest product ring, so associativity, commutativity, and distributivity are inherited. Functoriality and the ghost homomorphism property are then immediate from the definitions, completing the proof.

The additive and multiplicative structure is organized by three operators. The *Teichmüller lift* $\tau\colon A \to W(A)$, $\tau(a) = (a, 0, 0, \ldots)$, is multiplicative; the *Verschiebung* $V(a_0, a_1, \ldots) = (0, a_0, a_1, \ldots)$ is additive; and the *Frobenius* F is the ring endomorphism determined on ghost components by $w_n(Fx) = w_{n+1}(x)$. Their ghost identities $w_n(Vx) = p\, w_{n-1}(x)$ and $w_n(Fx) = w_{n+1}(x)$ give, by injectivity of the ghost map over a p-torsion-free ring and then specialization, the fundamental relation

$$VF = p \qquad \text{on } W(A)$$

When $A = k$ has characteristic p, the Frobenius is componentwise, $F(a_0, a_1, \ldots) = (a_0^p, a_1^p, \ldots)$, and if k is perfect F is an automorphism.

Theorem 1.5.3: Witt Vectors of a Perfect Field

Let k be a perfect field of characteristic p. Then $W(k)$ is a complete discrete valuation ring with maximal ideal $pW(k)$, uniformizer p, and residue field k; every element has a unique convergent expansion

$$x = \sum_{i=0}^{\infty} \tau(c_i)\, p^i, \qquad c_i \in k$$

In particular $W(\mathbb{F}_p) = \mathbb{Z}_p$, and $W(\mathbb{F}_q)$ for $q = p^f$ is the ring of integers of the unramified extension of $\mathbb{Q}_p$ of degree f.

***Proof*:**
Since k is perfect, F is an automorphism, so $pW(k) = VF\, W(k) = V\, W(k) = \{(a_0, a_1, \ldots) \mid a_0 = 0\}$, the vectors with vanishing first component. Reduction modulo this ideal is the first-component map $W(k) \to k$, a surjective ring homomorphism with kernel $pW(k)$, so $W(k)/pW(k) \cong k$.

Given x, the difference $x - \tau(x_0)$ has first component $x_0 - x_0 = 0$, hence lies in $pW(k)$; writing $x - \tau(x_0) = p\,x^{(1)}$ and iterating produces $c_i \in k$ with $x \equiv \sum_{i=0}^{n-1} \tau(c_i)p^i \pmod{p^n W(k)}$ for every n. As $W(k) = \varprojlim_n W(k)/p^n W(k)$ is p-adically complete (each $W(k)/p^n W(k) = W_n(k)$ depends only on the first n components), the series converges to x, and the c_i are unique because $\tau(c_0)$ is forced by x mod p and the rest by induction. Thus every nonzero x is $p^v \cdot (\text{unit})$ with v the least index with $c_v \neq 0$, so $W(k)$ is a complete DVR with uniformizer p and residue field k.

For $k = \mathbb{F}_p$ the expansion is $\sum \tau(c_i)p^i$ with $c_i \in \{0, 1, \dots, p-1\}$ Teichmüller digits, which is exactly the p-adic expansion, so $W(\mathbb{F}_p) = \mathbb{Z}_p$. For $k = \mathbb{F}_q$, $W(\mathbb{F}_q)$ is a complete DVR with uniformizer p (so absolutely unramified, $e = 1$) and residue field $\mathbb{F}_q$ of degree f over $\mathbb{F}_p$; by the classification of unramified extensions (proposition 3.3.6) its fraction field is the unramified extension of $\mathbb{Q}_p$ of degree f, and $W(\mathbb{F}_q)$ is its ring of integers, completing the proof.

The construction is canonical enough to characterize the rings it produces, which is the Cohen structure theorem.

Theorem 1.5.4: Cohen Structure Theorem

Let $\mathcal{O}$ be a complete discrete valuation ring with perfect residue field k of characteristic p.

1. If $\mathcal{O}$ has equal characteristic p, then $\mathcal{O} \cong k[[t]]$.
2. If $\mathcal{O}$ has mixed characteristic $(0, p)$, then there is a unique local homomorphism $W(k) \to \mathcal{O}$ lifting id_k; it is injective, and $\mathcal{O}$ is a totally ramified extension of $W(k)$. In particular, if $\mathcal{O}$ is absolutely unramified (p is a uniformizer), then $\mathcal{O} \cong W(k)$.

***Proof Mixed characteristic case*:**
Let $\mathcal{O}$ have mixed characteristic with perfect residue field k. The Teichmüller map $\omega\colon k \to \mathcal{O}$ of proposition 1.4.1 (available since k is perfect and $\mathcal{O}$ is complete) is a multiplicative section of reduction. Define $\Phi\colon W(k) \to \mathcal{O}$ on Witt vectors by $\Phi\big(\sum \tau(c_i)p^i\big) = \sum \omega(c_i)p^i$, the right side converging p-adically in $\mathcal{O}$. That Φ is a ring homomorphism follows because the Witt addition and multiplication polynomials are universal: the identities defining $+$ and $\cdot$ on $W(k)$ are matched by ω and the arithmetic of $\mathcal{O}$, both governed by the same w_n through the ghost components. It lifts id_k by construction and is unique because any lift must send $\tau(c)$ to the unique multiplicative representative $\omega(c)$. Injectivity holds because $W(k)$ is a DVR and $\Phi(p) = p \neq 0$, so the kernel, a proper ideal, is 0; and $\mathcal{O}$, being a complete DVR containing the absolutely unramified $W(k)$ with the same residue field, is generated over it by a uniformizer, hence a totally ramified extension. When p is already a uniformizer of $\mathcal{O}$, the extension has degree 1 and Φ is an isomorphism, completing the proof.

The equal-characteristic case is the analogous lifting with $k[[t]]$ in place of $W(k)$; the assignment $k \mapsto W(k)$ is the mixed-characteristic counterpart of $k \mapsto k[[t]]$. Applying the theory to the algebraic closure of the residue field produces the maximal unramified ring.

Example 1.5.5: Witt Vectors of $\overline{\mathbb{F}}_p$

The field $\overline{\mathbb{F}}_p$ is perfect, so $W(\overline{\mathbb{F}}_p)$ is an absolutely unramified complete DVR with residue field $\overline{\mathbb{F}}_p$. Its fraction field is the completion $\widehat{\mathbb{Q}_p^{\mathrm{nr}}}$ of the maximal unramified extension of $\mathbb{Q}_p$: the finite unramified extensions of $\mathbb{Q}_p$ are the $W(\mathbb{F}_{p^f})[1/p]$ (theorem 1.5.3), their union has residue field

$\overline{\mathbb{F}}_p = \bigcup \mathbb{F}_{p^f}$, and completing gives $W(\overline{\mathbb{F}}_p)[1/p]$. The Frobenius automorphism of $\overline{\mathbb{F}}_p$ lifts, through functoriality of W, to the Frobenius of $\widehat{\mathbb{Q}_p^{\mathrm{nr}}}/\mathbb{Q}_p$, the topological generator of its Galois group $\widehat{\mathbb{Z}}$ met in example 3.3.12.

Exercise 1.5.1

1. Using the Witt polynomials, compute the first two components of $\tau(a) + \tau(b)$ in $W(\mathbb{F}_p)$, and verify that the second is $-\frac{1}{p}\big((a+b)^p - a^p - b^p\big)$ reduced through the Witt addition.
2. Show that for k perfect of characteristic p, the ring $W(k)$ has characteristic 0, and contrast this with the truncation $W_1(k) = k$.
3. Prove that $VF = FV = p$ on $W(\mathbb{F}_p) = \mathbb{Z}_p$ recovers the identities $V(x) = px$ (shift) and $F = \mathrm{id}$, and reconcile this with F being componentwise p-power on $\mathbb{F}_p$.
4. Deduce from theorem 1.5.4 that two complete DVRs with the same perfect residue field of characteristic p and the same ramification index over $W(k)$ that are both absolutely unramified are isomorphic.

2

Hensel's Lemma and Newton Polygons

Having constructed the local fields and their valuations, we turn to the tool that makes them so useful for arithmetic: the ability to lift solutions from the residue field back to the ring itself. Hensel's lemma gives the exact condition under which a factorization modulo the maximal ideal lifts, and Newton polygons refine it by reading the valuations of the roots of a polynomial directly off its coefficients. Together they turn questions about roots over a local field into finite, combinatorial computations.

2.1 Hensel's Lemma

Hensel's lemma can be thought as the foundational result on why we want to work over local fields. Recall that given a polynomial $f(x) \in \mathbb{Z}[x]$, it is possible that $f(x)$ has a root a root mod each prime, but f need not have a root. For example recall that $x^4 + 1 \in \mathbb{Z}[x]$ is irreducible (use the fact that $\sqrt{2}$ is irrational in the proof). However, this polynomial is reducible mod every prime (see [10, chapter 11.3]). When working over local fields, it was shown in theorem 1.1.2 that a polynomial has a solution if it has a solution mod each p^n (or equivalently for $\varprojlim_n R/\mathfrak{m}^n$, for each $\mathfrak{m}^m$). Hensel's lemma says that under mild conditions, it suffices to check that a polynomial has a root mod p (resp. mod $\mathfrak{m}$) for it to have a root mod for each p^n (resp $\mathfrak{m}^n$)! In particular, Hensel's lemma give a way of lifting roots.

Let K be a complete field with respect to to some nonarchemedian absolute value $|-|$, and hence has a corresponding valuation. Let $\mathcal{O} \subseteq K$ be the valuation ring with corresponding unique maximal ideal $\mathfrak{p}$ and residue field $\kappa(\mathfrak{p}) = \mathcal{O}/\mathfrak{p}$. For example, $K = \mathbb{Q}_p$ and $\mathcal{O} = \mathbb{Z}_p$ with ideal (p) and

$\kappa((p)) = \mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z}$. Define a *primitive polynomial* $f(x) \in \mathcal{O}[x]$ if

$$f(x) \not\equiv 0 \bmod \mathfrak{p}$$

which similarly means

$$|f| = \max\{|a_0|_\mathfrak{p}, ..., |a_n|_\mathfrak{p}\} = 1$$

(recall that if $a_i = np^k$ for some $p \in \mathfrak{p},\ p \notin \mathfrak{p}^2$ then $|a_i|_\mathfrak{p} = p^{-k}$). If $\mathcal{O}$ is a Dedekind domain, this is equivalent to saying $\gcd(a_1), ..., (a_n)) = (1)$[1]. Note that it need not be monic. Then factorization mod $\mathfrak{p}$ gives factorization when lifted!

Theorem 2.1.1: Hensel's Lemma

Let K be a complete field with respect to a nonarchemedian absolute value $|-|$, and let $\mathcal{O}$ be the corresponding discrete valuation ring. Let $f(x) \in \mathcal{O}[x]$ be a primitive polynomial. Then if it admits a factorization modulo $\mathfrak{p}$:

$$f(x) \equiv \overline{g}(x)\overline{h}(x) \bmod \mathfrak{p}$$

into relatively prime polynomials $\overline{g}, \overline{h} \in \kappa[x]$, then $f(x)$ admits a factorization

$$f(x) = g(x)h(x)$$

for polynomials $g, h \in \mathcal{O}[x]$ such that the degree of at least g matches, and:

$$g(x) \equiv \overline{g}(x) \bmod \mathfrak{p} \qquad \text{and} \qquad h(x) \equiv \overline{h}(x) \bmod \mathfrak{p}$$

***Proof*:**

We shall inductively construct g, h. First, let $d = \deg(f)$, $\deg(\overline{g}) = m$, and hence $\deg(\overline{h}) \leq d - m$. Let $g_0, h_0 \in \mathcal{O}[x]$ be polynomial representing $\overline{g}$ and $\overline{h}$ respectively where $\deg(g_0) = \deg(\overline{g})$ and $\deg(h_0) \leq d - m$. As $(\overline{g}, \overline{h}) = 1$ are co-prime, there exits $a, b \in \mathcal{O}[x]$ such that $ag_0 + bh_0 = 1$, hence $ag_0 + bh_0 \equiv 1 \bmod \mathfrak{p}$. Now, the trick is to pick among the coefficients of $f - g_0h_0, ag_0bh_0 - 1 \in \mathfrak{p}[x]$ the one with minimum value with respect to $|-|$ (convince yourself this polynomials are in $\mathfrak{p}[x]$). Call this element π.

Using this element, we shall construct g, h that will be the factorizations through the following form:

$$\begin{aligned} g &= g_0 + p_1\pi + p_2\pi^2 + \cdots \\ h &= h_0 + q_1\pi + q_2\pi^2 + \cdots \end{aligned}$$

where $p_i, q_i \in \mathcal{O}[x]$ are polynomials of degree $< m$ and $\leq d - m$ respectively. These are indeed elements of $\mathcal{O}[x]$ when constructed as the limit:

$$\begin{aligned} g_{n-1} &= g_0 + p_1\pi + p_2\pi^2 + \cdots + p_{n-1}\pi^{n-1} \\ h_{n-1} &= h_0 + q_1\pi + q_2\pi^2 + \cdots + p_{n-1}\pi^{n-1} \end{aligned}$$

[1]Note that not all rings of integers have this conditions (in [1] rank-2 valuation on $\mathbb{Q}(x, y)$)

where
$$f \equiv g_{n-1}h_{n-1} \bmod \pi^n \tag{2.1}$$
as the term will always group up to form a polynomial of appropriate degree. What's to do is to show that we can find polynomials g_{n-1}, h_{n-1} that satisfy equation (2.1) which by a small generalization of theorem 1.1.2 shall give the desired polynomial.

For $n = 1$, the solution is immediately given by choice of π. Let us now inductively assume that it works for some $n-1$, and consider:

$$g_n = g_{n-1} + p_n\pi^n \qquad h_n = h_{n-1} + q_n\pi^n$$

Then:
$$f - g_{n-1}h_{n-1} \equiv (g_{n-1}q_n + h_{n-1}p_n)\pi^n \bmod \pi^{n+1}$$
As we are in an integral domain, inverses are well-defined and hence we may divide by π^n. This gives:
$$g_{n-1}q_n + h_{n-1}p_n \equiv (g_0q_n + h_0p_n) \equiv \pi^{-n}(f - g_{n-1}h_{n-1}) \equiv f_n \bmod \pi$$
where $f_n = \pi^{-n}(f - g_{n-1}h_{n-1}) \in \mathcal{O}[x]$. Next, since $g_0a + h_0b \equiv 1 \bmod \pi$, we have:

$$g_0af_n + h_0bf_n \equiv f_n \bmod \pi$$

If we can take $q_n = af_n$ and $p_n = bf_n$, we'd be done, however the degree's may be too big. Hence, write:
$$b(x)f_n(x) = q(x)g_0(x) + p_n(x)$$
where $\deg(p_n) < \deg(g_0) = m$. Since $g_0 \equiv \overline{g} \bmod \mathfrak{p}$ and $\deg(g_0) = \deg(\overline{g})$, the highest coefficient of g_0 i a unit, and so $q(z) \in \mathcal{O}[x]$. Thus, we get:

$$g_0(af_n + h_0q) + h_0p_n \equiv f_n \bmod \pi$$

Finally, omitting the coefficient divisible by π in the polynomial af_nh_0q, we get q_n such that $g_0q_n + h_0p_n \equiv f_n \bmod \pi$, where $\deg(f_n) \leq d$, $\deg(g_0) = m$, and $\deg(h_0p_n) < (d-m) + m = d$, meaning it has degree less than or to $d - m$, giving the induction step, and completing the proof.

Corollary 2.1.2: Hensel's Lemma – Lifting Roots

Let R be a complete DVR with maximal ideal $\mathfrak{p}$ and residue field $k = R/\mathfrak{p}$. Then if f is a monic polynomial whose reduction to $k[x]$ has a *simple* root $\overline{a} \in k$, then $\overline{a}$ can be lifted to a root of f in R.

***Proof*:**
If $\overline{a} \in k = R/\mathfrak{p}$ is a root, then by Hensel's lemma it lifts to a root in R.

Example 2.1.3: Hensel's Lemma

1. (roots of unity) Consider $x^{p-1} - 1 \in \mathbb{Z}_p[x]$. Over $\mathbb{F}_p \cong \mathbb{Z}_p/p\mathbb{Z}_p$, this polynomials splits completely into linear factors. Hence through repeated use of Hensel's lemma it splits in $\mathbb{Z}_p$. As such, the field $\mathbb{Q}_p$ of p-adic number contains the $(p-1)$ roots of unity!

These along with 0 can even from a system of representatives for the residue class field (the usual representatives are $0, 1, ..., p-1$).

2. Consider x^2+1. Then in mod 2, we have that $x^2+1 \equiv (x+1)^2 \bmod 2$, and hence Hensel's lemma cannot lift -1. By theorem 1.1.2, notice that $x^2+1 \bmod 4$ has no solutions, hence x^2+1 has no root in $\mathbb{Q}_2$.

3. Hensel's lemma is a sufficient, but not necessary condition. Consider x^2-p^2 over $\mathbb{Q}_p$. Then $x^2-p^2 \equiv x^2 \bmod p$ has multiple roots, hence Hensel's lemma does not apply. However, p of x^2-p^2 over $\mathbb{Q}_p$, namely $\pm p$. The problem can be seen as there being two possible lifting of $0 \bmod p$.

4. Show that $x^2-98 = x^2-2\cdot 7^2$ in $\mathbb{Q}_7$ has two roots (hint: quadratic reciprocity). However again it does not satisfy the condition of Hensel's lemma.

Corollary 2.1.4: Hensel's Lemma Giving Newtons Method

Let R be a complete DVR, $f \in R[x]$, and $a_0 \in R$ satisfies:

$$|f(a_0)| < |f'(a_0)|^2$$

implying $f'(a_0)$ divides $f(a_0)$. For any $n \geq 0$ define

$$a_{n+1} := a_n - \frac{f(a_n)}{f'(a_n)}$$

Then the sequence (a_n) is well-defined and converges to a unique root $a \in R$ for which

$$|a - a_0| \leq \epsilon := \frac{|f(a_0)|}{|f'(a_0)|^2}$$

Furthermore, $|f(a_n)| < \epsilon^{2n}|f'(a_0)|^2$ for all $n \geq 0$

***Proof*:**
look at Newton's method. (for ex.[12])

Corollary 2.1.5: Hensel-Kürchák Lemma

Let K be a complete field with respect to a non-Archimedean valuation $|-|$. Then for every irreducible polynomial $f(x) = a_0 + \cdots a_n x^n \in K[x]$ such that $a_0 a_n \neq 0$, we have:

$$|f| = \max\{|a_0|, |a_n|\}$$

In particular, if $a_n = 1$ and $a_0 \in \mathcal{O}$, then $f \in \mathcal{O}[x]$

This is a rather powerful result, as it limits the possible irreducible polynomials over such fields!

***Proof*:**
By multiplying by an appropriate element of K, assume that $f \in \mathcal{O}[x]$ and $|f| = 1$. Let a_r be the first coefficient where $|a_r| = 1$ so that:

$$f(x) \equiv x^r(a_r + a_{r+1}x + \cdots + a - nx^{n-r}) \bmod \mathfrak{p}$$

Then $\max\{|a_0|, |a_n|\} < 1$, $0 < r < n$, but then the congruence contradicts Hensel's lemma, as we sought to show.

2.2 Newton Polygons

Hensel's lemma lifts a root once its reduction is known, but it says nothing about where the roots of a polynomial sit when no root is visible modulo $\mathfrak{p}$. The Newton polygon fills this gap: from the valuations of the coefficients alone it reads off the valuations of all the roots, and it sorts the roots into groups of equal valuation, each group cut out by a genuine factor of the polynomial over K. It is the standard combinatorial instrument for p-adic factorization, and downstream it determines reduction types of elliptic curves and the slopes of L-functions. Throughout, K is complete with respect to a discrete valuation ν, extended uniquely to the algebraic closure $\overline{K}$ (theorem 3.1.7), and $f(T) = \sum_{i=0}^{n} a_i T^i \in K[T]$ has $a_0 a_n \neq 0$.

Definition 2.2.1: Newton Polygon

The *Newton polygon* of f is the lower convex hull of the set of points

$$\{(i, \nu(a_i)) \mid 0 \leq i \leq n,\ a_i \neq 0\} \subseteq \mathbb{R}^2$$

Its *slopes* are the slopes of the successive segments, read left to right, and the *horizontal length* of a segment is the difference of the x-coordinates of its endpoints.

Because the hull is convex, the slopes strictly increase from one segment to the next. The content of the theory is that these slopes are precisely the negatives of the valuations of the roots.

Theorem 2.2.2: Newton Polygon and Root Valuations

If the Newton polygon of f has a segment of slope $-\lambda$ and horizontal length ℓ, then f has exactly ℓ roots (counted with multiplicity) in $\overline{K}$ of valuation λ.

***Proof*:**
Let $\alpha_1, \ldots, \alpha_n \in \overline{K}$ be the roots of f, indexed so that $\nu(\alpha_1) \leq \cdots \leq \nu(\alpha_n)$, and write $f(T) = a_n \prod_j (T - \alpha_j)$. The coefficients are the elementary symmetric functions: $a_{n-k}/a_n = (-1)^k e_k(\alpha_1, \ldots, \alpha_n)$. Each monomial of e_k is a product of k distinct roots, so by the ultrametric inequality $\nu(e_k) \geq \nu(\alpha_1) + \cdots + \nu(\alpha_k)$, the right side being the smallest possible sum of k root valuations. Moreover, whenever $\nu(\alpha_k) < \nu(\alpha_{k+1})$ the monomial $\alpha_1 \cdots \alpha_k$ has strictly smaller valuation than every other monomial of e_k, so no cancellation occurs and

$$\nu(a_{n-k}) - \nu(a_n) = \nu(e_k) = \nu(\alpha_1) + \cdots + \nu(\alpha_k)$$

Thus at every index k where the root valuation jumps, the point $(n-k, \nu(a_{n-k}))$ lies on the lower convex hull, and between two consecutive jumps the points lie on or above the segment joining them. The segment of the hull spanning x-coordinates from $n-k'$ to $n-k$ (for a maximal run $\nu(\alpha_{k+1}) = \cdots = \nu(\alpha_{k'}) =: \lambda$) therefore has slope

$$\frac{\nu(a_{n-k'}) - \nu(a_{n-k})}{(n-k') - (n-k)} = \frac{(\nu(\alpha_1) + \cdots + \nu(\alpha_{k'})) - (\nu(\alpha_1) + \cdots + \nu(\alpha_k))}{k - k'} = \frac{(k'-k)\lambda}{-(k'-k)} = -\lambda$$

and horizontal length $k' - k$, the number of roots of valuation λ. Hence each segment of slope $-\lambda$ has horizontal length equal to the number of roots of valuation λ, completing the proof.

The theorem separates the roots by valuation, and this separation is realized by a factorization over K: distinct slopes cannot mix within an irreducible factor.

Corollary 2.2.3: Factorization by Slopes

Write the distinct slopes of the Newton polygon of f as $-\lambda_1 < \cdots < -\lambda_r$ with horizontal lengths $\ell_1, \ldots, \ell_r$. Then $f = a_n\, g_1 \cdots g_r$ with each $g_s \in K[T]$ monic of degree ℓ_s, all of whose roots have valuation λ_s. In particular, if the Newton polygon is a single segment then all roots of f share one valuation.

***Proof*:**
Group the roots by valuation: for each s, let $g_s = \prod_{\nu(\alpha_j)=\lambda_s}(T - \alpha_j)$, a factor of degree ℓ_s by theorem 2.2.2. The Galois group $\mathrm{Gal}_K(\overline{K})$ preserves the valuation, since for $\sigma \in \mathrm{Gal}_K(\overline{K})$ the map $\nu \circ \sigma$ is again an extension to $\overline{K}$ of the valuation of K, and such an extension is unique (theorem 3.1.7); so σ permutes the roots of each fixed valuation among themselves, each g_s is $\mathrm{Gal}_K(\overline{K})$-invariant, and hence $g_s \in K[T]$. The product is f/a_n, giving the factorization, as we sought to show.

A single segment of slope $-1/n$ recovers the Eisenstein criterion and more.

Corollary 2.2.4: Eisenstein via the Newton Polygon

Suppose f is monic of degree n with Newton polygon a single segment from $(0,1)$ to $(n,0)$ passing through no lattice point strictly between its endpoints, that is $\nu(a_0) = 1$ and $\nu(a_i) \geq 1$ for $0 < i < n$. Then f is irreducible over K, and each root generates a totally ramified extension of degree n.

***Proof*:**
By theorem 2.2.2 every root α has valuation $1/n$, so $\nu(\alpha) = 1/n$ forces the ramification index of $K(\alpha)/K$ to be a multiple of n; since $[K(\alpha) : K] \leq n$, it equals n and $K(\alpha)/K$ is totally ramified. Then $\deg f = n = [K(\alpha) : K]$ with α a root of f, so f is the minimal polynomial of α and is irreducible. The hypothesis $\nu(a_0) = 1$, $\nu(a_i) \geq 1$ $(0 < i < n)$, $\nu(a_n) = 0$ is exactly the Eisenstein condition, completing the proof.

Example 2.2.5: Slopes of $T^n - p$ and a Two-Slope Polynomial

1. For $f(T) = T^n - p$ over $\mathbb{Q}_p$ the points are $(0, 1)$ and $(n, 0)$, a single segment of slope $-1/n$; every root has valuation $1/n$, the polynomial is Eisenstein, and $\mathbb{Q}_p(p^{1/n})/\mathbb{Q}_p$ is totally ramified of degree n.

2. For $f(T) = T^3 + pT^2 + pT + p^3$ over $\mathbb{Q}_p$ the points are $(0, 3), (1, 1), (2, 1), (3, 0)$; the lower hull runs from $(0, 3)$ to $(1, 1)$ (slope -2) and from $(1, 1)$ to $(3, 0)$ (slope $-1/2$). So f has one root of valuation 2 and two roots of valuation $1/2$, and factors over $\mathbb{Q}_p$ as a linear times a quadratic, the quadratic being ramified.

Exercise 2.2.1

1. Draw the Newton polygon of $f(T) = T^4 + pT^3 + p^2T + p^5$ over $\mathbb{Q}_p$ and list the valuations of its roots.

2. Show that if the Newton polygon of a monic f has all integer slopes, then the valuations of the roots are integers, and conversely.

3. Use the Newton polygon to prove that $T^p - T - 1/p$ over $\mathbb{Q}_p$ is irreducible (its roots have negative valuation).

4. Let f be monic with a single Newton-polygon slope $-a/n$ in lowest terms with $\gcd(a, n) = 1$. Show that f is irreducible over K.

3

Extensions of Local Fields and Ramification

We now study how a local field sits inside its finite extensions. Because a single prime of the extension lies above the base prime, the whole arithmetic is governed by one ramification index and one residue degree, tied together by the fundamental identity $n = ef$. This chapter develops that theory: the extension of valuations, the local Dedekind-Kummer theorem, and the decomposition of an extension into its unramified and ramified parts.

3.1 Extension of Valuations and the Fundamental Identity

With the definition of the local fields, and the ground work for how to complete any discrete valuation, we now look at how valuations change under field extensions. In this whole section, we shall be consider the following scenario all the time:

Let A be an integral domain, K the field of fractions of A, L a finite extension of K, and B the integral closure of A in L. This setup shall be called the *AKLB setup*.

We shall put some different conditions on A depending on what we need. In number theory, we usually have A be a Dedekind domain.

Recall that each prime $\mathfrak{p} \subseteq A$ where A is a Dedekind domain determines a discrete valuation $\nu_\mathfrak{p} : \mathcal{I}_A \to \mathbb{Z}$ by giving the exponent $n_\mathfrak{p}$ form the unique factorization (i.e. $\nu_\mathfrak{p}(I) = n$ if and only if $IA_\mathfrak{p} = \mathfrak{p}^n A_\mathfrak{p}$). Thus for each prime $\mathfrak{p} \subseteq A$, we have the discrete valuation:

$$\nu_\mathfrak{p} : K \to \mathbb{Z} \qquad \nu_\mathfrak{p}(x) = \nu_\mathfrak{p}(xA)$$

and hence an absolute value $|x|_{\mathfrak{p}} := c^{\nu_{\mathfrak{p}}(x)}$ for $0 < c < 1$. If we have $AKLB$, primes $\mathfrak{P}/\mathfrak{p}$ also give rise to a discrete valuation $\nu_{\mathfrak{P}}$ on L, each restricts to a valuation on K, but the valuation need not be equal to $\nu_{\mathfrak{p}}$. Let us start by understanding the relation

Definition 3.1.1: Extending Valuations

Let L/K be a finite separable extension, and let ν, w be discrete valuation on K, L respectively. Then if $w|_K = e\nu$ for some $e \in \mathbb{N}_{>0}$, then w is said to *extend ν with index e*

Theorem 3.1.2: Extension of Valuation

Let $AKLB$, $\mathfrak{p} \subseteq A$ a prime, and $\mathfrak{P}/\mathfrak{p}$ a prime in B lying over $\mathfrak{p}$. Then there is a discrete valuation $\nu_{\mathfrak{P}}$ which extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}}$, and every discrete valuation on L that extends $\nu_{\mathfrak{p}}$ arises in this way. Thus, we have a bijection $\mathfrak{P} \to \nu_{\mathfrak{P}}$ from $\mathfrak{P}/\mathfrak{p}$ to the set of discrete valuation on L that extend $\nu_{\mathfrak{p}}$.

***Proof*:**
Let $\mathfrak{P}/\mathfrak{p}$ so that $\nu_{\mathfrak{P}}(\mathfrak{p}B) = e_{\mathfrak{P}}$ by definition, and if $\mathfrak{p} \neq \mathfrak{p}'$ then $\nu_{\mathfrak{P}}(\mathfrak{p}'B) = 0$ as $\mathfrak{P}$ lies above only one prime. Now if $I = \prod_{\mathfrak{p}'}(\mathfrak{p}')_{n_{\mathfrak{p}'}}$ is a nonzero fractional ideal then:

$$\nu_{\mathfrak{P}}(IB) = \nu_{\mathfrak{P}}\left(\prod_{\mathfrak{p}'}(\mathfrak{p}')_{n_{\mathfrak{p}'}}B\right) = \nu_{\mathfrak{P}}(\mathfrak{p}^{n_{\mathfrak{p}}}B) = \nu_{\mathfrak{P}}(\mathfrak{p}B)n_{\mathfrak{p}} = e_{\mathfrak{P}}n_{\mathfrak{P}} = e_{\mathfrak{P}}\nu_{\mathfrak{p}}(I)$$

Thus, $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}}$.

Furthermore, if $\mathfrak{P}, \mathfrak{P}'$ are two distinct primes lying over $\mathfrak{p}$, then neither contains the other, and for any $x \in \mathfrak{P} - \mathfrak{P}'$ we have

$$\nu_{\mathfrak{P}}(x) > 0 \geq \nu_{\mathfrak{P}'}(x)$$

and so $\nu_{\mathfrak{P}} \neq \nu_{\mathfrak{P}'}$, and so $\mathfrak{P} \mapsto \nu_{\mathfrak{P}}$ is injective.

To show that every extension of valuation arises in this way, let w be a discrete valuation on L that extends $\nu_{\mathfrak{b}}$ with index e. Let

$$W = \{x \in L \mid w(x) \geq 0\}$$

be the associated DVR and let $\mathfrak{m}$ be the associated maximal ideal. Since $w|_K = e\nu_{\mathfrak{p}}$ by definition of w, the discrete valuation on w is nonzero on A, and hence $A \subseteq W$. The elements of A with nonzero valuation are exactly the elements of $\mathfrak{p}$, and hence

$$\mathfrak{p} = \mathfrak{m} \cap A$$

Furthermore, the DVR W is integrally closed in its fraction field L (see [4, Valuation Ring Properties]), and so $B \subseteq W$ by minimality for B. Now, $\mathfrak{P}$ is prime (since $\mathfrak{m}$ is) and $\mathfrak{p} = \mathfrak{m} \cap A = \mathfrak{P} \cap A$, and so $\mathfrak{P}$ lies over $\mathfrak{p}$. The ring W contains $B_{\mathfrak{P}}$ and is contained in $\text{Frac}B_{\mathfrak{P}} = L$. But now, there are no intermediate rings between a DVR and its fraction field (see [4]), and so $W = B_{\mathfrak{P}}$, and $w = \nu_{\mathfrak{P}}$ with $e = e_{\mathfrak{P}}$, completing the proof.

With this, we return to upgrading our AKLB setup to local fields. We have that that the extension of complete DVR's is "closed". We shall next upgrade the following:

Lemma 3.1.3: Detecting Integral Elements, Complete DVR

Let A be a complete DVR with fraction field K, and L/K be a finite extension of degree n. Then $\alpha \in L$ is integral over A if and only if $N_{L/K}(\alpha) \in A$

Recall that in the case for number fields, this was *not* a sufficient condition [10, chapter 22.1].

***Proof*:**
Let $f = \sum_i^d a_i x^i \in K[x]$ be the minimal polynomial of α. If α is integral over A, then $f \in A[x]$, meaning all it's coefficients are in A including the constant, which is equal to $N_{L/K}(\alpha)^e$ ($e = [L : K(\alpha)]$). Conversely, if $N_{L/K}(\alpha) = (-1)^n f(0)^e \in A$, then $f(0) \in A$ since $f(0) \in K$ is the roots of $x^e - (-1)^n N_{L/K}(\alpha) \in A[x]$ and A is integrally closed. Thus, the constant coefficien of f lies in A, and as it is monic, so does f, and so by corollary 2.1.5 $f \in A[x]$, showing that α is integral, completing the proof.

Theorem 3.1.4: Extension of DVR's

Assume $AKLB$ and let A be a complete DVR with maximal ideal $\mathfrak{p}$. Then B is a DVR whose maximal ideal $\mathfrak{P}$ is necessarily the unique prime ideal above $\mathfrak{p}$.

***Proof*:**
Let's first show that the number of primes lying over $\mathfrak{p}$ is 1. There is at least one $\mathfrak{P} \subseteq B$ lying over $\mathfrak{p}$ as the factorization of $\mathfrak{p}B \not\subseteq B$ is non-trivial by theorem 3.1.2. Now, for the sake of contradiction say that there were $\mathfrak{P}, \mathfrak{P}'$ lying over $\mathfrak{p}$ where $\mathfrak{P} \neq \mathfrak{P}'$. Then choose $b \in \mathfrak{P} - \mathfrak{P}'$ and consider $A[b] \subseteq B$. The ideals $\mathfrak{P} \cap A[b]$ and $\mathfrak{P}' \cap A[b]$ are distinct primes ideals of $A[b]$ containing $\mathfrak{p}A[b]$. Furthermore, both are maximal since they are nonzero and $\dim A[b] = \dim(A) = 1$ (as $A[b]$ is integral over$A$, and therefore has the same dimension). The quotient ring $A[b]/\mathfrak{p}A[b]$ thus has *at least* two maximal ideals. Now let $f \in A[x]$ be the minimal polynomial of b over K and let $\overline{f} \in (A/\mathfrak{p})[x]$ be its reduction to the residue field $A/\mathfrak{p}$. Then we have:

$$\frac{(A/\mathfrak{p})[x]}{(\overline{f})} \cong \frac{A[x]}{(\mathfrak{p}, f)} \cong \frac{A[b]}{\mathfrak{p}A[b]}$$

Hence, the ring $(A/\mathfrak{p})[x]/(\overline{f})$ also has at least two maximal ideals, implying that $\overline{f}$ is divisible by two distinct irreducible polynomials (as $(A/\mathfrak{p})[x]$ is a PID). Thus, we get that $\overline{f} = \overline{g}\overline{h}$. Finally, by Hensel's lemma, these can be lifted to a non-trivial factorization $f = gh$ for $f \in A[x]$, contradicting the irreducibility of f.

To close the argument, recall that every maximal ideal of B lies above a maximal ideal of A, but A has only one maximal ideal $\mathfrak{p}$, so B has a unique (nonzero) maximal ideal $\mathfrak{P}$. Thus B is a local Dedekind domain, hence a local PID, and not a field, which is an equivalent definition to a DVR, completing the proof.

Note that completion is necessary for this lemma to work:

Example 3.1.5: Failure Without Completion

Take $A = \mathbb{Z}_{(5)}$ which is a DVR with maximal ideal (5) but is not complete. Let $K = \mathbb{Q}$ and $L = \mathbb{Q}(i)$. Then $\mathcal{O}_L = \mathbb{Z}_{(5)}[i]$, which is a PID, but is *not* a DVR, as the ideals $(1+2i)$ and $(1-2i)$ are both maximal and not equal. However, if we take $A = \mathbb{Z}_5$ and $K = \mathbb{Q}_5$ with $L = \mathbb{Q}_5(i) = \mathbb{Q}_5$, and $B = \mathbb{Z}_5[i] = \mathbb{Z}_5$, it is a DVR, since $x^2 + 1$ has roots in $\mathbb{F}_5$ and we can lift roots in $\mathbb{Z}_5$ via Hensel's lemma.

Notice that the factorization $5 = (1+2i)(1-2i)$ in $\mathbb{Z}_{(5)}[i]$ when replaced with $\mathbb{Z}_5$ we will get that (5) is the product of the maximal ideal and a unit. In fact, not matter what prime we localize at we get this result!

This immediately gives the following

Corollary 3.1.6: The Fundamental Identity For Local Fields

Let K be a local field and L/K a finite extension. Let $\mathfrak{P}$ be the maximal prime of $\mathcal{O}_L$ and $\mathfrak{p}$ the maximal prime of K. Then:

$$[L : K] = n = ef$$

where e and f are the ramification index and inertia degree respectfully.

***Proof*:**
As there is only one prime lying above, the result follow from the global fundamental identity (see [2, Fundamental Identity])

Finally, we show that completion is preserved too. First, recall that we may define a norm on a vector-space over k $\|-\| : V \to \mathbb{R}_{\geq 0}$, and that if k is complete then all norms on a finite dimensional k-vector-space are equivalent, including the maximal and euclidean norm. Now:

Theorem 3.1.7: Extension of DVR's, Completion

Let A be a complete DVR with maximal ideal $\mathfrak{p}$, $K = \mathrm{Frac}(A)$, and $\nu_{\mathfrak{p}}$ a discrete valuation which gives the absolute value $|x|_{\mathfrak{p}} := c^{\nu_{\mathfrak{p}}(x)}$ where $0 < c < 1$. Let L/K be a finite extension of degree n. Then the following are equivalent:

1. There is a unique absolute value $|x| := |N_{L/K}(x)|_{\mathfrak{p}}^{1/n}$ on L that extends $|-|_{\mathfrak{p}}$
2. The field L is complete with respect to $|-|$, and it's valuation ring $\{x \in L \mid |x| \leq 1\}$ is equal to the integral closure B of A in L

Furthermore, if L/K is separable, then B is a complete DVR whose maximal ideal $\mathfrak{P}$ induces:

$$|x| = |x|_{\mathfrak{P}} := c^{\frac{1}{e_{\mathfrak{P}}}\nu_{\mathfrak{P}}(x)}$$

where $e_{\mathfrak{P}}$ is the ramification index of $\mathfrak{P}$ (recall that $\mathfrak{p}B = \mathfrak{P}^{e_{\mathfrak{P}}}$).

***Proof*:**
Let's first show that $|-|$ is an absolute value. Certainly $|x| = 0$ if and only if $x = 0$, and $|-|$ is multiplicative as the norm is. To show the triangle inequality, it suffices to show that :

$$|x| \leq 1 \Rightarrow |x+1| \leq |x| + 1$$

since in general we always have:

$$|y+z| = |z||y/z+1| \qquad |y| + |z| = |z|(|y/z|+1)$$

Without loss of generality, we may assume that $|y| \leq |z|$. Then we can even show that $|x| \leq 1 \Rightarrow |x+1| \leq 1$ via;

$$\begin{aligned}
&|x| \leq 1 \\
&\Leftrightarrow |N_{L/K}(x)|_{\mathfrak{p}} \leq 1 \\
&\Leftrightarrow N_{L/K}(x) \in A \\
&\Leftrightarrow x \in B \\
&\Leftrightarrow x+1 \in B \\
&\Leftrightarrow |x+1| \leq 1
\end{aligned}$$

The first bi-conditional is from the definition of $|-|$, the second from the definition of $|-|_{\mathfrak{p}}$, the third from lemma 3.1.3, the fourth is since it is a ring, and the fith is from the first three by replacing x with $x+1$. This

Now, let $x \in K$. Then:

$$|x| = |N_{L/K}(x)|_{\mathfrak{p}}^{1/n} = |x^n|_{\mathfrak{p}}^{1/n} = |x|_{\mathfrak{p}}$$

showing it is indeed an extension. If $|-|_{\mathfrak{p}}$ is non-trivial, then $|x|_{\mathfrak{p}} \neq 1$ for some $x \in K^*$, and if $|x|^a = |x|_{\mathfrak{p}} = |x|$, then $a = 1$, showing that $|-|$ is the unique absolute value in its equivalence class extending $|-|_{\mathfrak{p}}$. As every norm on L induces the same topology, $|-|$ is the only absolute value on L that extends $|-|_{\mathfrak{p}}$.

This covers the equivalences, so now assume L/K is separable. Then B is a DVR by theorem 3.1.4. It is complete since it the valuation ring of L. Next, let $\mathfrak{P}$ be the unique maximal ideal of $\mathfrak{P}$. The valuation $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}}$, and so

$$\nu_{\mathfrak{P}}(x) = e_{\mathfrak{P}}\nu_{\mathfrak{p}}(x)$$

Then as $0 < c^{1/e_{\mathfrak{P}}} < 1$, we have the absolute value on L induced by $\nu_{\mathfrak{P}}$ to be:

$$|x|_{\mathfrak{P}} := (c^{1/e_{\mathfrak{P}}})^{\nu_{\mathfrak{P}}(x)}$$

This is indeed equal to $|-|$, for it extends $|-|_{\mathfrak{p}}$ which is know has a unique extension. Computing:

$$y|x|_{\mathfrak{P}} = c^{\frac{1}{e_{\mathfrak{P}}}\nu_{\mathfrak{P}}(x)} = c^{\frac{1}{e_{\mathfrak{P}}}\nu_{\mathfrak{P}}(x)} = c^{\nu_{\mathfrak{p}}(x)} = |x|_{\mathfrak{p}}$$

completing the proof.

By transitivity of the norm in towers, we can uniquely extend the absolute value on the fraction field K of a complete DVR to an algebraic closure $\overline{K}$.

Corollary 3.1.8: DVR Completion And Number Rings

Assume $AKLB$ and let A be a complete DVR with maximal ideal $\mathfrak{p}$. Let $\mathfrak{P}/\mathfrak{p}$. Then

$$\nu_{\mathfrak{P}}(x) = \frac{1}{f_{\mathfrak{P}}}\nu_{\mathfrak{p}}(N_{L/K}(x))$$

for all $x \in L$

***Proof*:**

$$\nu_{\mathfrak{p}}(N_{L/K}(x)) = \nu_{\mathfrak{p}}(N_{L/K}((x))) = \nu_{\mathfrak{p}}(N_{L/K}(\mathfrak{P}^{\nu_{\mathfrak{P}}(x)})) = \nu_{\mathfrak{p}}(\mathfrak{p}^{f_{\mathfrak{P}}\nu_{\mathfrak{P}}(x)}) = f_{\mathfrak{P}}\nu_{\mathfrak{P}}(x)$$

3.2 Dedekind-Kummer for Local Rings

Recall the Dedekind-Kummer Theorem ([10, chapter 22.4]) where a prime $\mathfrak{P}/\mathfrak{p}$ factors like the minimal polynomial of the finite separable extension L/K (i.e. $AKLB$) if the ring of integers of L is monogenic or the prime doesn't contain the conductor. In the case where the ring of integers of K is a DVR and the residue of the extension is finite, we get that B is always monogenic. This require Nakayama's lemma which we remind the reader of:

Lemma 3.2.1: Nakayama's Lemma

Let A be a local ring with maximal ideal $\mathfrak{p}$ and let M b e a finitely generated A-module. If the image of $x_1, ..., x_n \in M$ generate $M/\mathfrak{p}M$ as an $(A/\mathfrak{p})$-vector space, then $x_1, ..., x_n$ generate M as an A-module.

***Proof*:**

see [4, Nakayama's Lemma III]

This gives a local version of the Dedekind-Kummer theorem that doesn't even need that A, B be Dedekind domain

Corollary 3.2.2: Local Ring Maximal Ideal Extension

Let A be a local Noetherian ring with maximal ideal $\mathfrak{p}$, $g \in A[x]$ be a polynomial with quotient $\overline{g} \in (A/\mathfrak{p})[x]$, and let α be the image of x in the ring $B := A[x]/(g(x))$. Then the maximal ideals of B are $(\mathfrak{p}, g_i(\alpha))$ where $g_1, ..., g_m \in A[x]$ are lifts of the distinct irreducible polynomials $\overline{g}_i \in (A/\mathfrak{p})][x]$ that divide $\overline{g}$.

Nakayama will give the correspondence, and the fact that A has only a single maximal ideal will force that the extension have only so many possible maximal ideals:

***Proof*:**
By Nakayama's lemma, the quotient map $B \to B/\mathfrak{p}B$ gives a one-to-one correspondence between the maximal ideals of B and the maximal ideals of $B/\mathfrak{p}B$, and hence:

$$B/\mathfrak{p}B \cong A[x]/(\mathfrak{p}, g(x)) \cong (A/\mathfrak{p})[x]/(\overline{g}(x))$$

Now, each maximal ideal of the right hand side is the quotient of an irreducible divisor of $\overline{g}$ (you can use the Chinese remainder theorem if you're unconvinced) and hence as $(A/\mathfrak{p})[x]$ is a PID it is one of the $\overline{g_i}$, as we sought to show.

Theorem 3.2.3: Local Dedekind-Kummer Theorem

Take $AKLB$, where A, B are DVR's, and let $\kappa = A/\mathfrak{p}, \lambda = B/\mathfrak{P}$ be the residue fields given a prime $\mathfrak{P}$ over $\mathfrak{p}$. Then if λ/κ is separable,

$$B = A[\alpha]$$

for appropriate $\alpha \in B$. If L/K is unramified, this then holds for every lift α for any generator $\overline{\alpha}$ for $\lambda = \kappa(\alpha)$.

Hence, by [10, Corollary 22.4.8], every ring of integers extension $\mathcal{O}_L/\mathcal{O}_K$ is monogenic, and the [global] Dedekind-Kummer Theorem applies.

***Proof*:**
Let $\mathfrak{p}B = \mathfrak{P}^e$ be the factorization of $\mathfrak{P}/\mathfrak{p}$ and let $f = [\lambda : \kappa$ be the inertia degree so that $ef = n := [L : K]$. As the extension λ/κ is separable, there exists a primitive element

$$\lambda = \kappa(\overline{\alpha}_0)$$

for some $\overline{\alpha}_0 \in \lambda$ whose minimal polynomial $\overline{g}$ is separable of degree f. Let $g \in A[x]$ be the monic lift of $\overline{g}$ and let α_0 be any lift of $\overline{\alpha}_0$ to B. If $\nu_{\mathfrak{P}}(g(\alpha_0)) = 1$, let $\alpha := \alpha_0$. Otherwise, let π_+0 be any uniformizer for B and let $\alpha := \alpha_0 + \pi_0 \in B$. Then $\alpha \equiv \alpha_0 \bmod \mathfrak{P}$. Decomposing

$$g(x + \pi_0) = g(x) + \pi_0 g'(x) + \pi_0^2 h(x)$$

for appropriate $h \in A[x]$, we get:

$$\nu_{\mathfrak{P}}(g(\alpha)) = \nu_{\mathfrak{P}}(g(\alpha_0 + \pi_0)) = \nu_F B(g(x + \pi_0) = g(x) + \pi_0 g'(x) + \pi_0^2 h(x)) = 1$$

and hence $\pi := g(\alpha)$ is the uniformizer of B.

Now, we claim that $B = A[\alpha]$. This is the same as $1, \alpha, ..., \alpha^{n-1}$ generates B as an A-module. By Nakayama's lemma, this is equivalent to asking if $1, \alpha, ..., \alpha^{n-1}$ span $B/\mathfrak{p}B$ as a κ-vector space. As $\mathfrak{p} = \mathfrak{P}^e$, $\mathfrak{p}B = (\pi^e)$. This means we can represent each element of $B/\mathfrak{p}B$ as cosets:

$$b + \mathfrak{p}B = b_0 + b_1\pi + b_2\pi + \cdots + b_{e-1}\pi^{e-1} + \mathfrak{p}B$$

where $b_0, ..., b_{e-1}$ are given up to equivalence modulo πB. Now the field theory $1, \overline{\alpha}, ..., \overline{\alpha}^{f-1}$

forms a basis for $B/\pi B = B/\mathfrak{P}$ as a κ-vector space, so we can write this as:

$$\begin{aligned} b + \mathfrak{p}B = {} & (a_0 + a_1\alpha + \cdots + a_{f-1}\alpha^{f-1}) \\ & + (a_f + a_{f+1}\alpha + \cdots + a_{2f-1}\alpha^{f-1})g(\alpha) \\ & + \cdots \\ & + (a_{ef-f+1} + a_{ef-f+2}\alpha + \cdots + a_{ef-1}\alpha^{f-1})g(\alpha)^{e-1} + \mathfrak{p}B \end{aligned}$$

As $\deg g = f$ and $n = ef$, this expresses $b + \mathfrak{p}B$ in the form $b' + \mathfrak{p}B$ with b' in the A-span of $1, ..., \alpha^{n-1}$. Thus:

$$B = A[\alpha]$$

Finally, if L/K is unramified ,then λ/κ is separable, and $e = 1$, $f = n$, and so there there is no need to check if $g(\alpha)$ is a uniformizer which immediately gives $\alpha = \alpha_0$, as we sought to show.

3.3 Unramified, Tame, and Wild Extensions

Let L/K be a finite extension of local fields. This section shall explore how to decompose such an extension into an unramified extension and a completely ramified extension. To remind the reader:

Definition 3.3.1: Ramification Index And Inertia Degree Of Local Fields

Let K be a local field and L/K a finite extension. Let ν be the discrete valuation associated to K and w be extension of ν given by theorem 3.1.2. Then as $\nu(K^*) \subseteq w(L^*)$ define the *ramification index* of L with respect to K to be:

$$e(L : K) = [w(L^*) : \nu(K^*)]$$

and the *inertia degree* of L with respect to K to be

$$f(L : K) = [\lambda : \kappa]$$

where λ, κ are the residue fields of L and K respectively.

Notice that $w = \frac{1}{n}\nu \circ N_{L/K}$, and hence is discrete. If $\mathfrak{o}, \mathfrak{p}, \pi$ and $\mathcal{O}, \mathfrak{P}, \Pi$ are the valuation-ring/maximal-ideal/prime-element of K and L respectively, then:

$$e(L : K) = [w(\Pi)\mathbb{Z} : \nu(\pi)\mathbb{Z}]$$

Hence $\nu(\pi) = ew(\Pi)$ which gives:

$$\pi = u\Pi^e$$

for some unit $u \in \mathcal{O}^*$, which mirrors the result for the definition of ramification index in the case of global fields.

3.3.1 Unramified Extensions

With the fundamental identity property re-defined, let us upgrade the definition of unramified extensions from [10, chapter 22.4]:

Definition 3.3.2: Unramified Extension for local field

Let A be a complete DVR with finite residue field, K it's field of fractions, and L a finite extension. Then if:

$$[L : K] = [\kappa(\mathfrak{P}) : \kappa(\mathfrak{p})] = f$$

Then L/K is said to be an *unramified extension.*

Hence, when L/K is finite and unramified, the degree n of L/K has the same degree as the finite separable extension λ/κ

$$[L : K] = [\lambda : \kappa]$$

We may thus ask what is the relation of L/K and λ/κ. In particular, if we fix K with residue field κ, what is the relationship between the finite unramified extensions L/K of degree n and the finite separable extensions λ/κ of degree n. We already know that each L/K uniquely determined a corresponding λ/κ, but is the converse true too? In fact, is is almost as nice as it gets:

Theorem 3.3.3: Unramified Extension And Residue Field Category Equivalence

Let A be a complete DVR with $K = \mathrm{Frac}(A)$ its residue field. Then the category $\mathcal{C}_K^{nr}$ and $\mathcal{C}_\kappa^{\mathrm{sep}}$ are equivalent; the functor $\mathcal{F} : \mathcal{C}_K^{nr} \to \mathcal{C}_\kappa^{\mathrm{sep}}$ sends each unramified extension L/K to its residue field λ, and each K-algebra homomorphism $\varphi : L_1 \to L_2$ to the κ-algebra homomorphism $\overline{\varphi} : \lambda_1 \to \lambda_2$ given by $\overline{\varphi}(\overline{\alpha}) := \overline{\varphi(\alpha)}$ where α is any lift of $\overline{\alpha} \in \lambda_1 = B_1/\mathfrak{P}_1$ to B_1 and $\overline{\varphi}$ is the reduction of $\varphi(\alpha) \in B_2$ to $\lambda_2 := B_2/\mathfrak{P}_2$.

This means that $\mathcal{F}$ gives a bijection between the isomorphism classes in $\mathcal{C}_K^{nr}$, and $\mathcal{C}_\kappa^{\mathrm{sep}}$ and if L_1, L_2 have residue fields λ_1, λ_2, then $\mathcal{F}$ induces a bijection of finite sets:

$$\mathrm{Hom}_K(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_\kappa(\lambda_1, \lambda_2)$$

The most interesting part of the proof is $\mathrm{Hom}_K(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_\kappa(\lambda_1, \lambda_2)$, which will ultimately come down to Hensel's lemma after all the appropriate build-up.

***Proof*:**
Let us start with showing that $\mathcal{F}$ is well-defined. The object certainly map to the object, so we need to show that the morphism map is independent of lifts. Let $\varphi : L_1 \to L_2$ be a K-algebra homomorphism, and for $\overline{\alpha} \in \lambda_1$, let α, α' be two lifts to B_1. Then $\alpha - \alpha' \in \mathfrak{P}_1$, implying $\varphi(\alpha - \alpha') \in \varphi(FB_1) \stackrel{!}{=} \varphi(B_1) \cap \mathfrak{P}_2 \subseteq \mathfrak{P}_2$, and hence $\overline{\varphi(\alpha)} = \overline{\varphi(\alpha')}$, showing it is well-defined. The $\stackrel{!}{=}$ equality comes from φ restricts to the ring homomorphism $B_1 \to B_2$ and $B_2/\varphi(B_1)$ is a finite extension of DVRs in which $\mathfrak{P}_2$ lies over the prime $\varphi(\mathfrak{P}_1)$ of $\varphi(B_1)$. Checking that the identity is set to the identity and composition is well-defined now follows quickly.

Let us now show that $\mathcal{F}$ gives an equivalence of categories. Recall from [3, Characterization of Equivalences] that this means that:

1. $\mathcal{F}$ is essentially surjective: each separable extension λ/κ is isomorphic to the residue field of some unramified L/K
2. $\mathcal{F}$ is full and faithful, i.e. the hom-map is a bijection.

Starting with essential surjectivity, given a finite separable extension λ/κ, we may apply the primitive element theorem and get:

$$\lambda \cong k(\overline{\alpha}) \cong \frac{\kappa[x]}{(\overline{g}(x))}$$

For appropriate $\overline{\alpha} \in \lambda$ whose minimal polynomial $\overline{g}$ is necessarily monic, irreducible, separable, and of degree $n := [\lambda : \kappa]$. Take any monic lift $g \in A[x]$. Then g is also irreducible, separable, and of degree $n>$ Now let :

$$L := \frac{K[x]}{(g(x))} = K(\alpha)$$

Then L/K is a finite separable extensions. By corollary 3.2.2 $(\mathfrak{p}, g(\alpha))$ is the unique maximal ideal of $A[\alpha]$, and :

$$B/\mathfrak{P} \cong \frac{A[\alpha]}{(\mathfrak{p}, g(\alpha))} \cong \frac{A[x]}{(\mathfrak{p}, g(x))} \cong \frac{(A/\mathfrak{p})[x]}{(\overline{g}(x))} \cong \lambda$$

and hence:

$$[L : K] = \deg g = [\lambda : \kappa] = n \ [= f]$$

showing that L/K is unramified, namely by the fundamental identify the ramification index of $\mathfrak{P}$ is necessarily 1.

Next, to show that $\mathcal{F}$ is full and faithful, let us start with the induced maps:

$$\mathrm{Hom}_K(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_A(B_1, B_2) \to \mathrm{Hom}_\kappa(\lambda_1, \lambda_2)$$

The first map was shown in [10, Chapter 22.1]. It must be shown that the same applies for the second map.

First, notice that:

$$\mathrm{Hom}_A(B_1, B_2) = \mathrm{Hom}_A\left(\frac{A[x]}{(g(x))}, B_2\right) = \mathrm{Hom}_A(A(\alpha), B_2)$$

which comes from B_1 always being a monogenic extension of A (Theorem 3.2.3) and that each A-module homomorphism is uniquely determined by the image of x in B_2. This gives a bijection between $\mathrm{Hom}_A(B_1, B_2)$ and the roots of g in B_2. Now, with the right choice of α, we can give the map. By the primitive root theorem, $\lambda_1 = \kappa(\overline{\alpha})$. Lift $\overline{\alpha}$ to $\alpha \in B_1$. Then $L_1 = K(\alpha)$ since $[L_1 : K] = [\lambda_1 : \kappa]$, and these two quantities equal to the degree of the minimal polynomial of $\overline{g}$ of $\overline{\alpha}$ which cannot be less han the degree of the minimal polynomial g of α (both are monic, so no issues will come from characteristics). Then we saw in theorem 3.2.3 that $B)_1 = A[\alpha]$. Now, consider:

$$\mathrm{Hom}_\kappa(\lambda_1, \lambda_2) = \mathrm{Hom}_\kappa\left(\frac{\kappa[x]}{(\overline{g}(x))}, \lambda_2\right) = \mathrm{Hom}_\kappa(\kappa(\overline{\alpha}), \lambda_2)$$

where the equality again comes from the uniquely determined image of x in λ_2, and there is a bijection between $\mathrm{Hom}_\kappa(\lambda_1, \lambda_2)$ and the roots of $\overline{g}$ in λ_2. Now, $\overline{g}$ is separable, so by Hensel's lemma every root of $\overline{g}$ in $\lambda_2 = B_2/\mathfrak{P}_2$ lifts to unique roots of g in B_2. But then the map is a bijection, as we sought to show.

Note that for the bijection it was only necessary that L_1/K be unramified, so at least the bijection holds if L_2/K is unramified. Furthermore, using this bijection we can compute the galois group of automorphisms of a field by looking at the automorphisms of the residue field.

Corollary 3.3.4: Characterizing Unramified Extensions through residue field

Assume $AKLB$ where A is a complete DVR with residue field κ. Then L/K is unramified if and only if $B = A[\alpha]$ for some $\alpha \in L$ whose minimal polynomial has separable image in $\kappa[x]$.

***Proof*:**
The "if" was given by theorem 3.3.3. For the reverse, note that if g is the minimal polynomial, $\overline{g}$ must be irreducible, otherwise Hensel's lemma would give a lift to a non-trivial factorization of $\overline{g}$, so the residue field extension is separable and has the same degree as L/K, and hence L/K is unramified as $[L:K] = f$, completing the proof.

We shall now build-up to show that all unramified extensions where the residue field κ is finite will be isomorphic to a Cyclotomic extension. Starting with the following:

Lemma 3.3.5: Unramified Cyclotomic Extensions

Let A be a complete DVR with fraction field K and residue field κ, and let ζ_n be the primitive nth roots of unity in some algebraic closure of K with $(n, p) = 1$ where $\operatorname{char} \kappa = p$. Then $K(\zeta_n)/K$ is unramified

***Proof*:**
By the relation's of the root of $f(x) = x^n - 1$, The field $K(\zeta_n)$ is splitting filed of this polynomial over K, The image $\overline{f}$ is separable when $\nmid n$ as $\gcd(\overline{f}, \overline{f}') \neq 1$ if and only if $\overline{f}' = nx^{n-1}$ is zero, or equivalently when $p \big| n$. When $\overline{f}$ is separable, so are all of its divisors, iclusing the reduction of the minimal polynomial of ζ_n (which is a factor of f after dividing by $x - 1$), and it must be irreducible since otherwise we would contain a non-trivial factorization by Hensel's lemma. THus ,the residue field $\kappa(\zeta_n)$ is a separable extension of κ, and so by corollary 3.3.4 $K(\zeta_n)/K$ is unramified, as we sought to show.

Adding on the extra condition that the residue field is finite (which is always the case for local fields), we get to characterize unramified extnesions:

Proposition 3.3.6: Characterizing Unramified Extension Of Local Fields

Let A be a complete DVR with fraction field K and finite residue field $\mathbb{F}_q$. Let L/K be a degree n extension. Then L/K is unramified iff

$$L \cong K(\zeta_{q^n-1})$$

Furthermore, $A[\zeta_{q^n-1}]$ is the integral closure of A in L, and L/K is a galois extension with

$$\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$$

***Proof*:**
The above corollary showed the "only if" direction, so let us show the "if direction" by assuming L/K is unramified. Then $[\lambda : \kappa] = [L : K] = n$ where by the finiteness of the residue field $\lambda = \mathbb{F}_{q^n}$. By field theory, $\mathbb{F}_{q^n}$ has the multiplicative group of order $q^n - 1$ generated by some $\overline{\alpha}$, say $\overline{g}$ is its minimal polynomial. Then $\overline{g}$ divides $x^{q^n-1} - 1$, and sine $\overline{g}$ is irreducible, $(\overline{g})$ is coprime to $(x^{q^n-1} - 1)/\overline{g}$. By Hensel's lemma, we can lift this all to $A[\zeta_{q^n-1}]$ and show that g divides $x^{q^n-1} - 1 \in A[x]$, and again by Hensel's lemma we can lift $\overline{\alpha}$ to a root α of g which shows that α is a root of $x^{q^n-1} - 1$. Thus, it is a primitive $(q^n - 1)$-root of unity as the reduction is, which implies that

$$L \cong K(\zeta_{q^n-1})$$

Finally, by theorem 3.2.3 we have $B \cong A[\zeta_{q^n} - 1]$ and L is the splitting field of $x^{q^n-1} - 1$ (we can lift the factorization of this polynomial from $\mathbb{F}_{q^n}$ to L via Hensel's lemma). Hence L/K is galois, and the bijection between the $(q^n - 1)$-roots of unity in L and $\mathbb{F}_{q^n}$ induces $\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$, as we sought to show.

The fact that an extension is unramified if and only if $L \cong K(\zeta_{q^n-1})$ implies that we can also deduce when a cyclic extension *is* ramified:

Corollary 3.3.7: Ramification Of Cyclic Extension

Let A be a complete DVR with field of fraction K and finite residue field of characteristic p. Suppose that K does not contain a primitive pth root of unity. Then the extension $K(\zeta_m)/K$ is ramified if and only if $p \big| m$.

***Proof*:**
If $p \nmid m$, by lemma 3.3.5 $K(\zeta_m)/K$ is unramified. if $p \big| m$, then $K(\zeta_m)$ must contain $K(\zeta_p)$ which by proposition 3.3.6 is unramified if and only if $K(\zeta_p) \cong K(\zeta_{p^n-1})$ with $n = [K(\zeta_p) : K]$, which occurs if $p \big| p^n - 1$ (since $\zeta_p \notin K$ by assumption), which certainly is not the case. Hence, $K(\zeta_p)$ is ramified, implying $K(\zeta_m)$ is ramified, as we sought to show.

Now with a choice of complete DVR, we can fully understand the unramified extensions

Example 3.3.8: Local Number Fields
Let $A = \mathbb{Z}_p$ so that $K = \mathbb{Q}_p$ and $\kappa = \mathbb{F}_p$. Then for each n, the galois field $\mathbb{F}_{p^n}$ is the unique extension of degree n. Thus, by the above results, there is a unique unramified extension of degree n, $L/\mathbb{Q}_p$. This can be explicitly constructed by adjoining a primitive root of unity ζ_{p^n-1} to $\mathbb{Q}_p$.

Maximal Unramified Extension

Let us build-up to the notion of maximal unramified extension, which is an important object for the Artin map. First, the "3rd isomorphism theorem" property applies for unramified extensions

Proposition 3.3.9: Unramified Square

Let L/K and M/K be two extensions, with $\mathfrak{p} \subseteq K$, $\mathfrak{P} \subseteq L$, $\mathfrak{P}' \subseteq M$, $\mathfrak{P} \cap K = \mathfrak{P}' \cap K = \mathfrak{p}$. Then if L/K is unramified for $\mathfrak{P}$ over $\mathfrak{p}$ if LM/M is unramified for $\mathfrak{P}'$ over $\mathfrak{p}$.

Visually,

$$\begin{tikzcd}
 & LM & \\
L \arrow[ur] & \Rightarrow & M \arrow[ul, "\text{unramified}"'] \\
 & K \arrow[ul, "\text{unramified}"] \arrow[ur] &
\end{tikzcd}$$

***Proof*:**
Define:

$$\mathfrak{o}, \mathfrak{p}, \kappa' \qquad \mathcal{O}, \mathfrak{P}, \lambda, \qquad \mathfrak{o}', \mathfrak{p}', \kappa \qquad \mathcal{O}', \mathfrak{P}', \lambda'$$

to the ring of integers, prime, and residue field of K, L, M, LM respectively, and assume L/K is finite. Then λ/κ is also finite and separable, and hence generated by a primitive element $\overline{\alpha}$, $\lambda = \kappa(\overline{\alpha})$. Without loss of generality, we may have $\alpha \in \mathcal{O}$ by multiplying out the denominator, and hence there is a minimal polynomial for α, $f(x) \in \mathfrak{o}[k]$. Let $\overline{f} \in \kappa[x]$ be $\overline{f}(x) = f(x) \bmod \mathfrak{p}$. By minimality, the minimal polynomial of α in λ divides $\overline{f}$, hence:

$$[\lambda : \kappa] \leq \deg(\overline{f}) = \deg(f) = [K(\alpha) : K] \leq [L : K] = [\lambda : \kappa]$$

hence $[\lambda : \kappa] \leq [K(\alpha) : K] \leq [\lambda : \kappa]$, and so $L = K(\alpha)$, which also shows that $\overline{f}$ is the minimal polynomial of $\overline{\alpha}$ over κ.

Thus, $LM = M(\alpha)$. To show this extension is unramified, let $g(x) \in \mathcal{O}'[x]$ be the minimal polynomial of α over k', and let $\overline{g}(x) \equiv g(x) \bmod \mathfrak{p}' \in \kappa'[x]$. Then as a factor of $\overline{f}$, it is separable, hence irreducible over κ' (or else $g(x)$ is reducible by Hensel's lemma!). Thus we get:

$$[\lambda' : \kappa'] \leq [LM : M] = \deg(g) = \deg(\overline{g}) = [\kappa'(\overline{\alpha}) : \kappa'] \leq [\lambda' : \kappa']$$

Giving that LM/M is unramified.

Finally, if L/K is a extension of the unramified extension LM/K , then it follows that LM/L is unramified, and thus so is L/K by the degree formula, as we sought to show.

Corollary 3.3.10: Composite Of Unramified Extensions

The composite of two unramified extensions of L is again unramified

***Proof*:**
It suffices to show that if L/K, M/K, namely as L/K is unramified by proposition 3.3.9 LM/M is unramified, and hence LM/K is unramified as separability is transitive and the degrees are multiplicative.

Hence, the following is well-defined:

Definition 3.3.11: Maximally Unramified Extension (Hilbert Class Field)

Let L/K be an algebraic extensions. Then the composite of all unramified extensions M/K is called the *maximally unramified extension* of L/K.

If $\overline{K}$ is the algebraic closure of K,then K_{nr}/K is the maximal unramified extension of K.

The term K_{nr} is given by the french “non-ramifiée”.

Example 3.3.12: Maximal Unramified Extension For Local Fields

Take $K = \mathbb{Q}_p$. Then $\mathbb{Q}_p^{nr}$ will be an finite extension with galois group:

$$\mathrm{Gal}_{\mathbb{Q}_p}(\mathbb{Q}_p^{nr}) \cong \mathrm{Gal}_{\mathbb{F}_p}(\overline{\mathbb{F}}_f) = \varprojlim_n \mathrm{Gal}_{\mathbb{F}_p}(\mathbb{F}_{p^n}) \cong \varprojlim_n \mathbb{Z}/n\mathbb{Z} =: \hat{\mathbb{Z}}$$

where the last ring is called the *profinite completion* of $\mathbb{Z}$. The field has value group $\mathbb{Z}$ (it's image under valuation) and residue field $\overline{\mathbb{F}}_p$.

Corollary 3.3.13: Root Of Unity In Maximally Unramified Extension

Let K_{nr}/K be the maximally unramified extension. Then K_{nr} contains all the roots of unity of order M not divisible by the characteristic κ

If κ is a finite field, then the extension K_{nr}/K is generated by the roots of unity.

***Proof*:**
This is putting together the results we saw earlier. The polynomial $x^m - 1$ splits over $\overline{\kappa}_s$, and hence over K_{nr} by Hensel's lemma. Similarly, as the roots of unity generate $\overline{\kappa}_s/\kappa$, by Hensel's they also generate K_{nr}

This also shows then that K_{nr} is a Cyclotomic extension of K.

Proposition 3.3.14: Residue Field Of Maximally Unramified Extension

Let M/K be the maximal unramified extension of K. Then the residue class field of M is the separable closure of λ_s of κ.

***Proof*:**
This follows from the equivalence of categories presented earlier.

To wrap up this section, let us now look at decomposing an extension L/K into the ramified and unramified portion, and in the case where L/K is galois to find what it is:

Theorem 3.3.15: Tower Extension And Galois Group

Assume $AKLB$ with A a complete DVR and separable residue field extension λ/κ. Let e, f be the ramification index and residue field degree, and let $\mathfrak{P}$ be the unique prime of B. Then:

1. There is a unique intermediate field extension E/K that contains every unramified extension of K in L and has degree $[E:K] = f$
2. The extension L/E is totally ramified and has degree $[L:E] = e$
3. If L/K is galois, then $\mathrm{Gal}_K(L)$ is the decomposition group of $D_{\mathfrak{P}}$. $\mathrm{Gal}_E(L)$ is the inertia subgroup of $I_{\mathfrak{P}}$, and E/K is galois with $\mathrm{Gal}_K(E) \cong D_{\mathfrak{P}}/I_{\mathfrak{P}} \cong \mathrm{Gal}_\kappa(\lambda)$

***Proof*:**

1. exercise
2. tower argument
3. Take $D_{\mathfrak{P}} \subseteq \mathrm{Gal}_K(L)$. Then as there is only one prime, the entire group must be the stabilizer, and so $D_{\mathfrak{P}} = \mathrm{Gal}_K(L)$. Take now $\mathfrak{P}_E = \mathfrak{P} \cap E$. Then recall that we get:
$$I_{\mathfrak{P}_E} = \mathrm{Gal}_E(L) \cap I_{\mathfrak{P}}$$
then these thee groups all have order e, and will then have to be equal by their proprieties. As $I_{\mathfrak{P}}$ is a normal subgroup of $D_{\mathfrak{P}}$, E/K is normal, and as it is separable it is Galois. It follows then that:
$$\mathrm{Gal}_K(E) \cong D_{\mathfrak{P}}/I_{\mathfrak{P}} \cong \mathrm{Gal}_\kappa(\lambda)$$
completing the proof.

This gives the following "simplified" tower as compared to the global case:

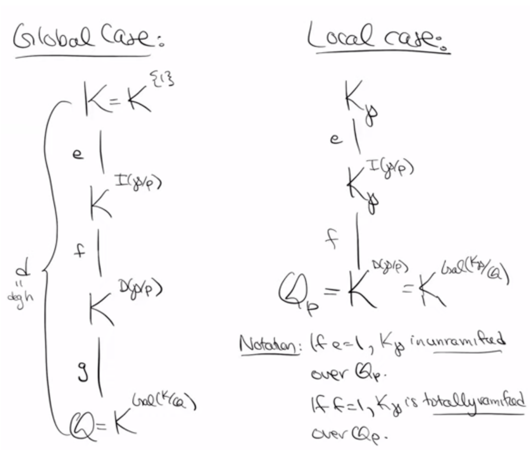


Figure 3.1: Varma: global-local field comparison

Exercise 3.3.1

1. Assume $AKLB$, with A a complete DVR with finite residue field. Then L/K is unramified if and only if $N_{L/K}(B^*) = A^*$. Is this the case for global fields?

3.3.2 Ramified Extensions

We are now in a position where if we have any finite separable extension L/K where A is a complete DVR, then we have a tower $L/E/K$ where E/K is unramified and L/E is totally ramified. We just classified the unramified extensions (or in particular the unramified extensions of field of fractions of complete DVRs), so let us now focus on the ramified case.

Totally ramified extensions of complete DVR

Recall that a polynomial is *Eisenstein* if for some prime $\mathfrak{p} \subseteq R$, the polynomial $f(x) \in R[x]$ has coefficients $a_i \in \mathfrak{p}$ but $a_i^2 \notin \mathfrak{p}$. In the case where the ring is a DVR, this implies that a_0 is a uniformizer, as $\nu_{\mathfrak{p}}(a_0) = 1$. As the ring is a DVR, it is a PID and hence a UFD, meaning we can apply Gauss's lemma and get that it is irreducible over the field of fractions.

Lemma 3.3.16: DVR Extension Using Eisenstein Polynomial

Let A be a DVR and let $f \in A[x]$ be an Eisenstein polynomial. Then $B = A[\pi] = A[x]/(f)$ is a DVR with uniformizer π, where π is the image of x in $A[x]/(f)$.

***Proof*:**
Let $\mathfrak{p}$ be the maximal ideal of A. Then $f \equiv x^n \bmod \mathfrak{p}$, and so by corollary 3.2.2 the ideal $\mathfrak{P}(\mathfrak{p}, x) = (\mathfrak{p}\pi)$ is the only maximal ideal of B. Let $f = \sum_i a_i x^i$. Then as f is Eisenstein, $\mathfrak{p} = (a_0)$, $\mathfrak{P} = (a_0, \pi)$ and $a_0 = -a_1\pi - f_2\pi^2 - \cdots - \pi^n \in (\pi)$, hence $\mathfrak{P} = (\pi)$. As the unique maximal ideal is principal, B is a DVR as we sought to show.

Theorem 3.3.17: Characterizing Totally Ramified Extensions

Assume $AKLB$ with A a complete DVR and π a uniformizer for B. Then the extension L/K is totally ramified if and only if $B = A[\pi]$ and the minimal polynomial of π is Eisenstein

***Proof*:**
By proposition theorem 3.2.3, $B = A[\pi]$. Let f be the minimal polynomial of π. If f is Eisenstein then by lemma 3.3.16 we have $\mathfrak{P} = \mathfrak{p}^n$, hence $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}} = n$ and L/K is totally ramified.

Conversely, if L/K is totally ramified, then $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index n, implying

$$\nu_{\mathfrak{P}}(K) = n\mathbb{Z}$$

Let's first find a K-basis for L. Take the set $\{1, \pi, \pi^2, ..., \pi^{n-1}\}$ where $\pi \in K$ is a uniformizer given by $\nu_{\mathfrak{p}}$. This set is linearly independent as their valuations are distinct modulo $\nu_{\mathfrak{P}}(K)$, namely if $\sum_{i=0}^{n-1} a_0\pi^i = 0$ then it must be the case that for some nonzero a_i, a_j $\nu_{\mathfrak{P}}(a_i\pi^i) = \nu_F B(a_j\pi^j)$ which is impossible. As this also spans L, we have that

$$L = K(\pi)$$

Let $f = \sum_{i=0}^{n} a - ix^i \in A[x]$ be the minimal polynomial of π. We must show it is Eisenstein. First note that $\nu_{\mathfrak{P}}(f(\pi)) = 0$ and $\nu_{\mathfrak{P}}(a_i\pi^i) = i \bmod n$ for $0 \leq i \leq n$. This is possible only if

$$\nu_{\mathfrak{P}}(a_0) = \nu_{\mathfrak{P}}(a_i\pi^0) = \nu_{\mathfrak{P}}(a_n\pi^n) = \nu_{\mathfrak{P}}(\pi^n) = n$$

and $\nu_{\mathfrak{P}}(a_i) \geq n$ for $0 \leq i < n$. But then this implies $\nu_{\mathfrak{p}}(a_0) = 1$ as $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index n, and $\nu_{\mathfrak{p}}(a_i) \geq 1$ for $0 \leq i < n$. But then f is Eisenstein. Then lemma 3.3.16 implies $A[\pi] \subseteq B$ is a DVR, and by maximality we get $B = A[\pi]$, completing the proof.

Example 3.3.18: Environment Title

I'd like to add example 11.6. I want to put in the exercise fomr the HW somewhere because it is actually a really cool exercise.

Definition 3.3.19: Tamely Ramified And Wildly Ramified

Assume $AKLB$ with A a complete DVR and a separable field extension of characteristic $p \geq 0$. Then:

1. The extension L/K is *tamely ramified* if $p \nmid e_{L/K}$ (which is always true if $p = 0$)
2. The extension L/K is *wildly ramified* if $p | e_{L/K}$

All unramified extensions are tamely ramified. A totally ramified extension L/K is totally tamely ramified if $p \nmid e_{L/K}$ and is *totally wildly ramified* if $e_{L/K}$ is a power of p. Note that a totally ramified extension that is wildly ramified need not be totally wildly ramified.

Proposition 3.3.20: Tower of Ramified Extensions

The tower of extensions of fields of fractions of complete DVR's with separable residue field extensions of all types of ramified extensions are transitive

***Proof*:**
exercise

The same is not the case for compositum's, for example the compositum of totally ramified extensions need not be totally ramified (see example 3.3.18).

Theorem 3.3.21: Characterizing Totally Tamely Ramified Extensions

Assume $AKLB$ with A a complete DVR and separable residue field extension of characteristic $p \geq 0$ not dividing $n := [L : K]$. Then the extension L/K is *totally tamely ramified* if and only if $L = K(\pi_A^{1/n})$ for some uniformizer π_A of A.

The main place the tame assumption shall come is when we need to use Hensel's lemma to lift a certain solution. The equivalent of this result in the global case required that the extension be a *Kummer Extension*, which is an extension with enough roots of unity and that there are certain properties satisfied by the galois group of K.

***Proof*:**
If $L = K(\pi_A^{1/n})$, then $\pi = \pi_A^{1/n}$ has minimal polynomial $x^n - \pi_A$, which his Eisenstein and so $A[\pi]$ is a DVR by lemma 3.3.16. By maximality of DVR's, $B = A[\pi]$, and by theorem 3.3.17 L/K is totally tamely ramified, as $p \nmid n$.

Conversely, assume L/K is totally tamely ramified and $\mathfrak{p}, \mathfrak{P}$ are the maximal ideals of A, B with uniformizer π_A, π_B respectfully. Then $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}} = n$ and $\nu_{\mathfrak{P}}(\pi_B^n) = n = \nu_{\mathfrak{P}}(\pi_A)$. Then $\pi_B^n = u\pi_A$ for some unit $u \in B^*$. As we are totally ramified, $f_{\mathfrak{P}} = 1$, so B and A have the *same* residue field, and so if we lift the image of $u \in B/\mathfrak{P} \cong A/\mathfrak{p}$, to a unit $u_A \in A$, and replace π_A with $u_A^{-1}\pi_A$, we can assume $u \equiv 1 \bmod \mathfrak{P}$ (i.e. it is a unit also in $B/\mathfrak{P}$. From this ,define

$$g(x) = x^n - u \in B[x] \qquad \Rightarrow \qquad \overline{g}(x) = x^n - 1 \in (B/\mathfrak{P})[x]$$

Then $\overline{g}'(1) = n \neq 0$ as $p \nmid n$, showing it is a primitive polynomial and so by Hensel's lemma we can lift the root of 1 to r for $g(x)$ in B. Say $\pi = \pi_B/r$. Then π is a uniformizer for B, hence by theorem 3.3.17 $B = A[\pi]$, so $L = K(\pi)$. As $\pi^n = \pi_B^n/r = \pi_B^n/u = \pi_A$, we have:

$$L = K(\pi_A^{1/n})$$

we completed the other direction, as we sought to show.

Proposition 3.3.22: Tower of Ramification

Let K be the field of fractions of a complete DVR and L/K a totally ramified extension. Then there is a unique intermediate field E/K that is totally tamely ramified and L/E is totally wildly ramified

***Proof*:**
See MIT notes p.101

Corollary 3.3.23: Tower of Ramification in Separable Case

Let K be the field of fractions of a complete DVR with separable residue field and L/K be a finite separable extension with separable residue field extensions. Then there is a unique intermediate field E such that E/K is tamely ramified and L/E is totally wildly ramified.

***Proof*:**
See MIT notes p.101

Exercise 3.3.2

1. Let L/K be a finite extension. Then L/K is a tamely ramified extension (not necessarily totally tamely ramified) if and only if $L = K(\sqrt[m_1]{a_1}, \ldots, \sqrt[m_r]{a_r})$ where $\gcd(m_i, p) = 1$ (generalize theorem 3.3.21).
2. Generalize proposition 3.3.22 to tamely ramified extensions. Use this to show there exists a maximally tamely ramified extension.

3.4 Higher Ramification Groups

The trichotomy of the previous section sorts every extension into unramified, tamely ramified, or wildly ramified. The first two are transparent: an unramified extension is governed entirely by its residue field, and a totally tamely ramified one by a single root of a uniformizer. Wild ramification is subtler. When $p\,|\,e$ the inertia group no longer injects into the residue field, and the single integer e no longer records how the Galois group acts near the prime. What is needed is a *filtration* of the inertia group, each step isolating the automorphisms that act trivially to higher and higher order on the ring of integers. This filtration, the higher ramification groups, is the finest invariant of ramification: it computes the different that measures wildness quantitatively, and through the conductor it controls the local constants attached to Galois representations.

Throughout this section L/K is a finite Galois extension of local fields with group $G = \mathrm{Gal}_K(L)$, and ν is the valuation on L normalized so that $\nu(L^\times) = \mathbb{Z}$. We write $\mathfrak{P}$ for the maximal ideal of $\mathcal{O}_L$ and $\lambda = \mathcal{O}_L/\mathfrak{P}$ for its residue field, of characteristic p.

Definition 3.4.1: Ramification Groups in the Lower Numbering

For an integer $i \geq -1$, the i-th *ramification group* of L/K is

$$G_i = \{\sigma \in G \mid \nu(\sigma(x) - x) \geq i + 1 \text{ for all } x \in \mathcal{O}_L\}$$

The condition tightens as i grows, so the groups form a descending chain

$$G = G_{-1} \supseteq G_0 \supseteq G_1 \supseteq \cdots$$

The first two steps are already familiar. For $i = -1$ the condition $\nu(\sigma(x) - x) \geq 0$ is automatic, so $G_{-1} = G$. For $i = 0$ the condition says $\sigma(x) \equiv x \bmod \mathfrak{P}$ for every $x \in \mathcal{O}_L$, that is, σ acts trivially on the residue field λ; thus $G_0 = I$ is precisely the inertia group of theorem 3.3.15. The higher groups $G_1, G_2, \ldots$ dissect the wild part of the inertia, and it is their behaviour that the rest of the section describes.

Verifying membership in G_i against every $x \in \mathcal{O}_L$ would be laborious; a single generator suffices.

Lemma 3.4.2: Ramification Groups from a Generator

Suppose $\mathcal{O}_L = \mathcal{O}_K[\alpha]$. Then for every $\sigma \in G$ and every integer $i \geq -1$,

$$\sigma \in G_i \iff \nu(\sigma(\alpha) - \alpha) \geq i + 1$$

In particular the integer $i_G(\sigma) := \nu(\sigma(\alpha) - \alpha)$ satisfies $G_i = \{\sigma \in G \mid i_G(\sigma) \geq i + 1\}$, and the resulting groups do not depend on the choice of generator α.

***Proof*:**
The forward implication is immediate, since $\alpha \in \mathcal{O}_L$. For the converse, suppose $\nu(\sigma(\alpha)-\alpha) \geq i+1$ and let $x \in \mathcal{O}_L$ be arbitrary. As $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ we may write $x = f(\alpha)$ for some $f \in \mathcal{O}_K[t]$. Then

$$\sigma(x) - x = f(\sigma(\alpha)) - f(\alpha) = (\sigma(\alpha) - \alpha)\, g(\sigma(\alpha), \alpha)$$

for a polynomial g with coefficients in $\mathcal{O}_K$, because $t - s$ divides $f(t) - f(s)$ in $\mathcal{O}_K[t, s]$. Both factors lie in $\mathcal{O}_L$, so $\nu(\sigma(x) - x) \geq \nu(\sigma(\alpha) - \alpha) \geq i + 1$. Thus $\sigma \in G_i$. Since the groups G_i were defined without reference to α, the characterization $G_i = \{\sigma \mid i_G(\sigma) \geq i + 1\}$ shows i_G records the same data for any generator, as we sought to show.

Proposition 3.4.3: The Ramification Filtration

Each G_i is a normal subgroup of G, the chain $G_0 \supseteq G_1 \supseteq \cdots$ is decreasing, and $G_i = 1$ for all sufficiently large i.

***Proof*:**
Recall that the valuation ν on L is preserved by every $\tau \in G$, since $\nu \circ \tau$ is another valuation extending that of K and the extension is unique (theorem 3.1.7).

Subgroup. Let $\sigma, \tau \in G_i$ and $x \in \mathcal{O}_L$. Then $\tau(x) \in \mathcal{O}_L$, so

$$\nu(\sigma\tau(x) - x) \geq \min\big(\nu(\sigma(\tau x) - \tau x),\, \nu(\tau x - x)\big) \geq i + 1$$

giving $\sigma\tau \in G_i$. For inverses, $\nu(\sigma^{-1}(x) - x) = \nu\big(\sigma(\sigma^{-1}x - x)\big) = \nu(x - \sigma(x)) \geq i + 1$, so $\sigma^{-1} \in G_i$.

Normality. For $\tau \in G$ and $\sigma \in G_i$, using $\nu \circ \tau = \nu$ and $\tau(x) \in \mathcal{O}_L$,

$$\nu\big(\tau^{-1}\sigma\tau(x) - x\big) = \nu\big(\sigma\tau(x) - \tau(x)\big) \geq i + 1$$

so $\tau^{-1}\sigma\tau \in G_i$.

Eventual triviality. Fix a generator $\mathcal{O}_L = \mathcal{O}_K[\alpha]$, which exists because the residue extension is separable (theorem 3.2.3). For $\sigma \neq 1$ we have $\sigma(\alpha) \neq \alpha$, since α generates L over K, so $i_G(\sigma) = \nu(\sigma(\alpha) - \alpha)$ is finite. As G is finite, $m := \max_{\sigma \neq 1} i_G(\sigma)$ is finite, and lemma 3.4.2 gives $G_i = 1$ for all $i \geq m$, completing the proof.

An integer i with $G_i \neq G_{i+1}$ is called a *ramification break* or *jump* of L/K; the breaks are the finitely many indices at which the filtration strictly decreases. The next theorem is the structural heart of the subject: it identifies each successive quotient with a piece of the residue field, and so pins down what kind of group the inertia can be.

Theorem 3.4.4: Ramification Quotients

Let π be a uniformizer of L and let $U^{(0)} = \mathcal{O}_L^\times$, $U^{(n)} = 1 + \mathfrak{P}^n$ for $n \geq 1$ denote the unit filtration of L. For each $i \geq 0$ the map

$$\theta_i \colon G_i/G_{i+1} \to U^{(i)}/U^{(i+1)}, \qquad \sigma \mapsto \frac{\sigma(\pi)}{\pi} \bmod U^{(i+1)}$$

is a well-defined injective homomorphism, independent of the choice of π. Consequently:

1. G_0/G_1 embeds into $\lambda^\times$; it is cyclic of order prime to p.
2. For $i \geq 1$, G_i/G_{i+1} embeds into the additive group λ^+; it is an elementary abelian p-group.

***Proof*:**
The groups G_i for $i \geq 0$ lie in $G_0 = I$, and their defining condition refers only to the action of σ on $\mathcal{O}_L$; it is therefore unchanged if K is replaced by the inertia field $E = L^{G_0}$ (theorem 3.3.15). Over E the extension L/E is totally ramified, so $\mathcal{O}_L = \mathcal{O}_E[\pi]$ for any uniformizer π (theorem 3.3.17), and we may compute with $\alpha = \pi$ in lemma 3.4.2: for $i \geq 0$,

$$\sigma \in G_i \iff \nu(\sigma(\pi) - \pi) \geq i + 1 \iff \nu\left(\frac{\sigma(\pi)}{\pi} - 1\right) \geq i \iff \frac{\sigma(\pi)}{\pi} \in U^{(i)}$$

and likewise $\sigma \in G_{i+1} \iff \sigma(\pi)/\pi \in U^{(i+1)}$. Thus θ_i carries G_i into $U^{(i)}$ and has kernel exactly G_{i+1}; injectivity of the induced map on G_i/G_{i+1} follows once we check it is a homomorphism.

Homomorphism. Let $\sigma, \tau \in G_i$ and write $\tau(\pi)/\pi = 1 + b$ with $\nu(b) \geq i$. Applying σ,

$$\frac{\sigma\tau(\pi)}{\pi} = \frac{\sigma(\pi)}{\pi} \cdot \sigma\left(\frac{\tau(\pi)}{\pi}\right) = \frac{\sigma(\pi)}{\pi}\,(1 + \sigma(b))$$

Now $b \in \mathcal{O}_L$ and $\sigma \in G_i$, so $\nu(\sigma(b) - b) \geq i+1$, whence $(1+\sigma(b))/(1+b) = 1 + (\sigma(b)-b)/(1+b) \in U^{(i+1)}$. Therefore

$$\theta_i(\sigma\tau) = \frac{\sigma(\pi)}{\pi} \cdot \frac{\tau(\pi)}{\pi} \bmod U^{(i+1)} = \theta_i(\sigma)\,\theta_i(\tau)$$

Independence of π. Any other uniformizer is $\pi' = u\pi$ with $u \in \mathcal{O}_L^\times$, and $\sigma(\pi')/\pi' = (\sigma(u)/u)(\sigma(\pi)/\pi)$. For $\sigma \in G_i$ we have $\nu(\sigma(u) - u) \geq i + 1$, so $\sigma(u)/u \in U^{(i+1)}$ and θ_i is unchanged modulo $U^{(i+1)}$.

Finally, the unit filtration was computed in proposition 1.2.2: reduction modulo $\mathfrak{P}$ gives $U^{(0)}/U^{(1)} \cong \lambda^\times$, and $1 + a\pi^i \mapsto a \bmod \mathfrak{P}$ gives $U^{(i)}/U^{(i+1)} \cong \lambda^+$ for $i \geq 1$. Composing with θ_i embeds G_0/G_1 into $\lambda^\times$, a finite subgroup of the multiplicative group of a field and hence cyclic, of order prime to $p = \operatorname{char} \lambda$; and it embeds G_i/G_{i+1} into λ^+, all of whose nonzero elements have order p, so G_i/G_{i+1} is an elementary abelian p-group, as we sought to show.

The consequences for the structure of the inertia group are immediate and decisive.

Corollary 3.4.5: Structure of the Inertia Group

Let $G_0 = I$ be the inertia group of L/K, of order e. Then:

1. G_1 is a p-group, normal in G_0, and is the unique Sylow p-subgroup of G_0; it is called the *wild inertia group*.
2. G_0/G_1 is cyclic of order prime to p, and $G_0 = G_1 \rtimes C$ for a cyclic subgroup C of order prime to p.
3. L/K is tamely ramified if and only if $G_1 = 1$.
4. G is solvable.

***Proof*:**
By theorem 3.4.4 the chain $G_1 \supseteq G_2 \supseteq \cdots \supseteq 1$ has successive quotients that are elementary abelian p-groups, so G_1 is a p-group; it is normal in G_0 by proposition 3.4.3. Since G_0/G_1 has order prime to p, the order of G_1 is the full power of p dividing $|G_0|$, so G_1 is the unique Sylow p-subgroup. This proves (1), and (2) follows: the quotient G_0/G_1 is cyclic of order prime to p by theorem 3.4.4, and the Schur-Zassenhaus theorem ([5]) splits the extension $1 \to G_1 \to G_0 \to G_0/G_1 \to 1$, providing the complement $C \cong G_0/G_1$.

For (3), L/K is tamely ramified exactly when $p \nmid e = |G_0|$, which by (1) holds if and only if the Sylow p-subgroup G_1 is trivial. For (4), G_1 is a p-group, hence solvable; G_0/G_1 is abelian; and $G/G_0 \cong \mathrm{Gal}_\kappa(\lambda)$ is cyclic (theorem 3.3.15). A group with a solvable normal subgroup and solvable quotient is solvable, applied twice, so G is solvable, completing the proof.

The classical fact that Galois groups of local fields are solvable, opaque from the definitions, is thus a direct reading of the ramification filtration. The two examples below show the filtration: the first is the wild cyclotomic tower, the reference example for the whole theory; the second is its positive-characteristic analogue, where a single Artin-Schreier equation produces a single wild break.

Example 3.4.6: Ramification of the Cyclotomic Tower $\mathbb{Q}_p(\zeta_{p^n})$

Let $L = \mathbb{Q}_p(\zeta_{p^n})$ with $n \geq 1$, so $L/\mathbb{Q}_p$ is totally ramified of degree $e = \varphi(p^n) = p^{n-1}(p-1)$, with uniformizer $\pi = \zeta - 1$ where $\zeta = \zeta_{p^n}$, and $G = \mathrm{Gal}_{\mathbb{Q}_p}(L) \cong (\mathbb{Z}/p^n\mathbb{Z})^\times$ via $\sigma_a(\zeta) = \zeta^a$. To compute the filtration, evaluate

$$i_G(\sigma_a) = \nu(\sigma_a(\pi) - \pi) = \nu(\zeta^a - \zeta) = \nu(\zeta^{a-1} - 1)$$

using $\nu(\zeta) = 0$. If $a \equiv 1 \bmod p^k$ but $a \not\equiv 1 \bmod p^{k+1}$, then ζ^{a-1} is a primitive p^{n-k}-th root of unity, and since $\zeta_{p^{n-k}} - 1$ is a uniformizer of $\mathbb{Q}_p(\zeta_{p^{n-k}})$ over which L has ramification index $\varphi(p^n)/\varphi(p^{n-k}) = p^k$,

$$i_G(\sigma_a) = \nu(\zeta_{p^{n-k}} - 1) = p^k$$

Therefore $\sigma_a \in G_i$ if and only if $p^{v_p(a-1)} \geq i + 1$, that is, $a \equiv 1 \bmod p^k$ where $k = \lceil \log_p(i+1) \rceil$. Writing this out, for $p^{k-1} \leq i \leq p^k - 1$,

$$G_i = \left\{ \sigma_a \;\middle|\; a \equiv 1 \bmod p^k \right\} = \mathrm{Gal}_{\mathbb{Q}_p(\zeta_{p^k})}(L)$$

with $G_0 = G$ and $G_i = 1$ for $i \geq p^{n-1}$. The breaks occur at $i = p^k - 1$ for $k = 0, 1, \ldots, n-1$. The quotients match theorem 3.4.4: $G_0/G_1 \cong (\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$ prime to p, while for $k \geq 1$ each $G_{p^{k-1}}/G_{p^k}$ is cyclic of order p, an elementary abelian p-group.

Example 3.4.7: An Artin-Schreier Extension

Let $K = \mathbb{F}_p((t))$ and let $L = K(y)$ where $y^p - y = t^{-m}$ with $m \geq 1$ prime to p. This is a cyclic totally (wildly) ramified extension of degree p, with $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/p\mathbb{Z}$ generated by $\sigma\colon y \mapsto y+1$. From $y^p - y = t^{-m}$, whose right side has valuation $-mp$ while the left has valuation $p\,\nu_L(y)$, it follows that $\nu_L(y) = -m$ and $\nu_L(t) = p$. Since $\gcd(m,p) = 1$, choose integers a, b with $-ma + pb = 1$ and set $\pi = y^a t^b$, a uniformizer of L; reducing $-ma + pb = 1$ modulo p shows $p \nmid a$. For $1 \leq j \leq p-1$,

$$\sigma^j(\pi) - \pi = t^b\big((y+j)^a - y^a\big)$$

and the lowest-valuation term of $(y+j)^a - y^a = aj\,y^{a-1} + \cdots$ is $aj\,y^{a-1}$, of valuation $-m(a-1)$ because $p \nmid aj$. Hence

$$i_G(\sigma^j) = \nu_L(\sigma^j(\pi) - \pi) = pb - m(a-1) = (pb - ma) + m = m+1$$

independent of j. Therefore

$$G_0 = G_1 = \cdots = G_m = G \cong \mathbb{Z}/p\mathbb{Z}, \qquad G_{m+1} = 1$$

so the extension has a single break, at $i = m$. There is no tame part ($G_0 = G_1$), as befits a wildly ramified extension of degree p, and the lone quotient $G_m/G_{m+1} \cong \mathbb{Z}/p\mathbb{Z}$ sits inside λ^+ exactly as theorem 3.4.4 predicts.

Exercise 3.4.1

1. Compute the ramification filtration of $\mathbb{Q}_2(\sqrt{2})/\mathbb{Q}_2$: find $i_G(\sigma)$ for the nontrivial automorphism σ, and list the groups G_i.
2. Show that the integer $i_G(\sigma) = \nu(\sigma(\alpha) - \alpha)$ of lemma 3.4.2 is independent of the choice of generator α with $\mathcal{O}_L = \mathcal{O}_K[\alpha]$.
3. Let L/K be totally and tamely ramified of degree e (so $p \nmid e$). Show directly from theorem 3.4.4 that $G_0 = G$ is cyclic of order e and $G_1 = 1$, so the only break is at $i = 0$.
4. Using corollary 3.4.5, show that if $[L : K]$ is a prime $\ell \neq p$ and L/K is totally ramified, then L/K is tame and cyclic.

3.5 The Herbrand Functions and Upper Numbering

The lower-numbering filtration has one defect: it does not pass to quotients. If $H \trianglelefteq G$ and $K' = L^H$, the ramification groups of $\mathrm{Gal}_K(K') = G/H$ are *not* the images G_iH/H of the groups of G; the wild ramification of L/K' distorts the indices. For a subgroup the lower numbering behaves perfectly, but a quotient is exactly what one needs to descend from a large extension to a small one, and ultimately to make sense of ramification in an infinite abelian extension. Herbrand's theorem repairs this by reindexing the filtration through a piecewise-linear change of variable, the function $\varphi_{L/K}$; in the resulting *upper numbering* the ramification groups pass to quotients exactly. This is the numbering that class field theory attaches to characters, and in which the Hasse-Arf theorem asserts that the jumps of an abelian extension are integers.

Throughout, L/K is a finite Galois extension with group $G = \mathrm{Gal}_K(L)$, lower-numbering ramification groups G_i of orders $g_i = |G_i|$, and ν the normalized valuation on L. The lower numbering restricts flawlessly to subgroups, the one compatibility that comes for free.

Proposition 3.5.1: Subgroups and the Lower Numbering

Let $H \le G$ and $K' = L^H$, so $H = \mathrm{Gal}_{K'}(L)$. Then the ramification groups of L/K' are $H_u = H \cap G_u$ for all $u \ge -1$.

***Proof*:**
The extensions L/K' and L/K share the ring $\mathcal{O}_L$ and the valuation ν, and the defining condition of a ramification group refers only to these. Thus $H_u = \{s \in H \mid \nu(s(x) - x) \ge u+1 \text{ for all } x \in \mathcal{O}_L\} = H \cap G_u$, as we sought to show.

Definition 3.5.2: The Herbrand Function

The *Herbrand function* of L/K is $\varphi_{L/K}\colon [-1,\infty) \to [-1,\infty)$,

$$\varphi_{L/K}(u) = \int_0^u \frac{dt}{(G_0 : G_t)}$$

where $G_t = G_{\lceil t \rceil}$ for $t > 0$ and $G_t = G_0$ for $-1 \le t \le 0$. Its inverse is $\psi_{L/K} = \varphi_{L/K}^{-1}$.

Proposition 3.5.3: Properties of the Herbrand Function

The function $\varphi = \varphi_{L/K}$ is continuous, piecewise linear, strictly increasing and concave, with $\varphi(u) = u$ on $[-1, 0]$. On $(m-1, m]$ for an integer $m \ge 1$ its slope is g_m/g_0, so for integer $m \ge 0$

$$\varphi(m) = \frac{1}{g_0}(g_1 + g_2 + \cdots + g_m)$$

It is a homeomorphism of $[-1,\infty)$ onto itself, and $\psi = \varphi^{-1}$ is continuous, piecewise linear, strictly increasing, convex, with $\psi(v) = v$ on $[-1, 0]$.

***Proof*:**
On $(m-1, m]$ the integrand $1/(G_0 : G_t) = g_m/g_0$ is constant, so φ is continuous and piecewise linear with the stated slopes; the slopes are positive and nonincreasing since the g_m decrease, giving strict monotonicity and concavity. Integrating the constant slopes yields $\varphi(m) = \sum_{i=1}^m g_i/g_0$, and on $[-1,0]$ the integrand is 1, so $\varphi(u) = u$ there. As φ is a strictly increasing continuous surjection onto $[-1,\infty)$ (its slope is at least $1/g_0 > 0$, so it is unbounded), it is a homeomorphism, and $\psi = \varphi^{-1}$ carries the reciprocal slopes, hence is convex and equals v on $[-1, 0]$, as we sought to show.

Definition 3.5.4: Ramification Groups in the Upper Numbering

For $v \geq -1$, the *upper-numbering ramification group* is $G^v = G_{\psi_{L/K}(v)}$; equivalently $G^{\varphi_{L/K}(u)} = G_u$ for all $u \geq -1$.

Since $\varphi(-1) = -1$ and $\varphi(0) = 0$, the two numberings agree at the bottom: $G^{-1} = G$ and $G^0 = G_0$ is the inertia group. They encode the same filtration through the substitution $v = \varphi(u)$; a *jump* of the upper numbering (a v at which $G^v \neq G^{v+\epsilon}$ for every $\epsilon > 0$) is the φ-image of a lower ramification break.

We now turn to the quotient. Fix $H \trianglelefteq G$, set $K' = L^H$ and $\bar{G} = G/H = \mathrm{Gal}_K(K')$, and write $H_i = H \cap G_i$ (proposition 3.5.1), $h_i = |H_i|$, and $e' = e_{L/K'} = h_0$. For $\bar{\sigma} \in \bar{G}$ put $i_{\bar{G}}(\bar{\sigma}) = \nu_{K'}(\bar{\sigma}(y') - y')$, computed with a generator $\mathcal{O}_{K'} = \mathcal{O}_K[y']$. The bridge between the two filtrations is the following identity, whose proof leans on the transitivity of the different.

Lemma 3.5.5: The Quotient Identity

For $\bar{\sigma} \in \bar{G}$ with $\bar{\sigma} \neq 1$, and any lift $\sigma \in G$,

$$e'\, i_{\bar{G}}(\bar{\sigma}) = \sum_{s \in \sigma H} i_G(s)$$

***Proof*:**
Step 1: the inequality. Let $\mathcal{O}_L = \mathcal{O}_K[x]$ and $\mathcal{O}_{K'} = \mathcal{O}_K[y']$. Since L/K' is Galois with group H, the minimal polynomial of x over K' is $h(T) = \prod_{\tau \in H}(T - \tau x) \in \mathcal{O}_{K'}[T]$. Applying σ to its coefficients gives $\sigma h(T) = \prod_{\tau \in H}(T - \sigma\tau x)$, and since $h(x) = 0$,

$$\prod_{\tau \in H}(x - \sigma\tau x) = \sigma h(x) - h(x) = \sum_k (\sigma c_k - c_k)\, x^k$$

where $c_k \in \mathcal{O}_{K'}$ are the coefficients of h. Writing $c_k = P_k(y')$ with $P_k \in \mathcal{O}_K[T]$, each $\sigma c_k - c_k = P_k(\sigma y') - P_k(y')$ is divisible by $\sigma y' - y'$, so $\sigma y' - y'$ divides the whole sum. Hence

$$\sum_{s \in \sigma H} i_G(s) = \sum_{\tau \in H} \nu(x - \sigma\tau x) = \nu\Big(\prod_{\tau \in H}(x - \sigma\tau x)\Big) \geq \nu(\sigma y' - y') = e'\, i_{\bar{G}}(\bar{\sigma})$$

the last equality because $y' \in K'$ and $\nu|_{K'} = e'\,\nu_{K'}$.

Step 2: equality by summation. Sum the inequality over all $\bar{\sigma} \neq 1$; the left total is $\sum_{s \in G \setminus H} i_G(s)$. By the ramification formula for the different (theorem 3.6.3) and its transitivity (theorem 3.6.7), together with $\nu(\mathfrak{d}_{L/K'}) = \sum_{i \geq 0}(h_i - 1) = \sum_{s \in H \setminus 1} i_G(s)$ (from proposition 3.5.1),

$$\sum_{s \in G \setminus 1} i_G(s) = \nu(\mathfrak{d}_{L/K}) = \nu(\mathfrak{d}_{L/K'}) + e'\,\nu_{K'}(\mathfrak{d}_{K'/K}) = \sum_{s \in H \setminus 1} i_G(s) + e' \sum_{\bar{\sigma} \neq 1} i_{\bar{G}}(\bar{\sigma})$$

Subtracting the sum over $H \setminus 1$ leaves $\sum_{s \in G \setminus H} i_G(s) = e' \sum_{\bar{\sigma} \neq 1} i_{\bar{G}}(\bar{\sigma})$. Thus the left totals equal the right totals of Step 1 while each left term is at least its right term; equality of the totals forces equality term by term, which is the claim, completing the proof.

To read off the groups, the coset sum is evaluated in terms of the filtration of H.

Theorem 3.5.6: Herbrand's Theorem

Let $H \trianglelefteq G$ and $K' = L^H$. Then for all $u \geq -1$,

$$(G/H)_{\varphi_{L/K'}(u)} = G_u \, H/H$$

***Proof*:**
Fix $\bar{\sigma} \neq 1$ and choose a lift $\sigma_0 \in \sigma H$ with $m := i_G(\sigma_0) = \max_{s \in \sigma H} i_G(s)$. The subgroup property of the G_j gives $i_G(ab) \geq \min(i_G(a), i_G(b))$ for all a, b, with equality when $i_G(a) \neq i_G(b)$: if $\min = i_G(a) < i_G(b)$ and $i_G(ab) > i_G(a)$ then $a = (ab)b^{-1}$ would lie in $G_{i_G(a)}$, a contradiction. Hence for $\tau \in H$, $i_G(\sigma_0 \tau) = \min(m, i_H(\tau))$: when $i_H(\tau) \neq m$ this is the equality case, and when $i_H(\tau) = m$ we get $i_G(\sigma_0\tau) \geq m$ while maximality of m over the coset forces $i_G(\sigma_0\tau) \leq m$. Here $i_H(\tau) = i_G(\tau)$ for $\tau \in H$, both equal to $1 + \max\{j \mid \tau \in H_j\}$ by proposition 3.5.1. As τ runs over H, $\sigma_0\tau$ runs over σH, so

$$\sum_{s \in \sigma H} i_G(s) = \sum_{\tau \in H} \min(m, i_H(\tau)) = \sum_{\tau \in H} \#\{j \mid 0 \leq j \leq m-1, \ \tau \in H_j\} = \sum_{j=0}^{m-1} h_j$$

using $\min(m, i_H(\tau)) = \#\{j \mid 0 \leq j \leq m-1, \ \tau \in H_j\}$ (the groups H_j decrease, so $\tau \in H_j$ for $j < i_H(\tau)$ and no further). By lemma 3.5.5 and $e' = h_0$, and since $\varphi_{L/K'}(m-1) = \sum_{j=1}^{m-1} h_j/h_0$,

$$i_{\bar{G}}(\bar{\sigma}) = \frac{1}{h_0} \sum_{j=0}^{m-1} h_j = 1 + \frac{1}{h_0} \sum_{j=1}^{m-1} h_j = 1 + \varphi_{L/K'}(m-1)$$

Thus $i_{\bar{G}}(\bar{\sigma}) - 1 = \varphi_{L/K'}(m-1)$. Now, for $v = \varphi_{L/K'}(u)$,

$$\bar{\sigma} \in (G/H)_v \iff i_{\bar{G}}(\bar{\sigma}) - 1 \geq v \iff \varphi_{L/K'}(m-1) \geq v \iff m - 1 \geq u$$

since $\varphi_{L/K'}$ is increasing with $\varphi_{L/K'}(u) = v$. The last condition says some $s \in \sigma H$ has $i_G(s) \geq u+1$, that is, σH meets G_u, equivalently $\bar{\sigma} \in G_u H/H$. Therefore $(G/H)_{\varphi_{L/K'}(u)} = G_u H/H$, completing the proof.

The same reindexing makes the Herbrand function transitive, and with it the upper numbering descends to quotients cleanly.

Theorem 3.5.7: Transitivity of the Herbrand Function

For a tower $L/K'/K$ with K'/K Galois,

$$\varphi_{L/K} = \varphi_{K'/K} \circ \varphi_{L/K'}, \qquad \psi_{L/K} = \psi_{L/K'} \circ \psi_{K'/K}$$

***Proof*:**
Both $\varphi_{L/K}$ and $\varphi_{K'/K} \circ \varphi_{L/K'}$ vanish at 0 and are continuous and piecewise linear, so it suffices to compare right derivatives at a noninteger $u > 0$. Write $\bar{G} = G/H$ and $w = \varphi_{L/K'}(u)$. By the chain rule and proposition 3.5.3,

$$\left(\varphi_{K'/K} \circ \varphi_{L/K'}\right)'(u) = \frac{1}{(\bar{G}_0 : \bar{G}_w)} \cdot \frac{1}{(H_0 : H_u)}$$

Herbrand's theorem (theorem 3.5.6) gives $\bar{G}_w = G_u H/H$, so $|\bar{G}_w| = |G_u|/|G_u \cap H| = g_u/h_u$ and, at $u = 0$, $|\bar{G}_0| = g_0/h_0$; hence $(\bar{G}_0 : \bar{G}_w) = (g_0 h_u)/(h_0 g_u)$. With $(H_0 : H_u) = h_0/h_u$, the product is

$$\frac{h_0 g_u}{g_0 h_u} \cdot \frac{h_u}{h_0} = \frac{g_u}{g_0} = \frac{1}{(G_0 : G_u)} = \varphi'_{L/K}(u)$$

So the two functions have equal derivatives and agree at 0, hence coincide; inverting gives the identity for ψ, completing the proof.

Corollary 3.5.8: Herbrand's Theorem in the Upper Numbering

For $H \trianglelefteq G$ and $K' = L^H$, $(G/H)^v = G^v H/H$ for all $v \geq -1$.

***Proof*:**
By definition $(G/H)^v = (G/H)_{\psi_{K'/K}(v)}$. Taking $u = \psi_{L/K}(v)$ and using transitivity (theorem 3.5.7) gives $\varphi_{L/K'}(u) = \varphi_{L/K'}(\psi_{L/K'}(\psi_{K'/K}(v))) = \psi_{K'/K}(v)$, so Herbrand's theorem (theorem 3.5.6) yields $(G/H)_{\psi_{K'/K}(v)} = G_u H/H = G_{\psi_{L/K}(v)} H/H = G^v H/H$, as we sought to show.

The upper numbering is the natural home for one of the deepest facts about abelian ramification. Its jumps, a priori rational numbers, are in fact integers.

Theorem 3.5.9: Hasse-Arf

Let L/K be a finite *abelian* extension of local fields. Then every jump of the upper-numbering filtration (G^v) is an integer: if $G^v \neq G^{v+\epsilon}$ for all $\epsilon > 0$, then $v \in \mathbb{Z}$.

The proof is a delicate analysis of the norm map and the different, self-contained but long enough that inserting it here would break the thread; it is given in full in appendix A. The integrality is what allows the upper numbering to be transported to a character of G: the conductor of an abelian character, defined as the least integer $v + 1$ with G^v acting trivially, is well defined precisely because the jumps are integral, and it is this conductor that indexes the local factors of an L-function. Without Hasse-Arf the upper-numbering jumps could be fractional and the notion of conductor would collapse.

Example 3.5.10: Upper Numbering of the Cyclotomic Tower
For $L = \mathbb{Q}_p(\zeta_{p^n})/\mathbb{Q}_p$ the lower breaks lie at $i = p^k - 1$ for $k = 0, 1, \ldots, n-1$ (example 3.4.6), where $G_i = \left\{\sigma_a \;\middle|\; a \equiv 1 \bmod p^k\right\}$ has order $g_i = p^{n-k}$. Grouping the sum defining φ by the value

of g_i,

$$\varphi(p^k-1) = \frac{1}{g_0}\sum_{i=1}^{p^k-1} g_i = \frac{1}{p^{n-1}(p-1)}\sum_{k'=1}^{k}\left(p^{k'}-p^{k'-1}\right)p^{n-k'} = \frac{1}{p^{n-1}(p-1)}\sum_{k'=1}^{k}(p-1)p^{n-1} = k$$

So the upper breaks fall at the integers $v = 0, 1, \ldots, n-1$, and

$$G^v = \left\{\sigma_a \;\middle|\; a \equiv 1 \bmod p^{\lceil v\rceil}\right\} \qquad (0 \le v \le n-1)$$

with $G^v = 1$ for $v > n-1$. The integrality of these breaks is Hasse-Arf made explicit, the extension being abelian.

Exercise 3.5.1

1. Compute $\varphi_{L/\mathbb{Q}_2}$ for $L = \mathbb{Q}_2(\sqrt{2})$ and determine the upper-numbering filtration; verify that the unique jump is at an integer, as Hasse-Arf requires for this abelian extension.
2. Show that $G^{-1} = G$ and $G^0 = G_0$ directly from the definitions, and that for a totally *tamely* ramified extension of degree e one has $\varphi_{L/K}(u) = u/e$ for $u \ge 0$ and a single upper break at $v = 0$.
3. For the tower $\mathbb{Q}_p(\zeta_{p^2})/\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$, compute $\varphi_{L/K'}$, $\varphi_{K'/K}$, and $\varphi_{L/K}$ explicitly and verify the transitivity $\varphi_{L/K} = \varphi_{K'/K} \circ \varphi_{L/K'}$.
4. Deduce from corollary 3.5.8 that if $G^v = 1$ then $(G/H)^v = 1$ for every $H \trianglelefteq G$; interpret this as "the conductor can only drop in a quotient."

3.6 The Different and Discriminant

The ramification groups of the previous section record how the Galois group acts near the prime, but they are a group-theoretic invariant of a Galois extension. The *different* repackages the same information as an ideal of $\mathcal{O}_L$, defined for any finite separable extension whether or not it is Galois, and its valuation counts ramification directly: it equals $e-1$ exactly when the extension is tame, and the excess over $e-1$ measures wildness. Taking norms down to K produces the *discriminant*, the invariant that detects which primes ramify in a global field. This section defines both, computes the different from the ramification filtration, and records the tower formula that makes the two multiplicative.

Throughout, L/K is a finite separable extension of local fields of degree n, with $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ a monogenic ring of integers (theorem 3.2.3) and $g \in \mathcal{O}_K[T]$ the minimal polynomial of α. The trace form $(x, y) \mapsto \mathrm{tr}_{L/K}(xy)$ is nondegenerate because L/K is separable ([2]), which is what makes the following dual object a fractional ideal.

Definition 3.6.1: Different and Codifferent

The *codifferent* (or inverse different) of L/K is

$$\mathcal{C}_{L/K} = \{x \in L \mid \operatorname{tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{O}_K\}$$

a fractional ideal of $\mathcal{O}_L$ containing $\mathcal{O}_L$. Its inverse, the *different*, is the integral ideal

$$\mathfrak{d}_{L/K} = \mathcal{C}_{L/K}^{-1} \subseteq \mathcal{O}_L$$

That $\mathcal{C}_{L/K}$ contains $\mathcal{O}_L$ is immediate, since $\operatorname{tr}_{L/K}(\mathcal{O}_L) \subseteq \mathcal{O}_K$; thus $\mathfrak{d}_{L/K} \subseteq \mathcal{O}_L$ is a genuine ideal, and it is the unit ideal precisely when $\operatorname{tr}_{L/K}(\mathcal{O}_L) = \mathcal{O}_K$. In the monogenic case the codifferent has a closed form, which reduces every computation of the different to a derivative.

Proposition 3.6.2: The Different of a Monogenic Extension

With $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ and g the minimal polynomial of α,

$$\mathcal{C}_{L/K} = \frac{1}{g'(\alpha)}\mathcal{O}_L, \qquad \mathfrak{d}_{L/K} = \big(g'(\alpha)\big)$$

***Proof*:**
Let $\alpha = \alpha_1, \ldots, \alpha_n$ be the conjugates of α in a splitting field, so $g(T) = \prod_i (T - \alpha_i)$. Euler's interpolation identity states that

$$\sum_{i=1}^{n} \frac{g(T)}{(T-\alpha_i)\, g'(\alpha_i)}\, \alpha_i^k = T^k \qquad (0 \le k \le n-1)$$

both sides being polynomials of degree $\le n-1$ agreeing at the n points $T = \alpha_i$. Reading off the trace, the elements $\beta_0, \ldots, \beta_{n-1} \in L$ defined by $\frac{g(T)}{(T-\alpha)g'(\alpha)} = \sum_j \beta_j T^j$ satisfy $\operatorname{tr}_{L/K}(\alpha^k \beta_j) = \delta_{jk}$; that is, $\{\beta_j\}$ is the trace dual basis of $\{\alpha^k\}$. Since $g(T)/(T-\alpha) = \sum_j \gamma_j T^j$ has coefficients $\gamma_j \in \mathcal{O}_K[\alpha] = \mathcal{O}_L$ (they are polynomials in α with coefficients built from those of g), we get $\beta_j = \gamma_j / g'(\alpha) \in \frac{1}{g'(\alpha)}\mathcal{O}_L$.

Now $x \in \mathcal{C}_{L/K}$ iff $\operatorname{tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{O}_K$, iff $\operatorname{tr}_{L/K}(x\alpha^k) \in \mathcal{O}_K$ for all k, iff x lies in the $\mathcal{O}_K$-span of the dual basis $\{\beta_j\}$. That span is $\bigoplus_j \mathcal{O}_K \beta_j = \frac{1}{g'(\alpha)} \bigoplus_j \mathcal{O}_K \gamma_j = \frac{1}{g'(\alpha)}\mathcal{O}_L$, the last equality because $\{\gamma_j\}$ is an $\mathcal{O}_K$-basis of $\mathcal{O}_L$ (the γ_j are unit-triangular in the $\{\alpha^k\}$, as $\gamma_{n-1} = 1$). Hence $\mathcal{C}_{L/K} = \frac{1}{g'(\alpha)}\mathcal{O}_L$ and $\mathfrak{d}_{L/K} = (g'(\alpha))$, as we sought to show.

For a Galois extension the derivative $g'(\alpha)$ is a product of differences of conjugates, and each difference is measured by the function i_G of the previous section. The valuation of the different is therefore exactly the sum that also computes the orders of the ramification groups.

Theorem 3.6.3: The Different and the Ramification Filtration

Let L/K be a finite Galois extension of local fields with group G and ramification groups G_i of orders $g_i = |G_i|$. Then

$$\nu_L\left(\mathfrak{d}_{L/K}\right) = \sum_{s\in G,\, s\neq 1} i_G(s) = \sum_{i\geq 0}(g_i - 1)$$

***Proof*:**
Take α with $\mathcal{O}_L = \mathcal{O}_K[\alpha]$. As L/K is Galois, the conjugates of α are the $s\alpha$ for $s \in G$, so $g(T) = \prod_{s\in G}(T - s\alpha)$ and

$$g'(\alpha) = \prod_{s\neq 1}(\alpha - s\alpha)$$

By proposition 3.6.2, $\mathfrak{d}_{L/K} = (g'(\alpha))$, so taking valuations and using $\nu_L(\alpha - s\alpha) = \nu_L(s\alpha - \alpha) = i_G(s)$ (lemma 3.4.2, since α generates $\mathcal{O}_L$ over $\mathcal{O}_K$),

$$\nu_L(\mathfrak{d}_{L/K}) = \sum_{s\neq 1}\nu_L(\alpha - s\alpha) = \sum_{s\neq 1} i_G(s)$$

For the second equality, count each $s \neq 1$ once for every $i \geq 0$ with $s \in G_i$; since $s \in G_i \iff i_G(s) \geq i+1 \iff i \leq i_G(s) - 1$, there are exactly $i_G(s)$ such indices, and so

$$\sum_{s\neq 1} i_G(s) = \sum_{s\neq 1}\#\left\{i \geq 0 \mid s \in G_i\right\} = \sum_{i\geq 0}\#\left\{s \neq 1 \mid s \in G_i\right\} = \sum_{i\geq 0}(g_i - 1)$$

completing the proof.

Corollary 3.6.4: The Different Detects Tameness

Let L/K be finite Galois with ramification index e. Then $\nu_L(\mathfrak{d}_{L/K}) \geq e - 1$, with equality if and only if L/K is tamely ramified. In general

$$\nu_L(\mathfrak{d}_{L/K}) = (e-1) + \sum_{i\geq 1}(g_i - 1)$$

and the wild term $\sum_{i\geq 1}(g_i - 1)$ is positive exactly when $p \mid e$.

***Proof*:**
By theorem 3.6.3, $\nu_L(\mathfrak{d}_{L/K}) = (g_0 - 1) + \sum_{i\geq 1}(g_i - 1)$, and $g_0 = |G_0| = e$ is the order of the inertia group. Each term $g_i - 1 \geq 0$, so $\nu_L(\mathfrak{d}_{L/K}) \geq e - 1$, with equality iff $g_i = 1$ for all $i \geq 1$, that is, iff the wild inertia G_1 is trivial. By corollary 3.4.5 this happens exactly when L/K is tamely ramified, equivalently $p \nmid e$, as we sought to show.

Descending from L to K turns the different into an ideal of the base. Recall that the norm of ideals satisfies $N_{L/K}(\mathfrak{P}^m) = \mathfrak{p}^{fm}$, where f is the residue degree.

Definition 3.6.5: Discriminant

The *discriminant* of L/K is the ideal $\mathrm{disc}_{L/K} = N_{L/K}(\mathfrak{d}_{L/K})$ of $\mathcal{O}_K$.

Proposition 3.6.6: Valuation of the Discriminant

If L/K is Galois with a single prime $\mathfrak{P}$ of residue degree f, then $\nu_K(\mathrm{disc}_{L/K}) = f \cdot \nu_L(\mathfrak{d}_{L/K})$. In particular, for a totally ramified extension $\nu_K(\mathrm{disc}_{L/K}) = \nu_L(\mathfrak{d}_{L/K})$, and L/K is unramified if and only if $\mathrm{disc}_{L/K} = \mathcal{O}_K$.

***Proof*:**
Writing $\mathfrak{d}_{L/K} = \mathfrak{P}^d$ with $d = \nu_L(\mathfrak{d}_{L/K})$, the norm is $N_{L/K}(\mathfrak{P}^d) = (N_{L/K}\mathfrak{P})^d = \mathfrak{p}^{fd}$, so $\nu_K(\mathrm{disc}_{L/K}) = fd$. For totally ramified $f = 1$. Finally $\mathrm{disc}_{L/K} = \mathcal{O}_K$ iff $d = 0$ iff $\mathfrak{d}_{L/K} = \mathcal{O}_L$ iff every $g_i - 1 = 0$ including $g_0 = e = 1$, which is unramifiedness (an unramified extension has $G_0 = 1$), completing the proof.

Finally, both invariants are multiplicative in towers, the property that lets a global discriminant be assembled from local contributions and that will drive the quotient theory of the next section.

Theorem 3.6.7: Transitivity of the Different

For a tower $L/K'/K$ of finite separable extensions of local fields,

$$\mathfrak{d}_{L/K} = \mathfrak{d}_{L/K'} \cdot \big(\mathfrak{d}_{K'/K}\mathcal{O}_L\big)$$

and consequently $\nu_L(\mathfrak{d}_{L/K}) = \nu_L(\mathfrak{d}_{L/K'}) + e_{L/K'}\,\nu_{K'}(\mathfrak{d}_{K'/K})$.

***Proof*:**
Transitivity of the trace, $\mathrm{tr}_{L/K} = \mathrm{tr}_{K'/K} \circ \mathrm{tr}_{L/K'}$, gives the corresponding statement for codifferents: for $x \in L$,

$$x \in \mathcal{C}_{L/K} \iff \mathrm{tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{O}_K \iff \mathrm{tr}_{K'/K}\big(\mathrm{tr}_{L/K'}(x\mathcal{O}_L)\big) \subseteq \mathcal{O}_K$$

The inner trace $\mathrm{tr}_{L/K'}(x\mathcal{O}_L)$ is a fractional ideal of $\mathcal{O}_{K'}$, and it lands in $\mathcal{C}_{K'/K}$ exactly when the outer condition holds; unwinding, $x \in \mathcal{C}_{L/K}$ iff $\mathrm{tr}_{L/K'}(x\mathcal{O}_L) \subseteq \mathcal{C}_{K'/K}$, iff $x \cdot \mathcal{C}_{K'/K}^{-1}\mathcal{O}_L \subseteq \mathcal{C}_{L/K'}$, that is $\mathcal{C}_{L/K} = \mathcal{C}_{L/K'} \cdot (\mathcal{C}_{K'/K}\mathcal{O}_L)$. Inverting gives $\mathfrak{d}_{L/K} = \mathfrak{d}_{L/K'} \cdot (\mathfrak{d}_{K'/K}\mathcal{O}_L)$. Applying ν_L and using $\nu_L|_{K'} = e_{L/K'}\nu_{K'}$ yields the valuation formula, completing the proof.

Example 3.6.8: Different of the Cyclotomic Fields
For $L = \mathbb{Q}_p(\zeta_p)$ the extension $L/\mathbb{Q}_p$ is totally and tamely ramified of degree $e = p-1$, so corollary 3.6.4 gives $\nu_L(\mathfrak{d}_{L/\mathbb{Q}_p}) = e - 1 = p - 2$; since it is totally ramified, $\nu_p(\mathrm{disc}_{L/\mathbb{Q}_p}) = p - 2$ as well, recovering $\mathrm{disc}(\mathbb{Q}(\zeta_p)) = \pm p^{p-2}$. For $L = \mathbb{Q}_p(\zeta_{p^2})$ the ramification groups computed in example 3.4.6 have $g_0 = p(p-1)$ and $g_1 = \cdots = g_{p-1} = p$, whence

$$\nu_L(\mathfrak{d}_{L/\mathbb{Q}_p}) = (g_0 - 1) + (p-1)(g_1 - 1) = \big(p(p-1) - 1\big) + (p-1)^2 = p(2p-3)$$

in agreement with the general formula $\nu_L(\mathfrak{d}_{\mathbb{Q}_p(\zeta_{p^n})/\mathbb{Q}_p}) = p^{n-1}(np - n - 1)$, which gives $p-2$ at $n=1$ and $p(2p-3)$ at $n=2$.

Example 3.6.9: Different of an Eisenstein Extension

Let $L = K(\pi)$ with π a root of an Eisenstein polynomial $g(T) = T^e + a_{e-1}T^{e-1} + \cdots + a_0$, so L/K is totally ramified of degree e and $\mathcal{O}_L = \mathcal{O}_K[\pi]$ (theorem 3.3.17). Then $\mathfrak{d}_{L/K} = (g'(\pi))$ with $g'(\pi) = e\pi^{e-1} + (e-1)a_{e-1}\pi^{e-2} + \cdots + a_1$. When $p \nmid e$ the leading term $e\pi^{e-1}$ has the strictly smallest valuation $e-1$ (the others have valuation $\geq e$, as $\nu_L(a_j) \geq e$ for $j \geq 1$ while $\nu_L(\pi^{j-1}) = j-1$), so $\nu_L(\mathfrak{d}_{L/K}) = e-1$, matching corollary 3.6.4. When $p|e$ the term $e\pi^{e-1}$ is no longer dominant and the different jumps above $e-1$, detecting the wild ramification.

Exercise 3.6.1

1. Compute $\nu_L(\mathfrak{d}_{L/\mathbb{Q}_2})$ for $L = \mathbb{Q}_2(\sqrt{2})$ directly from the minimal polynomial, and check it against $\sum_{i\geq 0}(g_i - 1)$ using the ramification groups found in section 3.4.
2. Show that $\nu_L(\mathfrak{d}_{L/K}) = 0$ if and only if L/K is unramified, working directly from theorem 3.6.3.
3. Let L/K be totally tamely ramified of degree e (so $p \nmid e$). Using an Eisenstein uniformizer, prove $\mathfrak{d}_{L/K} = (\pi^{e-1})$ and $\mathrm{disc}_{L/K} = \mathfrak{p}^{e-1}$.
4. Using theorem 3.6.7, verify for the tower $\mathbb{Q}_p(\zeta_{p^2})/\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$ that the middle and top differents add up (weighted by $e_{L/K'}$) to the different of $\mathbb{Q}_p(\zeta_{p^2})/\mathbb{Q}_p$.

3.7 The Hilbert Symbol

The different and the ramification groups measure how an extension sits over its base. The Hilbert symbol measures something dual: whether an element is a norm. It packages the question "is a a norm from $K(\sqrt{b})$?" into a pairing (a,b) taking values ± 1, and the payoff is that this pairing is bilinear, so norm conditions can be manipulated by linear algebra over $\mathbb{F}_2$. The symbol is the local atom of quadratic reciprocity, the invariant that classifies quaternion algebras over K, and the object that [7] reads from here to compute local Brauer groups. This section develops the quadratic symbol from scratch; the n-th symbol for general n, defined through the norm-residue map of local class field theory, and the explicit reciprocity laws belong to the sequel *Class Field Theory*.

Throughout, K is a local field of characteristic $\neq 2$, with valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{p}$, uniformizer π, and residue field $\mathbb{F}_q$.

Definition 3.7.1: The Hilbert Symbol

For $a, b \in K^\times$, the *Hilbert symbol* is

$$(a,b) = \begin{cases} 1 & \text{if } ax^2 + by^2 = z^2 \text{ has a solution } (x,y,z) \neq (0,0,0) \text{ in } K, \\ -1 & \text{otherwise.} \end{cases}$$

The symbol is unchanged if a or b is multiplied by a square, since substituting $x \mapsto cx$ turns ac^2x^2 back into ax^2; thus (a,b) depends only on the classes of a,b in $K^\times/(K^\times)^2$. It is also symmetric,

$(a, b) = (b, a)$, by exchanging the roles of x and y. The elementary identities below hold over any field of characteristic $\neq 2$.

Proposition 3.7.2: Elementary Identities

For all $a, b \in K^\times$:

1. $(a, b) = 1$ if a or b is a square;
2. $(a, -a) = 1$ and $(a, 1 - a) = 1$ (the latter when $a \neq 1$);
3. $(a, b) = 1$ if and only if a is a norm from $K(\sqrt{b})$, when b is not a square.

***Proof*:**

1. If $b = c^2$, then $(x, y, z) = (0, 1, c)$ solves $ax^2 + by^2 = z^2$; symmetrically for a.
2. The projective point $(1, 1, 0)$ solves $ax^2 - ay^2 = z^2$, and $(1, 1, 1)$ solves $ax^2 + (1-a)y^2 = z^2$, since $a + (1 - a) = 1$.
3. Suppose b is not a square, so $K(\sqrt{b})/K$ is a quadratic extension with norm $N(x + y\sqrt{b}) = x^2 - by^2$. If $(a, b) = 1$, take a nonzero solution of $as^2 + bt^2 = w^2$; the case $s = 0$ would force $b = (w/t)^2$ a square, so $s \neq 0$, and dividing by s^2 gives $a = (w/s)^2 - b(t/s)^2 = N\big((w/s)+(t/s)\sqrt{b}\big)$. Conversely, if $a = x^2 - by^2$ is a norm, then $(x, y, 1)$ solves $z^2 = a\cdot 1 + by^2$ after rearranging $x^2 = a + by^2$, so $(1, y, x)$ is a nonzero solution and $(a, b) = 1$.

This exhausts the cases, as we sought to show.

The symbol can be evaluated in general because $K^\times$ modulo squares is small. Because K has odd residue characteristic, Hensel's lemma makes every principal unit a square: for $u \in 1 + \mathfrak{p}$, the polynomial $X^2 - u$ reduces to $X^2 - 1 = (X - 1)(X + 1)$ with simple roots, which lift. Combined with the decomposition $K^\times = \pi^{\mathbb{Z}} \times \mu_{q-1} \times (1 + \mathfrak{p})$ of proposition 1.3.5, this gives the following.

Lemma 3.7.3: Squares in a Local Field

If K has odd residue characteristic, then $K^\times/(K^\times)^2$ is a group of order 4, represented by $\{1, u_0, \pi, u_0\pi\}$ where $u_0 \in \mu_{q-1}$ is any non-square unit. The class of a unit u is trivial if and only if its residue $\bar{u}$ is a square in $\mathbb{F}_q^\times$.

***Proof*:**
By proposition 1.3.5 every element is $\pi^\alpha \zeta u$ with $\zeta \in \mu_{q-1}$ and $u \in 1 + \mathfrak{p}$; the principal unit u is a square by the Hensel argument above, and π^α contributes α mod 2. The cyclic group μ_{q-1} has even order $q - 1$, so its squares form the unique subgroup of index 2, and ζ is a square iff $\bar{\zeta}$ is a square in $\mathbb{F}_q^\times$. Hence $K^\times/(K^\times)^2$ is generated by the classes of π and of a non-square unit u_0, each of order 2, giving a group of order 4, as we sought to show.

The values on units and uniformizers now determine everything. Write $\left(\frac{u}{\mathfrak{p}}\right) = 1$ if $\bar{u}$ is a square in $\mathbb{F}_q^\times$ and -1 otherwise, the quadratic residue symbol of the residue field.

Theorem 3.7.4: The Tame Hilbert Symbol

Let K have odd residue characteristic. For units $u, w \in \mathcal{O}^\times$,

$$(u,w) = 1, \qquad (\pi, u) = \left(\frac{u}{\mathfrak{p}}\right), \qquad (\pi,\pi) = \left(\frac{-1}{\mathfrak{p}}\right)$$

Consequently, writing $a = \pi^\alpha u$ and $b = \pi^\beta w$ with $u, w \in \mathcal{O}^\times$,

$$(a,b) = \left(\frac{-1}{\mathfrak{p}}\right)^{\alpha\beta} \left(\frac{u}{\mathfrak{p}}\right)^{\beta} \left(\frac{w}{\mathfrak{p}}\right)^{\alpha}$$

***Proof*:**
Units. Reducing $uX^2 + wY^2 = Z^2$ modulo $\mathfrak{p}$ gives a nondegenerate conic over $\mathbb{F}_q$; as the two sets $\left\{\bar{u}\,x^2 \,\middle|\, x \in \mathbb{F}_q\right\}$ and $\left\{\bar{z}^2 - \bar{w}\,y^2 \,\middle|\, y, z \in \mathbb{F}_q\right\}$ each have more than $q/2$ elements, they meet, yielding a nonzero $(\bar{x}, \bar{y}, \bar{z})$ with $\bar{u}\bar{x}^2 + \bar{w}\bar{y}^2 = \bar{z}^2$ and, after relabeling, $\bar{z} \neq 0$. Then $f(Z) = ux^2 + wy^2 - Z^2$ has $f(z) \equiv 0$ and $f'(z) = -2z \not\equiv 0 \bmod \mathfrak{p}$, so Hensel lifts z to a root in $\mathcal{O}$; hence $(u,w) = 1$.

Uniformizer against a unit. If $\bar{u}$ is a square then u is a square (lemma 3.7.3) and $(\pi, u) = 1$ by proposition 3.7.2. If $\bar{u}$ is not a square, suppose $\pi x^2 + uy^2 = z^2$ had a nonzero solution, scaled so that $\min(\nu(x), \nu(y), \nu(z)) = 0$. Reducing modulo $\mathfrak{p}$ kills πx^2 (valuation ≥ 1) and leaves $\bar{z}^2 = \bar{u}\,\bar{y}^2$; were $\bar{y} \neq 0$ this would make $\bar{u}$ a square, so $\bar{y} = 0$ and then $\bar{z} = 0$, forcing $\nu(y), \nu(z) \geq 1$. But then $\nu(\pi x^2) = \nu(z^2 - uy^2) \geq 2$, so $\nu(x) \geq 1$, contradicting the normalization. Thus $(\pi, u) = -1 = \left(\frac{u}{\mathfrak{p}}\right)$.

Uniformizer against itself. By proposition 3.7.2, $(\pi, -\pi) = 1$, so multiplying $b = \pi$ by the square-free part reveals $(\pi,\pi) = (\pi, -\pi)\,(\pi, -1) = (\pi, -1) = \left(\frac{-1}{\mathfrak{p}}\right)$ by the previous case applied to the unit -1.

The formula. Both sides depend only on a, b modulo squares, so by lemma 3.7.3 it suffices to check the four generators, which is exactly the three values above (with $(u,w) = 1$ handling $\alpha = \beta = 0$). Since the right-hand side is multiplicative in each of (α, u) and (β, w) and agrees with (a,b) on generators, it equals (a,b), completing the proof.

Corollary 3.7.5: Bilinearity and Nondegeneracy

For K of odd residue characteristic, the Hilbert symbol is a nondegenerate symmetric bilinear pairing

$$K^\times/(K^\times)^2 \times K^\times/(K^\times)^2 \to \{\pm 1\}$$

That is, $(aa', b) = (a,b)(a',b)$ for all a, a', b, and for each $a \notin (K^\times)^2$ there is a b with $(a,b) = -1$.

***Proof*:**
Bilinearity is visible from the explicit formula of theorem 3.7.4, whose right-hand side is a product of the multiplicative residue symbols and the exponents α, β combined additively. For

nondegeneracy, view $K^\times/(K^\times)^2$ as an $\mathbb{F}_2$-vector space with basis the classes of π and u_0, and write the symbol additively by sending $1 \mapsto 0$ and $-1 \mapsto 1$ in $\mathbb{F}_2$. Then the pairing has Gram matrix $\left(\begin{smallmatrix} \epsilon & 1 \\ 1 & 0 \end{smallmatrix}\right)$, where $\epsilon = 0$ if $q \equiv 1 \bmod 4$ and $\epsilon = 1$ if $q \equiv 3 \bmod 4$, coming from $(\pi, \pi) = \left(\frac{-1}{\mathfrak{p}}\right)$, $(\pi, u_0) = (u_0, \pi) = -1$, and $(u_0, u_0) = 1$. This matrix has determinant $-1 = 1 \neq 0$ in $\mathbb{F}_2$, so the form is nondegenerate; in particular every nonzero class pairs nontrivially with some class, completing the proof.

3.7.6 The Dyadic Case and the n-th Symbol When the residue characteristic is 2 (for instance $K = \mathbb{Q}_2$) the group $K^\times/(K^\times)^2$ is larger, of order 8 for $\mathbb{Q}_2$, and the explicit formula acquires correction terms involving the second-order behaviour of units; the symbol remains a nondegenerate symmetric bilinear pairing, proved by the same strategy of computing on generators. For $n > 2$, and for a full account of the norm-residue symbol $(a, b) = \sigma_a(\sqrt[n]{b})/\sqrt[n]{b}$ defined through the local reciprocity map, see *Class Field Theory*. The tame formula above is the odd-residue-characteristic case of the general explicit reciprocity law.

The symbol assembles across the places of a global field into the statement from which quadratic reciprocity falls out.

Proposition 3.7.7: The Product Formula

For $a, b \in \mathbb{Q}^\times$, the local symbols $(a, b)_v$ (one for each prime v and for $v = \infty$) satisfy $(a, b)_v = 1$ for all but finitely many v, and

$$\prod_v (a, b)_v = 1$$

***Proof*:**
By theorem 3.7.4, $(a, b)_p = 1$ whenever p is odd and a, b are both units at p, which excludes all but the finitely many primes dividing $2ab$; so the product is finite. Bilinearity (corollary 3.7.5) reduces the identity to the generators of $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, namely -1 and the primes, and for these the equality $\prod_v (a, b)_v = 1$ is precisely the law of quadratic reciprocity together with its two supplements, proved in [2]. The finiteness and the reduction are the local input developed here; the reciprocity law is the cited arithmetic input, and the extension to an arbitrary global field is carried out in *Class Field Theory*, completing the proof.

Example 3.7.8: Hilbert Symbols over $\mathbb{Q}_p$

Over $\mathbb{Q}_p$ with p odd, $(p, p) = \left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$, so $(p, p) = 1$ for $p \equiv 1 \bmod 4$ and -1 for $p \equiv 3 \bmod 4$. For a unit u, $(p, u) = 1$ exactly when u is a quadratic residue mod p; thus $(p, u) = -1$ detects that $\mathbb{Q}_p(\sqrt{u})$ is the unramified quadratic extension while $\mathbb{Q}_p(\sqrt{p})$ is ramified. Two units always pair trivially: $(u, w) = 1$, reflecting that a conic in two unit coefficients already has a smooth point over $\mathbb{F}_p$.

Exercise 3.7.1

1. Compute all Hilbert symbols (a, b) over $\mathbb{Q}_5$ for $a, b \in \{2, 5, 10\}$, using $\left(\frac{2}{5}\right) = -1$.

2. Using bilinearity (corollary 3.7.5) and proposition 3.7.2, show that $(a, -ab) = (a, b)$ for all $a, b \in K^\times$.
3. Prove that for K of odd residue characteristic, $(a, b) = 1$ for all $b \in K^\times$ if and only if a is a square, giving a second proof of nondegeneracy.
4. Let u be a unit of K (odd residue characteristic). Show that $K(\sqrt{u})/K$ is unramified and that $(\pi, u) = -1$ if and only if this extension is nontrivial, linking the symbol to the ramification of the quadratic extension it cuts out.

4

The Local-Global Correspondence

The local fields were introduced as the completions of a global field at its primes. This chapter closes the circle: the arithmetic of a global field is faithfully reflected in the family of its completions, and constructions carried out locally can be reassembled into global ones. The analytic engine is Krasner's lemma, which controls how the roots of a polynomial move under small perturbations of its coefficients; with it we prove the local-global correspondence, deduce the Kronecker-Weber theorem, and settle the Hasse principle for conics.

4.1 Krasner's Lemma and Continuity of Roots

Recall that if $f \in \mathbb{R}[x]$ or $\mathbb{C}[x]$ is a separable polynomial, then it is always possible to "jiggle" the roots around and f remains separable, or similarly the coefficients of f can be "jiggled" and the polynomial remains separable. In fact, more is true: the coefficients can be jiggled and the field extension $K(f(\alpha))/K$ and $k(g(\alpha))/K$ (for appropriate $K = \mathrm{Frac}(R)$ for complete DVR R) are isomorphic. This is the result aimed to be proven in this section. Such results shall come up a few times in the book it is required to use a separable polynomial f which has certain necessary roots but has some "bad" property by making a bad choice (whether a forced choice or arbitrary choice), and so another polynomial with the desired properties is found.

Lemma 4.1.1: Automorphism Preserves Absolute Value

Let K be a fraction field of a complete DVR with algebraic closure $\overline{K}$, and extend the absolute value $|-|$ to $\overline{K}$. Then for all $\alpha \in \overline{K}$ and $\sigma \in \mathrm{Aut}_K(\overline{K})$

$$|\sigma(\alpha)| = |\alpha|$$

***Proof*:**
First, $\alpha, \sigma(\alpha)$ have the same minimal polynomial, for if f is the minimal polynomial of α:

$$f(\sigma(\alpha)) = \sigma(f(\alpha)) = \sigma(0) = 0$$

Thus, from number theory ([2]) we have that

$$N_{K(\alpha)/K}(\alpha) = f(0) = N_{K(\sigma(\alpha))/K}(\sigma(\alpha))$$

Then by theorem 3.1.7, setting $n = \deg f$ we get:

$$|\sigma(\alpha)| = |N_{K(\sigma(\alpha))/K}(\sigma(\alpha))|^{1/n} = |N_{K(\alpha)/K}(\alpha)|^{1/n} = |\alpha|$$

completing the proof.

Definition 4.1.2: Belonging

Let K be the field of fraction of a complete DVR with absolute value $|-|$ extended to $\overline{K}$. Then for any $\alpha, \beta \in \overline{K}$, we say β *belongs* to α if

$$|\beta - \alpha| < |\beta - \sigma(\alpha)| \qquad \forall \sigma \in \mathrm{Aut}_K(\overline{K}) \setminus \{\sigma \mid \sigma(\alpha) = \alpha\}$$

In other words, β is strictly closer to α than all its conjugates.

As all nonarchimedean triangles are isosceles[1], it is equivalent to

$$|\beta - \alpha| < |\alpha - \sigma(\alpha)|$$

Theorem 4.1.3: Krasner's Lemma

Let K be a field of fraction of a complete DVR, $\alpha, \beta \in \overline{K}$, and α is separable over K. Then if β belongs to α, $K(\alpha) \subseteq K(\beta)$ (i.e. $\alpha \in K(\beta)$).

***Proof*:**
For the sake of contradiction, suppose β belongs to α but $\alpha \notin K(\beta)$. Then $K(\alpha, \beta)/K(\beta)$ is separable and nontrivial, thus here exists an automorphism $\sigma \in \mathrm{Aut}_{K(\beta)}(K(\alpha, \beta)$ where $\sigma(\alpha) \neq \alpha$. Applying lemma 4.1.1 to $\beta - \alpha \in \overline{K}$:

$$|\beta - \alpha| = |\sigma(\beta - \alpha)| = |\sigma(\beta) - \sigma(a)| = |\beta - \sigma(\alpha)|$$

where σ fixes α by definition. But this not contradicts that β belongs to α, completing the proof.

4.1.4 Hensel-like characterization Sutherland comments how this is similar to Hensel's lemma in that it also characterizes Henselian fields, and that the lemma was proven by Ostrowski.

[1] If one side is shorter than another, it is shorter than both

Example 4.1.5: Krasner's Lemma
Give an example of it failing for a non-Henselian field.

Definition 4.1.6: L1 Norm

Let K be a field with absolute value $|-|$, and let $f = \sum_i f_i x^i \in k[x]$. Then

$$\|f\|_1 = \sum_i |f_i|$$

The reader should verify that this is indeed a norm on the K-vector space $K[x]$.

Lemma 4.1.7: Roots Smaller Than Polynomial

Let K be a field with absolute value $|-|$, $f = \prod_i (x_i - \alpha_i) \in K[x]$ be a monic polynomial with $\alpha_1, ..., \alpha_n \in L$ where L/K is a field with absolute value extending $|-|$. Then for every root α_i of f:

$$|\alpha_i| < \|f_1\|_1$$

***Proof*:**
If $n = 1$, then:

$$|\alpha| < |\alpha| + 1 = \|f\|_1$$

So assume $n \geq 2$. If $\|f\|_1 = 1$ then $f = x^n$ so $\alpha = |\alpha| = 0$. If $\|f\|_1 > 1$, given $|\alpha| \leq 1$ the lemma still holds ($\|f\|_1 \geq 1 + |\alpha| > |\alpha|$). So consider the case $|\alpha| > 1$. Then:

$$0 = |f(\alpha)| = \left|\alpha^n + \sum_i^{n-1} f_i \alpha^i\right| \overset{(1)}{\geq} |\alpha|^n - \sum_i^{n-1} |f_i||\alpha|^i \overset{(|\alpha|>1)}{\geq} |\alpha|^n - |\alpha|^{n-1} \sum_i^{n-1} |f_i| \overset{(2)}{\geq} |\alpha| - (\|f\|_1 - 1)$$

where $\overset{(1)}{\geq}$ comes form some absolute value manipulation, namely:

$$|a| = |a + b - b| \leq |a + b| + |-b| = |a + b| + |b| \ \Rightarrow \ |a + b| \geq |a| - |b|$$

and $\overset{(2)}{\geq}$ comes from dividing by $|\alpha|^{n-1} \geq 1$. Re-ararnging:

$$\|f_1\| - 1 \geq |\alpha| \qquad \text{i.e.} \qquad \|f\|_1 \geq |\alpha| + 1 > |\alpha|$$

as we sought to show.

Theorem 4.1.8: Continuity of Roots

Let K be a field of fraction of a complete DVR and $f \in K[x]$ a monic irreducible separable polynomial. Then there exists a $\delta = \delta \in \mathbb{R}_{>0}$ such that for every monic polynomial $g \in K[x]$ where $\|f - g\|_1 < \delta$,

$$\text{every roof of } \beta \text{ of } g \text{ belongs to a root } \alpha \text{ of } f \text{ for which } K(\alpha) = K(\beta)$$

In particular, every such g is irreducible, separable, and has the same splitting field as f

***Proof*:**
First, given the theorem is true and δ gives the desired result, then we can choose $\delta' = \min\{1, \delta\}$ so then any monic polynomial g satisfying $\|f - g\|_1 < \delta \leq 1$ is forced to have the same degree as f; thus we can assume $\deg f = \deg g$.

To prove the theorem, first fixed an algebraic closure $\overline{K}$ and extend $|-|$. Let $\alpha_1, ..., \alpha_n$ be the roots of f in $\overline{K}$ so that:

$$f = \prod_i^n (x - \alpha_i) = \sum_i^n f_i x^i$$

Next, let

$$\epsilon = \min_{\alpha_i \neq \alpha_j} \{1, |\alpha_i - \alpha_j|\}$$

Then define δ with respect to f to be:

$$\delta = \left(\frac{\epsilon}{2(|f|_1 + 1)}\right)^n > 0$$

As f is monic, $|f|_1 \geq 1$ so as $\epsilon \leq 1$, $\delta < 1$. Now suppose $g = \sum_i^n g_i x^i$ is a monic polynomial st $|f - g|_1 < \delta$. Then by adding $f - f$ and the triangle inequality

$$|g|_1 \leq |f|_1 + |f - g|_1 < |f|_1 + \delta$$

Furthermore, for any root β of g:

$$|f(\beta)| = |f(\beta) - 0| = |f(\beta) - g(\beta)| = |(f - g)(\beta)| = |\left|\sum_i^n (f_i - g_i)\beta^i\right| \leq \sum_i^n |f_i - g_i||\beta|^i$$

To bound this further, by lemma 4.1.7 $|\beta| < |g|_1$ and as $|g|_1 \geq 1$, $|\beta|^i < |g|_i^1 \leq |g|_1^n$. Thus:

$$|f(\beta)| < |f - g|_1 |g|_1^n < \delta(|f|_1 + \delta)^n < \delta(|f|_1^n + 1)^n \leq \left(\frac{\epsilon}{2}\right)^n$$

Squishing these inequalities, we get:

$$\prod_i^n |\beta - \alpha_i| = f(\beta) < \left(\frac{\epsilon}{2}\right)^n$$

Then it must be that for at least one α_i:

$$|\beta - \alpha_i| < \frac{\epsilon}{2}$$

In fact, this α_i is the unique root that has this property, for by the triangle inequality

$$|\alpha_i - \alpha_j| \geq \epsilon \qquad i \neq j$$

Thus, letting $\alpha = \alpha_i$, β belongs to α. By Krasner's lemma (theorem 4.1.3), $K(\alpha) \subseteq K(\beta)$, and as $n \leq [K(\alpha) : K] \leq [K(\beta) : K] \leq n$ we in fact have $K(\alpha) = K(\beta)$. Then as $n = \deg(g) = [K(\beta) : K]$, g must be the minimal polynomial (and hence irreducible) for β. Furthermore, as $K(\alpha)$ is separable, g must be separable.

To show they have the same splitting field, note that if a root β of g belongs to a root α of f, then for any $\sigma \in \mathrm{Aut}_K(\overline{K})$ such that $\sigma(\alpha) \neq \alpha$:

$$|\tau(\beta) - \tau(\alpha)| = |\tau(\alpha - \beta)| = |\beta - \alpha| < |\alpha - \sigma(\alpha)| = |\tau(\alpha - \sigma(\alpha))| = |\tau(\alpha) - \tau(\sigma(\alpha))|$$

(note that $\sigma(\alpha) \neq \alpha \Leftrightarrow \tau(\sigma(\alpha)) \neq \tau(\alpha)$). Thus, as β belongs to α, $\tau(\beta)$ belongs to $\tau(\alpha)$. As $\mathrm{Aut}_K(\overline{K})$ acts transitively on the roots of f and g, *every* root β of g belongs to a dissent root α of f, and so $K(\alpha) = K(\beta)$ each root. It follows that f, g have the same splitting field, completing the proof.

4.2 The Local-Global Correspondence

Let $\hat{L}$ be a local field. Then either by definition or by proposition 1.3.2 we know that $\hat{L}$ is a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$ for some $p \leq \infty$ or some $q = p^r$. Furthermore, any completion of a global field with any nontrivial absolute value is a local field. It is thus reasonable to ask if we can complete the square; where $\hat{L}$ is the completion of a corresponding global field L that is a finite extension of $\mathbb{Q}$ or $\mathbb{F}_q(t)$. In the local number field case:

$$\begin{tikzcd}
\mathbb{Q}_p \arrow[r, "\text{finite ext.}"] & \hat{L} \\
\mathbb{Q} \arrow[u, "|-|_p \text{ compl.}"] \arrow[r, dashed, red, "\text{finite ext.}"] & L \arrow[u, "|-|_\mathfrak{p} \text{ compl.}"']
\end{tikzcd}$$

Generalizing the question, given a fixed global field K and a local field $\hat{K}$ that is the completion of K with respect to a nontrivial absolute value $|-|$, if it the case that every finite extension of local field $\hat{L}/\hat{K}$ corresponds to an extension of global field L/K where $\hat{L}$ is the completion of L with respect to to an absolute value whose restriction to K must be equivalent to $|-|$. This is indeed the case! We shall show this for the separable case as it is what we need and the inseparable case offers more complication

Theorem 4.2.1: Local to Global Correspondence

Let K be a global with with a non-trivial absolute value $|-|$ and $\hat{K}$ the completion of K with respect to $|-|$. Then every finite separable extension $\hat{L}/\hat{K}$ is the completion of a finite separable extension of L of K with respect to an absolute value that restricts to $|-|$. Furthermore, L can be chosen so that

$$[L : K] = [\hat{L} : \hat{K}]$$

in which case $\hat{L} = \hat{K} \cdot L$.

***Proof*:**
Let $\hat{L}/\hat{K}$ be a separable extension of degree n. Let K have an absolute value. If $|-|$ is Archimedean, then K is a number field and $\hat{K} \underset{\sim}{\in} \{\mathbb{R}, \mathbb{C}\}$, in which case these cases are left to the reader.

So assume $|-|$ is nonarchimedean so that the valuation ring of $\hat{K}$ is a complete DVR and $|-|$ is induced by its discrete valuation. As the extension is separable, by the primitive element theorem, we have a monic irreducible and separable polynomial $f \in \hat{K}[x]$ such that

$$\hat{L} = \frac{\hat{K}[x]}{f}$$

Let us now find an appropriate $g \in K[x]$. As K is dense in its completion $\hat{K}$, we can find a monic $g \in K[x]$ such that $\|g - f\|_1 < \delta$ for any $\delta > 0$. Then by theorem 4.1.8, g is separable and

$$\hat{L} = \frac{\hat{K}[x]}{g}$$

Then $\hat{L}$ is a finite separable extension of the field of fraction of a complete DVR, and so it is a field of fraction of a complete DVR and has a unique absolute value that extends the absolute value $|-|$ on $\hat{K}$.

Now, consider $L := K[x]/(g)$. The polynomial is irreducible in $\hat{K}[x]$, and hence in $K[x]$, and so

$$[L : K] = \deg g = [\hat{L} : \hat{K}]$$

Note the field $\hat{L}$ contains both $\hat{K}$ and L, and by construction is the smallest field that does this (note that g is irreducible in $\hat{K}[x]$), and so

$$\hat{L} = \hat{K}L$$

Finally, the absolute value on $\hat{L}$ restricts to an absolute value on L extending the absolute value $|-|$ on K , and hence $\hat{L}$ is complete and hence contains the completion of L with respect to $|-|$. On the other hand, the completion of L with respect to $|-|$ contains L and $\overline{K}$, and so must be $\hat{L}$, as we sought to show.

Let us now assume that $\hat{L}/\hat{K}$ is galois. Then is the corresponding global extension L/K galois, and are their galois groups isomorphic? As it turns out, it need not even be galois:

Example 4.2.2: Corresponding Global Extension Not Galois

Let $K = \mathbb{Q}$ and $\hat{K} = \mathbb{Q}_7$. Take $\hat{L} = \hat{K}[x]/(x^3 - 2)$. This extension is galois. The extension by definition contains the cube-root of 2, $\sqrt[3]{2}$. The other roots are this root times ζ_3, which is contains in $\mathbb{Q}_7$. This is true as $x^2 + x + 1 \in \mathbb{F}_7[x]$ has two roots which by Hensel's lemma both lift to roots in $\mathbb{Q}_7[x]$. However, $L = K[x]/(x^3 - 2)$ is *not* a galois extension, containing only one root.

However, if we replace K with $\mathbb{Q}(\zeta_3)$, this doesn't change the completion to $\hat{K}$ (take the completion of K with respect to the absolute value induced by a prime above 7), and certainly doesn't change $\hat{L}$, but now $L = K[x]/(x^3 - 2)$ *is* a galois extension.

Is it always possible to give a nice enough adjustment in the base global field K to get a correspondence of galois groups? This turns out to be the case.

Corollary 4.2.3: Local To Global Correspondence, Galois Groups

Let $\hat{L}/\hat{K}$ be a finite galois extension of local fields. Then there exists a finite galois extension of global fields L/K and an absolute value $|-|$ on L such that $\hat{L}$ is the completion of L with respect to $|-|$, $\hat{K}$ is the completion of K with respect to to the restriction of $|-|$ to K, and

$$\mathrm{Gal}_K(L) \cong \mathrm{Gal}_{\hat{K}}(\hat{L})$$

***Proof*:**

The Archimedean case is immediate so assume $|-|$ is nonarchimedean on $\hat{L}$ and it restricts to an absolute value on $\hat{K}$. By theorem 4.2.1 we may assume that this field is the completion of a global field K with respect to the restriction of $|-|$. As in the proof of that theorem, take a monic separable irreducible polynomial $g \in K[x]$ that has all those properties in $\hat{K}[x]$ such that $\hat{L} = \hat{K}[x]/(g)$ and define again $L = K[x]/(g)$ so that $\hat{L} = \hat{K}L$.

This time, let M/K be the splitting field of g. As $\hat{L}/\hat{K}$ is galois, $\hat{L}$ contains the roots too, and so by the universal property of splitting fields $\hat{L}$ contains a subextension in of K isomorphic to M; we may identify this extension with M.

Now, note that $\hat{L}$ is equal to the completion of M with respect to to the restriction of $|-|$ to M. From this we have a group homomorphism induce by restriction:

$$\varphi : \mathrm{Gal}_{\hat{K}}(\hat{L}) \to \mathrm{Gal}_K(M)$$

This map is injective, as each $\sigma \in \mathrm{Gal}_{\hat{K}}(\hat{L})$ is determined by its action on the roots of g in M. Now, replace K by a fixed field of the image of φ, and replace L with M. The completion of K with respect to to the restriction of $|-|$ is still $\hat{K}$, and similarity for L and $\hat{L}$, and so:

$$\mathrm{Gal}_{\hat{K}}(\hat{L}) \cong \mathrm{Gal}_K(L)$$

as we sought to show.

Let us now put this all together and consider the $AKLB$ case and look a primes $\mathfrak{P}$ lying over $\mathfrak{p}$:

Theorem 4.2.4: Global To Local Correspondence

Assume AKLB[a], $\mathfrak{p} \subseteq A$, $\mathfrak{p}B = \prod_{\mathfrak{P}/\mathfrak{p}} \mathfrak{P}^{e_\mathfrak{P}}$. Let $K_\mathfrak{p}$ be the completion of K with respect to $|-|_\mathfrak{p}$ which has maximal ideal $\hat{\mathfrak{p}}$ in its valuation ring. Similarly, for each $\mathfrak{P}/\mathfrak{p}$, let $L_\mathfrak{P}$ denote the completion of L with respect to $|-|_\mathfrak{P}$ and let $\hat{\mathfrak{P}}$ denote the maximal ideal of the corresponding maximal ideal of the valuation ring. Then:

1. Each $L_\mathfrak{P}$ is a finite separable extension of $K_\mathfrak{p}$ with $[L_\mathfrak{P} : K_\mathfrak{p}] \leq [L : K]$.
2. Each $\hat{\mathfrak{P}}$ is the unique prime of $L_\mathfrak{P}$ lying over $\hat{\mathfrak{p}}$.
3. Each $\hat{\mathfrak{P}}$ has ramification index $e_{\hat{\mathfrak{P}}} = e_\mathfrak{P}$ and residue field degree $f_{\hat{\mathfrak{P}}} = f_\mathfrak{P}$.
4. $[L_\mathfrak{P} : K_\mathfrak{p}] = e_\mathfrak{P} f_\mathfrak{P}$
5. The map $L \otimes_K K_\mathfrak{p} \to \prod_{\mathfrak{P}/\mathfrak{p}} L_\mathfrak{P}$ given by $\ell \otimes x \mapsto (\ell x, ..., \ell x)$ is an isomorphism of finite étale $K_\mathfrak{p}$ algebras
6. If L/K is galois, then each $L_\mathfrak{P}/K_\mathfrak{p}$ is galois and we have isomorphisms of decomposition groups and inertia groups:

$$D_\mathfrak{P} \cong D_{\hat{\mathfrak{P}}} = \mathrm{Gal}_{K_\mathfrak{p}}(L_\mathfrak{P}) \qquad I_\mathfrak{P} \cong I_{\hat{\mathfrak{P}}}$$

[a] Recall L/K is finite and separable

***Proof*:**
(2) follows from theorem 3.1.4, (3) from simple computations, (4) from (2) and (3) and the fundamental identity. We'll show 1, 5, and 6

1. For each $\mathfrak{P}/\mathfrak{p}$, the embedding $K \hookrightarrow L$ induces an embedding $K_\mathfrak{p} \to L_\mathfrak{P}$ by mapping the equivalence classes, which is well-defined as a sequence which is Cauchy in K with respect to $|-|_\mathfrak{p}$ is also Cauchy in L with respect to $|-|_\mathfrak{P}$ (the valuation is extended). Thus $K_\mathfrak{p}$ can be viewed as a topological subfield of $L_\mathfrak{P}$. To show $[L_\mathfrak{P} : K_\mathfrak{p}] \leq [L : K]$, first any K-basis $b_1, ..., b_m$ for $L \subseteq L_\mathfrak{P}$ spans $L_\mathfrak{P}$ as a $K_\mathfrak{p}$-vector space. To see this, if $y = (y_n)$ is a Cauchy sequence of elements in L, then we can write for each y_n position:

$$y_n = x_{1,n} b_1 + \cdots + x_{m,n} b_m \qquad x_{i,n} \in K$$

   these all will give Cauchy-sequences, as linear maps of finite dimensional normed space are uniformly continuous, and thus preserve Cauchy sequences.

   Next, L is a finite étale K-algebra as L/K is separable. Thus the change of basis $L \otimes_K K_\mathfrak{p}$ to $K_\mathfrak{p}$ is a finite étale $K_\mathfrak{p}$-algebra (see [4]). Now, consider the $K_\mathfrak{p}$-algebra homomorphism:

$$\varphi_\mathfrak{P} : L \otimes_K K_\mathfrak{p} \to L_\mathfrak{P} \qquad \ell \otimes x \mapsto \ell x$$

   We have that $\varphi_\mathfrak{P}(b_i \otimes 1) = b_i$ for each K-basis element $b_i \in L$, and as $b_1, ..., b_m$ span $L_\mathfrak{P}$ as a $K_\mathfrak{p}$-vector space, this map is surjective. Next, $L \otimes_K K_\mathfrak{p}$ is by definition finite étale algebraic, hence it is isomorphic to a finite product of finite separable extensions of $K_\mathfrak{p}$. But then $L_\mathfrak{P}$ is isomorphic to a subproduct, and thus is also a finite étale $K_\mathfrak{p}$ algebra, implying $L_\mathfrak{P}/K_\mathfrak{p}$ is separable and finite.

5. From (1), combine each $\varphi_{\mathfrak{P}}$ to define the $K_{\mathfrak{p}}$-algebra homomorphism:

$$\varphi = \prod_{\mathfrak{P}/\mathfrak{p}} \varphi_{\mathfrak{P}} \qquad \varphi : L \otimes_K K_{\mathfrak{p}} \to \prod_{\mathfrak{P}/\mathfrak{p}} L_{\mathfrak{P}}$$

As dimension of étale algebras is base-change invariant (again see [4]):

$$\dim_{K_{\mathfrak{p}}}(L \otimes_K K_{\mathfrak{p}}) = \dim_K L = [L : K] = \sum_{\mathfrak{P}/\mathfrak{p}} e_{\mathfrak{P}} f_{\mathfrak{P}} = \sum_{\mathfrak{P}/\mathfrak{p}} [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = \dim_{K_{\mathfrak{p}}} \prod_{\mathfrak{P}/\mathfrak{p}} L_{\mathfrak{P}}$$

To show φ is an isomorphism, it suffices to show the map is surjective (as the dimensions line up). If we can pick a $K_{\mathfrak{p}}$-basis $\mathcal{B}$ for $L \otimes_K K_{\mathfrak{p}}$ and show that $\det([\varphi]_{\mathcal{B}}) \neq 0$, the result is proven. To that end, pick a basis $\mathcal{B}$ of tuples $\beta_i = (\beta_{i,\mathfrak{P}})_{\mathfrak{P}/\mathfrak{p}}$ for the codomain. To find a basis that maps to it, by theorem 4.1.8 pick $\alpha_i \in L$ such that

$$|\alpha_i - \beta_{i,\mathfrak{P}}|_{\mathfrak{P}} < \epsilon \qquad \forall \mathfrak{P}/\mathfrak{p}$$

Thus, given the product the sup-norm, each $\varphi(\alpha_i \otimes 1)$ is "close" to β_i; in particular the matrix M where each column expresses $\varphi(\alpha_j \otimes 1)$ in terms of β_i is closed to the identity matrix (note the determinant is continuous!), and hence the determinant is close to 1. Then for ϵ small enough, it must be that $\det(M) \neq 0$, and so φ is surjective, and thus an isomorphism.

6. Assume L/K is galois. Then each $\sigma \in D_{\mathfrak{P}}$ that acts on L respect $v_{\mathfrak{P}}$ as it fixes $\mathfrak{P}$, and so σ preserves Cauchy-sequences, showing that σ is an automorphism of $L_{\mathfrak{P}}$ and fixes $K_{\mathfrak{P}}$. We thus get a well-defined map:

$$\varphi : D_{\mathfrak{P}} \to \mathrm{Aut}_{K_{\mathfrak{p}}}(L\mathfrak{P})$$

This map is injective, for if σ acts trivially on $L_{\mathfrak{P}}$, certainly acts trivially on L. For surjectivity, note that :

$$e_{\mathfrak{P}} f_{\mathfrak{P}} = |D_{\mathfrak{P}}| \leq \#\mathrm{Aut}_{K_{\mathfrak{p}}}(L\mathfrak{P}) \leq [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = e_{\mathfrak{P}} f_{\mathfrak{P}}$$

and so $\#\mathrm{Aut}_{K_{\mathfrak{p}}}(L\mathfrak{P}) \leq [L_{\mathfrak{P}} : K_{\mathfrak{p}}]$ showing that $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is galois and that φ is surjective, and hence an automorphism. Next, as $\hat{\mathfrak{P}}$ is the only prime of $L_{\mathfrak{P}}$, it must be fixed by every $\sigma \in \mathrm{Gal}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}})$ and so

$$D_{\mathfrak{P}} \cong D_{\hat{\mathfrak{P}}} = \mathrm{Gal}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}})$$

Finally, the inertia groups $I_{\mathfrak{P}}$ and $\hat{I}_{\mathfrak{P}}$ both have the same order and φ restricts to an (injective) homomorphism between these groups, and hence they are isomorphic.

Corollary 4.2.5: Global To Local Correspondence, Norm and Trace

Assume AKLB and let $\mathfrak{p}$ be a prime of A. Then for every $\alpha \in L$, we have:

$$N_{L/K}(\alpha) = \prod_{\mathfrak{P}/\mathfrak{p}} N_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(\alpha) \qquad \mathrm{tr}_{L/K}(\alpha) = \sum_{\mathfrak{P}/\mathfrak{p}} \mathrm{tr}_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(\alpha)$$

***Proof*:**
The norm and trace defined by the determinant and trace of the K-linear map $L \xrightarrow{\times\alpha} L$ is unchained when tensoring with $K_{\mathfrak{p}}$, hence by the isomorphism in part (5) of the above theorem we get the desired result.

Corollary 4.2.6: Global To Local Correspondence, Decomposition

Assume AKLB,$\mathfrak{p}$ a prime of A, $\mathfrak{p}B = \prod_{\mathfrak{P}/\mathfrak{p}} \mathfrak{P}^{e_{\mathfrak{P}}}$, $\hat{A}_{\mathfrak{p}}$ the completion of A with respect to $|-|_{\mathfrak{p}}$ and for each $\mathfrak{P}/\mathfrak{p}$ let $\hat{B}_{\mathfrak{P}}$ denote the completion of B with respect to $|-|_{\mathfrak{P}}$. Then

$$B \otimes_A \hat{A}_{\mathfrak{p}} \cong_{\hat{A}_{\mathfrak{p}}} \prod_{\mathfrak{P}/\mathfrak{p}} \hat{B}_{\mathfrak{P}}$$

***Proof*:**
Choose some prime $\mathfrak{p} \subseteq A$. Then $B_{\mathfrak{p}}$ is a free A-module of rank $n = [L : K] = \sum_{\mathfrak{P}/\mathfrak{p}} e_{\mathfrak{P}} f_{\mathfrak{P}}$, so $B \otimes_A \hat{A}_{\mathfrak{p}}$ is a free $\hat{A}_{\mathfrak{p}}$-module of rank n. Next, as $\hat{A}_{\mathfrak{p}}$ and $\hat{B}_{\mathfrak{P}}$ are DVR's with respect to their respective primes, by theorem 4.2.4(4) $\prod_{\mathfrak{P}/\mathfrak{p}} B_{\mathfrak{P}}$ is a free $\hat{A}_{\mathfrak{p}}$-module of rank n. Finally, by theorem 4.2.4 these two $\hat{A}_{\mathfrak{p}}$-modules both lie in isomorphic finite étale $K_{\mathfrak{p}}$-algebras $L \otimes_K K_{\mathfrak{p}} \cong \prod_{\mathfrak{P}/\mathfrak{p}} L_{\mathfrak{P}}$, and it is clear this $\hat{K}_{\mathfrak{p}}$-isomorphism restricts to a $\hat{A}_{\mathfrak{p}}$-isomorphism, completing the proof.

4.3 The Kronecker-Weber Theorem

We are now in a position to prove the famous Kronecker-Weber Theorem using local fields. Let $K/\mathbb{Q}$ be a finite abelian extension so that $\mathrm{Gal}_{\mathbb{Q}}(K) = G$. Then we know that G can be represented as:

$$G \cong (\mathbb{Z}/r_1\mathbb{Z})^{n_1} \times \cdots \times (\mathbb{Z}/r_k\mathbb{Z})^{n_k}$$

We can embed G into $\mathbb{C}$ with some clever multiplication tricks. The question becomes if we can do the same with the field extensions:

Theorem 4.3.1: Global Kronecker-Weber Theorem

Every finite abelian extension of $\mathbb{Q}$ lies in a Cyclotomic field $\mathbb{Q}(\zeta_m)$

***Proof*:**
soon

This result can be readily deduced from the local result:

Theorem 4.3.2: Local Kronecker-Weber Theorem

Every finite abelian extension of $\mathbb{Q}_p$ lies in a Cyclotomic field $\mathbb{Q}_p(\zeta_m)$

***Proof*:**
soon

Proposition 4.3.3: Local Implies Global Kronecker-Weber

The Local Kronecker-Weber Theorem implies the Global Kronecker-Weber Theorem

***Proof*:**
Let $K/\mathbb{Q}$ be a finite abelian extension. Then for each ramified prime p of $\mathbb{Q}$, pick a prime $\mathfrak{p}/p$ and let $K_\mathfrak{p}$ be the completion of K at $\mathfrak{p}$. As $K/\mathbb{Q}$ is galois, it makes no difference which $\mathfrak{p}$ is picked. Then we know by theorem 4.2.4 that:

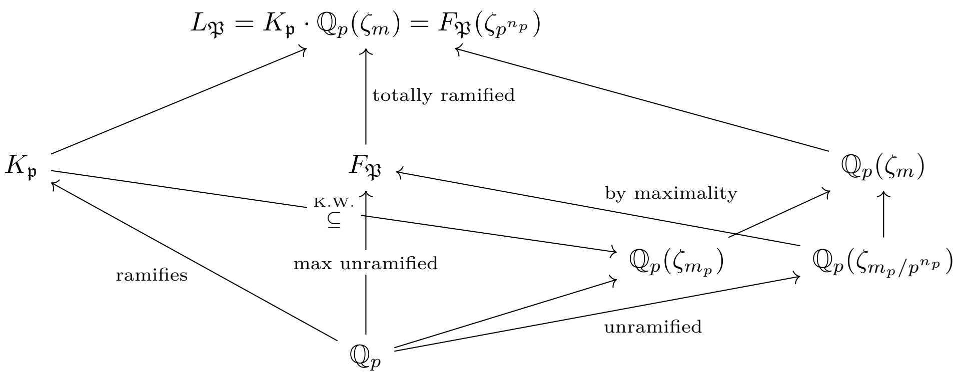
$$\mathrm{Gal}_{\mathbb{Q}_p}(K_\mathfrak{p}) \cong D_\mathfrak{p} \subseteq \mathrm{Gal}_\mathbb{Q}(K)$$

Hence $K_\mathfrak{p}$ i san abelian extension of $\mathbb{Q}_p$. By the local Kronecker-Weber Theorem we have that $K_\mathfrak{p} \subseteq \mathbb{Q}_p(\zeta_{m_p})$ for some $m_p \in N_{>0}$. Now, let $n_p = \nu_p(m_p)$, and define:

$$m = \prod_p p^{n_p}$$

which is finite as there are only finitely many ramified primes. Define $L = K(\zeta_m)$. Then we will show that $L = \mathbb{Q}(\zeta_m)$, showing that $K \subseteq \mathbb{Q}(\zeta_m)$.

First, the field $L = K \cdot \mathbb{Q}(\zeta_m)$ is the compositum of galois extensions of $\mathbb{Q}$, and hence is galois over $\mathbb{Q}$ with galois group isomorphic to a subgroup of $\mathrm{Gal}_\mathbb{Q}(K) \times \mathrm{Gal}_\mathbb{Q}(\mathbb{Q}(\zeta_m))$, and so is abelian. Let $\mathfrak{P}$ be a prime of L lying above the ramified prime $\mathfrak{p}/p$. Then the completion $L_\mathfrak{P}$ is a finite abliean extension of $\mathbb{Q}_p$ as $L/\mathbb{Q}$ is finite abelian. By minimality, $L_\mathfrak{P} = K_\mathfrak{p} \cdot \mathbb{Q}_p(\zeta_m)$. Now, let $F_\mathfrak{P}$ be the maximal unramified extension of $\mathbb{Q}_p$ in $L_\mathfrak{P}$. Then $L_\mathfrak{P}/F_\mathfrak{P}$ is totally ramified, giving $\mathrm{Gal}_{F_\mathfrak{P}}(L_\mathfrak{P}) \cong I_\mathfrak{P} = I_p$ (by theorem 4.2.4). Now, since $K_\mathfrak{p} \subseteq \mathbb{Q}_p(\zeta_{m_p})$ and $\mathbb{Q}_p(\zeta_{m_p/p^{n_p}})$ is unramified by lemma 3.3.5, we have $L_\mathfrak{P} = F_\mathfrak{P}(\zeta_{p^{n_p}})$ (since $\mathbb{Q}_p(\zeta_{p^{n_p}})$ is unramified). Hence $K_\mathfrak{p} \subseteq F_\mathfrak{P}(\zeta_{p^{n_p}})$. There are a lot of terms going around, so we may visualize them in the following diagram:



Furthermore, $F_\mathfrak{P} \cap \mathbb{Q}_p(\zeta_{p^{n_p}}) = \mathbb{Q}_p$ as $\mathbb{Q}_p(\zeta_{p^{n_p}})$ is totally ramified, hence:

$$I_p \cong \mathrm{Gal}_{F_\mathfrak{P}}(L_\mathfrak{P}) \cong \mathrm{Gal}_{\mathbb{Q}_p}(\mathbb{Q}_p(\zeta_{p^{n_p}}) \cong (\mathbb{Z}/p^{n_p}\mathbb{Z})^*$$

Now, let I be the group generated by the union of the groups $I_p \subseteq \mathrm{Gal}_{\mathbb{Q}}(L)$ for $p\big|m$. Then since $\mathrm{Gal}_{\mathbb{Q}}(L)$ is abelian, $I \subseteq \prod_p I_p$, and so:

$$\#I \leq \prod_{p\big|m} \#I_p = \prod_{p\big|m} \#(\mathbb{Z}/p^{n_p}\mathbb{Z}) = \prod_{p\big|m} \varphi(p^{n_p}) = \varphi(m) = [\mathbb{Q}(\zeta_m) : \mathbb{Q}]$$

Now, recall that each inertia field L^{I_p} is unramified at p, and $L^I \subseteq L^{I_p}$, and so $L^I/\mathbb{Q}$ is unramified, but then by definition of L^i, $L^I = \mathbb{Q}$. Then:

$$[L : \mathbb{Q}] = [L : L^I] = \#I \leq [\mathbb{Q}(\zeta_m) : \mathbb{Q}]$$

giving $\mathbb{Q}(\zeta_m) \subseteq L$, which must mean that $L = \mathbb{Q}(\zeta_m)$, but hen $K \subseteq L = \mathbb{Q}(\zeta_m)$, as we sought to show.

Before proving the main result, we shall prove a quick important lemma:

Lemma 4.3.4: Kronecker Weber Theorem, Lemma

For any prime p,

$$\mathbb{Q}_p\left((-p)^{1/(p-1)}\right) = \mathbb{Q}_p(\zeta_p)$$

***Proof*:**

Let $\alpha = (-p)^{1/(p-1)}$. Then α is the root of the Eisenstein polynomial $x^{p-1} + p$, and hence the extension $\mathbb{Q}_p(\alpha)$ is totally ramified of degree $p-1$ and α is a uniformizer. Let $\pi = \zeta_p - 1$. The minimal polynomial of π is:

$$f(x) = \frac{(x+1)^p - 1}{x} = x^{p-1} + px^{p-2} + \cdots + p$$

which his Eisenstein, so $\mathbb{Q}_p(\pi) = \mathbb{Q}_p(\zeta_p)$ and is totally ramified of degree $p-1$ with π as a uniformizer. Then we have $u = -\pi^{p-1}/p \equiv 1 \bmod \pi$ so u is a unit in the ring of integers of $\mathbb{Q}_p(\zeta_p)$. Setting $g(x) = x^{p-1} - u$ then $g(1) \equiv 0 \bmod \pi$ and $g'(1) = p - 1 \not\equiv 0 \bmod \pi$, and so by Hensel's lemma we can lift 1 to a root β of $g(x)$ in $\mathbb{Q}_p(\zeta_p)$.

We now have $p\beta^{p-1} = pu = -\pi^{-1}$, hence $(\pi/\beta)^{p-1} + p = 0$, and so $\pi/\beta \in \mathbb{Q}_p(\zeta_p)$ is a root of the minimal polynomial of α. As $\mathbb{Q}_p(\zeta_p)$ is galois, $\alpha \in \mathbb{Q}_p(\zeta_p)$. And since both $\mathbb{Q}_p(\alpha)$ and $\mathbb{Q}_p(\zeta_p)$ both have degree $p-1$, the fields are equal, as we sought to show.

***Proof*:**
of the local Kronecker Weber Theorem

Let us first reduce the proof to cyclic extensions of prime-power degree. This comes down to the correspondence between galois groups and compositum's. Namely, if $L_1/K, L_2/K$ are extensions where $L_1 \cap L_2 = K$, then their galois group is the product of the two galois groups. Then by the classification of finite abelian groups, we may decompose the abelian extension L/K into a compositum $L = L_1 \cdots L_n$ of linearly disjoint cyclic extensions L_i/K of prime-power degree. Then if each L_i lies in a Cyclotomic extension $K(\zeta_{m_i})$, so does L since

$$L \subseteq K(\zeta_{m_1}) \cdots K(\zeta_{m_n}) = K(\zeta_m) \qquad m = m_1 \cdots m_n$$

Hence, we shall focus on cyclic extensions $K/\mathbb{Q}_p$ of prime power l^r. This breaks up into two cass: $\ell = p$ and $\ell \neq p$.

The $\ell \neq p$ case

Let F be the maximal unramified extension of $\mathbb{Q}_p$ in K. Then by proposition 3.3.6 we have $F = \mathbb{Q}_p(\zeta_n)$. Then K/F is totally ramified, and as $\ell \neq p$ it must be tamely ramified. Then by theorem 3.3.21 we have

$$K = F(\pi^{1/e})$$

for some uniformizer π with $e = [K : F]$. Take $\pi = -pu$ for some $u \in \mathcal{O}_F^\times$, as $F/\mathbb{Q}_p$ is unramified[a]. Then the field $K = F(\pi^{1/e}) \subseteq F((-p)^{1/e}) \cdot F(u^{1/e})$. We shall show both lie in a Cyclotomic extension of $\mathbb{Q}_p$.

Now, since $\nu_{\mathfrak{P}}(\mathrm{disc}(x^e - u)) = 0$ for $p \nmid e$, $F(u^{1/e})/F$ is unramified, so $F(u^{1/e})/\mathbb{Q}_p$ is unramified, hence $F(u^{1/e}) = \mathbb{Q}_p(\zeta_k)$ for some $k \in \mathbb{N}_{>0}$. Take the compositum $K(u^{1/e}) = K\mathbb{Q}_p(\zeta_k)$, which is abelian and contains the subextension $\mathbb{Q}_p((-p)^{1/e})/\mathbb{Q}_p$ which is galois as it lies in an abelian extension. It is totally ramified since it is an Eisenstein extension. Next, the field $\mathbb{Q}_p((-p)^{1/e})$ containd ζ_e by taking the ratios of roots of $x^e + p$, but $\mathbb{Q}_p(\zeta_e)/\mathbb{Q}_p$ is unramified as $p \nmid e$, so it must be that

$$\mathbb{Q}_p(\zeta_e) = \mathbb{Q}_p$$

Thus, $e\big|(p-1)$. By lemma 4.3.4:

$$\mathbb{Q}_p((-p)^{2/r}) \subseteq \mathbb{Q}_p((-p)^{1/(p-1)}) = \mathbb{Q}_p(\zeta_p)$$

Then $F((-p)^{1/e}) = F \cdot \mathbb{Q}_p((-p)^{1/e}) \subseteq \mathbb{Q}_p(\zeta_n)\mathbb{Q}_p(\zeta_p) \subseteq \mathbb{Q}_p(\zeta_{np})$. Hence:

$$K \subseteq F(u^{1/e}) \cdot F((-p)^{1/e}) \subseteq \mathbb{Q}(\zeta_k) \cdot \mathbb{Q}(\zeta_{np}) \subseteq \mathbb{Q}(\zeta_{knp})$$

completing the $\ell \neq p$ case.

The $\ell = p > 2$ case

Let us start with a cyclic extension $K/\mathbb{Q}_p$ of odd degree p^r. Then there are two candidates:

1. $\mathbb{Q}_p(\zeta_{p^{p^r}-1})$, which is unramified of degree p^r
2. the index $p-1$ subfield of $\mathbb{Q}_p(\zeta_{p^{r+1}})$ which is a totally ramifed extension of degree p^r

If K is contained in the compositum of these two fields, we are done. Otherwise, the field $K(\zeta_m($ is a galois extension of $\mathbb{Q}_p$ with:

$$\mathrm{Gal}_{\mathbb{Q}_p}(K(\zeta_m)) \cong \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/p^s\mathbb{Z}$$

for some $s > 0$, in particular the first factor is form the galois group of $\mathbb{Q}_p(\zeta_{p^{p^r}-1})$, the second two from the factors from the galois group of $\mathbb{Q}_p(\zeta_{p^{r+1}})$ (note that $\mathbb{Q}_p(\zeta_{p^{p^r}-1}) \cap \mathbb{Q}_p(\zeta_{p^{r+1}}) = \mathbb{Q}_p$) and the last factor comes from the fact that by assumption $K \not\subseteq \mathbb{Q}_p(\zeta_m)$ so $\mathrm{Gal}_{\mathbb{Q}_p(\zeta_m)}(K(\zeta_m))$ is nontrivial and must have order p^s for some $1 \leq s \leq r$.

Now, the abelian group $\mathrm{Gal}_{\mathbb{Q}_p}(K(\zeta_m))$ has a quotient isomorphic to $(\mathbb{Z}/p\mathbb{Z})^3$, and the subfield of $K(\zeta_m)$ corresponding to this quotient is an abelian extension of $\mathbb{Q}_p$ with galois group $(\mathbb{Z}/p\mathbb{Z})^3$. I claim there is no such fields.

First, show that every totally wildly ramified galois extension of $\mathbb{Q}_p$ is cyclic for every odd p. Now, if $G = \mathrm{Gal}_{\mathbb{Q}_p}(K) \cong (\mathbb{Z}/p\mathbb{Z})^3$, then write G as an internal direct sum of the inertia subgroup

$I \leqslant G$ and a cyclic subgroup $H \leqslant G$ (as L^i is unramified), and hence a cyclic extension of $\mathbb{Q}_p$ with galois group isomorphic to $G/I \cong H$. But then L^H is totally wildly ramified abelian extension of $\mathbb{Q}_p$ whose galois group G/H is *not* cyclic, which completes the proof for the $\ell = p > 2$ case.

The $\ell = p = 2$ case

We must isolate the $p = 2$ case as there is an extension of $\mathbb{Q}_2$ with galois group isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$, namely the Cyclotomic field $\mathbb{Q}_2(\zeta_{24}) = \mathbb{Q}_2(\zeta_3) \cdot \mathbb{Q}_2(\zeta_8)$. We apply the following analogous argument

First, the unramified Cyclotomic field $\mathbb{Q}_2(\zeta_{2^{2^r}-1})$ has galois group $\mathbb{Z}/2^r\mathbb{Z}$ and the totally ramified Cyclotomic field $\mathbb{Q}_2(\zeta_{2^r+2})$ has galois group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^r\mathbb{Z}$ (up to isomorphic). Let $m = (2^{2^r} - 1)(2^{r+2})$. Now, if $K \not\subseteq \mathbb{Q}_2(\zeta_m)$, we must have:

$$\mathrm{Gal}_{\mathbb{Q}_2}(K(\zeta_m) \cong \begin{cases} \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}/2^r)^2 \times \mathbb{Z}/2^s\mathbb{Z} & 1 \leq s \leq r \\ (\mathbb{Z}/2^r)^2 \times \mathbb{Z}/2^s\mathbb{Z} & 2 \leq s \leq r \end{cases}$$

and thus must admit a quotient isomorphic to $(\mathbb{Z}/2\mathbb{Z})^4$ or $(\mathbb{Z}/4\mathbb{Z})^3$. We shall show no extension of $\mathbb{Q}_2$ has either of these galois groups, showing that K lies in $\mathbb{Q}_2(\zeta_m)$ which will complete the proof.

To do this, first show that there are exactly 7 quadratic extensions of $\mathbb{Q}_2$, and so there is no extension of $\mathbb{Q}_2$ that has galois group $(\mathbb{Z}/2\mathbb{Z})^4$ since this groups has 15 subgroups of index 2 whose fixed fields yields 15 distinct quadratic extensions of $\mathbb{Q}_2$.

Next, there are only finitely many extensions of $\mathbb{Q}_2$ for of degree d for a fixed d, and these can be enumerated by the Eisenstein polynomials in $\mathbb{Q}_2[x]$ of degree d divign the relation given by Krasner's lemma. Then there are 59 quartic extensions of $\mathbb{Q}_2$ of which 12 are cyclic. This would be very tedious to show directly, and can be verified given this link: LMFDB, 2-adic cyclic quadratic fields. Thus there is no extension of $\mathbb{Q}_2$ which has galois group $(\mathbb{Z}/4\mathbb{Z})^3$, as this group has 28 subgroups whose fixed fields gives 28 distinct cyclic quartic extensions of $\mathbb{Q}_2$, but then we have exhausted all possibilities, completing the proof.

[a] namely if $\mathfrak{P}/p$ is the maximal ideal of $\mathcal{O}_F$, then the valuation $\nu_{\mathfrak{P}}$ extends ν_p with index 1 so $\nu_{\mathfrak{P}}(-pu) = \nu_p(-p) = 1$

4.4 *Application: The Hasse Principle for Conics

This section is not necessary for the rest of this book, but it demonstrates some of the usefulness of local fields. It shall quickly skim over the necessary algebraic curve theory, the reader can checkout [10, chapter 21], particularly section 6, for more details.

Using the rational root test ([10, chapter 11.3]) finding rational solutions to any degree polynomial with 1 variable is simple, and there are necessarily finitely many. Once 2 variables are introduced, the complexity of finding rational solutions increases greatly, and the degree plays an important role in compartmentalizing the complexity. Degree 1 polynomials in two variables are linear, and hence are immediately solvable using linear algebra. Degree 3 polynomials in two variables are *elliptic curves*[2] and the complexity of finding rational solutions jumps greatly: see [6] for more details. Degree 4 and higher have only finitely many rational solutions by *Faltings's Theorem*, but as of writing this book

[2] or are singular elliptic curves

there is no *constructive method* of finding these roots.

In the case of degree 2 polynomials with two variables, we can find all rational solutions using the *Hasse Principle*. We start by defining our terms:

Definition 4.4.1: Conic

Let

$$C : ax^2 + by^2 + cxy + dx + ey + f = 0$$

be an arbitrary equation of degree 2. Then if C is smooth, it is called a *conic*.

The case where C is not smooth is be reduced to the smooth case by normalizing $C^\nu \to C$ and showing the results hold after projecting. Naturally, projectivising creates a homogeneous polynomial in 3 variables of degree 2, and hence we can also think of a conic as:

$$C : F(x_1, x_2, x_3) + \sum f_{ij} x_i x_j = 0$$

where $f_{ij} = f_{ji}$ and $\det f_{ij} \neq 0$. In [10, chapter 21], it was left as an exercise to show that over any perfect field any conic has a liner-isomorphism making it equivalent to a curve of the form:

$$f_1 x_1^2 + f_2 x_2^2 + f_3 x_3^2 = 0$$

for $f_i \neq 0$.

The main idea was that we can always diagonalize the coefficient matrix.

We can use some algebraic geometry to deduce the following:

Lemma 4.4.2: Conic – Single Solution

Let C be a degree 2 be a conic. Then C either has no rational solutions, or infinitely many

***Proof*:**

Let $f(x, y)$ be the equation for C and assume $(a, b) \in \mathbb{Q}^2$ such that $f(a, b) = 0$. Construct the rest of the solutions by taking all rational-slopped lines passing through (a, b). This in facts creates a bi-rational map between C and $\mathbb{P}^1_{\mathbb{Q}}$, showing that this is in fact *all* such solutions.

We shall show that we can find all rational solutions of a degree 2 curve in two variables

Theorem 4.4.3: Local-to-global For Conics

Let C be a conic over $\mathbb{Q}$. Then C has a rational point (a $\mathbb{Q}$-solution to C) if and only if C has a $\mathbb{Q}_p$-solution for each prime p and an $\mathbb{R}$-solution.

Note that if a solution exists over $\mathbb{Q}$, as $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$ for each p and $\mathbb{Q} \hookrightarrow \mathbb{R}$, a solution certainly exists over each local field (this also motivates why $\mathbb{Q}$ is called a global field: a solution in this field is trivially a solution over each local field). Thus, the converse will be proven. The key result in this proof shall be a clever application of Minkowski's theorem which recall here

Lemma 4.4.4: Minkowski's Lattice Point Theorem

Let Γ be a discrete complete lattice in the Euclidean vector space V of dimension n and let X be a centrally symmetric convex subset of V. Then if

$$\mathrm{vol}(X) > 2^n \mathrm{vol}(\Gamma)$$

then X contains at least one nonzero lattice point $\gamma \in \Gamma$

***Proof*:**
see [2, Minkowski's Lattice Point Theorem]

***Proof*:**
of theorem 4.4.3

Let $F(x_1, x_2, x_3)$ represent the conic C in projective coordinates and assume a nontrivial[a] solution exists for each $\mathbb{Q}_p$ and $\mathbb{R}$. As mentioned earlier, any conic has a linear isomorphism to a curve of the form:

$$f_1x_1^2 + f_2x_2^2 + f_3x_3^2$$

where each $f_i \neq 0$. By multiplying out the denominator we can assume each $f_i \in \mathbb{Z}$. Let us first show that we can reduce to the case where $f_1f_2f_3 \in \mathbb{Z}$ is square-free. For, if the product was not square free, if $p^2 \big| f_i$, then apply $x_i \to \frac{1}{p}x_i$ until f_i is square free. If two of the factors have a p factor, send the third to p times itself. Then the whole expression can be divided by p^3. Then the magnitude of $|f_1f_2f_3|$ must eventually terminate, giving the desired square-free reduction.

With this, let us setup how Minkowski's theorem will be key: we shall show there exists a lattice $\Lambda \subseteq \mathbb{Z}^3$ of index $m = 4|f_1f_2f_3|$ where

$$F(x_1, x_2, x_3) \equiv 0 \bmod \Lambda$$

Given such a lattice exists, we can define[b]

$$S = \left\{(x_1, x_2, x_3) \in \mathbb{Q}^3 \;\middle|\; |f_1|x_1^2 + |f_2|x_2^2 + |f_3|x_3^2 < m\right\}$$

Then:

$$V(S) > \frac{\pi}{3}2^3 4|f_1f_2f_3| > 4|f_1f_2f_3| = m$$

so by Minkowski's theorem there exists a $(x, y, z) \in S \cap \Lambda$ such that

$$F(x, y, z) \equiv 0 \bmod \Lambda \qquad \text{and} \qquad |F(x, y, z)| \leq |f_1|x_1^2 + |f_2|x_2^2 + |f_3|x_3^2 < m$$

which implies a rational solution exists!

Thus, we reduce to showing such a Λ exists. First, by assumption there exists points $0 \neq (a_1, a_2, a_3) \in \mathbb{Q}_p^3$ such that $F(a_1, a_2, a_3) = 0$, as well as the real case. Certainly all multiples of (a_1, a_2, a_3) are also solution, hence without loss of generality we may assume $\max_i |a_i|_p = 1$. Thus, $(a_1, a_2, a_3) \in \mathbb{Z}_p^3$. Now, we split up the proof in 3 cases:

$(p \neq 2,\ p \big| f_1 f_2 f_3)$: Without loss of generality, say $p \big| f_1$ so that $p \nmid f_2 f_3$. Then $|f_1 a_1^2|_p < 1$. Now, if it were the case that $|a_1|_p < 1$, then by the ultra-metric inequality:

$$|f_3 a_3^2|_p = \left|f_1 a_1^2 + f_2 a_2^2\right| < 1$$

and so $|a_3|_p < 1$, but then:

$$|f_1 a_1^2|_p = |f_2 a_2^2 + f_3 a_3^2|_p < p^{-2}$$

implying $|a_1|_p < 1$ – contradicting our earlier normalization. Thus,

$$|a_2|_p = |a_3|_p = 1$$

and hence these are units in $\mathbb{Q}_p$. Then:

$$|f_2 a_2^2 + f_3 a_3^2| < 1$$

Now, dividing by a_2^2 (as it is a unit) we can deduce there is some $r_p \in \mathbb{Z}$ and f_p such that

$$f_2 + r_p^2 f_3 \equiv 0 \bmod p$$

From this, we can now find a solution to our equation. By requiring $x_3 \equiv r_p x2 \bmod p$, we get

$$F(x_1, x_2, x_3) \equiv f_1 x_1^2 + f_2 x_2^2 + f_3 x_3^2 \equiv (f_2 r_p^2 f_3) x_3^2 \equiv 0 \bmod p$$

giving our first case

1. $(p = 2,\ p \nmid f_1 f_2 f_3)$ By examining the coefficients $\bmod 2$, exactly two of the coefficients must be units, without loss of generality say a_1, a_2. Then as $t^2 \equiv 0, 1 \bmod 4$, it must be that:

$$f_2 + f_3 \equiv 0 \bmod 4$$

Then by requiring

$$x_1 \equiv 0 \bmod 2 \qquad \text{and} \qquad x_2 \equiv x_3 \bmod 2$$

we get $F(x_1, x_2, x_3) \equiv 0 \bmod 4$, giving our second case.

2. $(p = 2,\ p \big| f_1 f_2 f_3)$ Without loss of generality say $2 \big| f_1$, so $2 \nmid f_2, f_3$. Then like in the first case we get $|a_2|_2 = |a_3|_2 = 1$. When $a \in \mathbb{Z}$ is odd, $a^2 \equiv 1 \bmod 8$. So we get the following two possibilities:

$$f_2 + f_3 \equiv 0 \bmod 8 \text{ iff } 2 \big| a_1$$

or:

$$f_1 + f_2 + f_3 \equiv 0 \bmod 8 \text{ iff } 2 \nmid a_1$$

In the first case, let $s = 0$ and in the second let $s = 1$. Then require:

$$x_2 \equiv x_3 \bmod 4 \qquad \text{and} \qquad x_1 \equiv s x_3 \bmod 2$$

Then $F(x_1, x_2, x_3) \equiv 0 \bmod 8$.

Now, define the lattice Λ in $\mathbb{Z}^3$ which satisfies these relations. Then by construciton its index is

$$m = 4|f_1 f_2 f_3|$$

Hence

$$F(x_1, x_2, x_3) \equiv 0 \bmod m$$

But then, by Minkowski's theorem, we get a rational point that satisfies this eqution, as we sought to show.

[a]In projective coordiantes, $[0:0:0]$ does not exist; this would be the trivial solution

[b]Note that the Minkowsk's theorem was originally used with sublattices of number rings given by primes, however part of its strengh in ability to generally conclude the existnece of intersection points of lattices, just like is being done here

5

Places: Generalizing Primes

As we saw, primes uniquely correspond to discrete valuations (or equivalence classes of absolute values). Form this perspective, the Archimedean prime could be thought of as the "prime at infinity". This can in fact be made very precise, an shall even need to be taken properly into account when working looking at *ray fields* in the sequel *Class Field Theory*. We thus dive deeper into this connection

5.1 Places and Completions

Definition 5.1.1: Places

Let K be a field. Then a *place* of K is an equivalence class of nontrivial absolute values on K. The set of places of K shall be denoted M_K

Recall that the completions of K are in correspondence with absolute values and are invariant under equivalent absolute values, and hence completions of K are in correspondence with places. If $|-|_\nu$ is some representative absolute value of a place, then K_ν shall denote its completion. If a place is Archimedean, the field K_ν shall be called Archimedean, and nonarchimedean otherwise. If K is a global field, then the completion at a place gives a local field, which if ν is Archimedean we know that $K_\nu \cong \mathbb{R}$ or $K_\nu \cong \mathbb{C}$, and it is nonarchimedean then the absolute value of K_ν is induced by a discrete valuation, which we shall denote ν (this motivated the symbol for general fields, as the discrete valuation is independent of the choice of absolute value representative).

Example 5.1.2: Places

1. When $K = \mathbb{Q}$, this gives the familiar places given by the primes as well as ∞.
2. When $K = \mathbb{F}_q[t]$ for $q = p^r$, the places correspond to the irreducible polynomials of $\mathbb{F}_q[t]$ along with a *nonarchimedean* place at infinity $|-|_\infty = q^{\deg(-)}$.

The other places that shall be considered are extensions of these fields.

Definition 5.1.3: Types of Places

Let K be a global field and ν a place. Then:

1. $K_\nu \cong \mathbb{R}$, ν is said to be a *real place*
2. $K_\nu \cong \mathbb{C}$, ν is said to be a *complex place*
3. If $|-|_\nu$ is induced by a discrete valuation $\nu_\mathfrak{p}$ for a prime $\mathfrak{p} \subseteq K$, then ν is a *finite place*. Otherwise it is an *infinite place*.

Every finite place must be nonarchimedean, every infinite place is Archimedean in characteristic 0 and nonarchimedean otherwise, every Archimedean place is an infinite place, but nonarchimedean may be finite or infinite (as demonstrated in example 5.1.2).

Definition 5.1.4: Extending Places

Lt L/K be an extension of global fields. Then for every ω of L, any absolute value $|-|_w$ that represents the equivalence class w restricts to an absolute value on K that represents a place ν of K independent of choice of $|-|_w$. Then we write $w|\nu$ and say that w *extends* ν or that w *lies above* ν

Here now comes the most important results about places that generalize the Archimedean embedding and theorem 4.2.4(5):

Theorem 5.1.5: Extending Global Ring Tensoring Local Ring

let L/K be a finite separable extension of global fields and let ν be a place of K. Then:

$$L \otimes_K K_\nu \cong \prod_{w|\nu} L_w$$

given by

$$\ell \otimes x \mapsto (\ell x, ..., \ell x)$$

Note that this was already proven for the nonarchimedean places intheorem 4.2.4. The following proof shows how the ideas work for places in general:

***Proof*:**
Let ν be a place on K. First off, the separable extension L/K is a finite étale K-algebra, so the base change $L \otimes_K K_v$ is a finite étale K_v-algebra, and is therefore isomorphic to a finite product $\prod_{i \in I} L_i$ of finite separable extensions L_i of K_v, each of which is a local field (any finite extension of a local field is a local field). Thus, it suffices to show that there is a bijective correspondence between the sets of local fields and localizations:

$$\{L_i : i \in I\} \qquad \leftrightarrow \qquad \{L_w : w|v\}.$$

To that end, fix an absolute value $|-|_v$ on K_v. As L_i is a local field extending K_v, when ν is nonarchimedean by theorem 3.1.7 there is a unique absolute value $|-|_w$ that restricts to $|-|_v$; when v is Archimedean it is immediate since then either $K_v \cong L_w$ or $K_v \cong \mathbb{R} \subseteq \mathbb{C} \cong L_w$ and the Euclidean absolute value on $\mathbb{R}$ is the restriction of the Euclidean absolute value on $\mathbb{C}$. Then the map $L \hookrightarrow L \otimes_K K_v \cong \prod_i L_i \twoheadrightarrow L_i$ allows us to view L as a subfield of each L_i ($L \subseteq L_i$), so $|-|_w$ on L_i restricts to L that uniquely determines a place $w|v$. There is thus a well-defined map

$$\varphi : \{L_i : i \in I\} \to \{L_w : w|v\}$$

Then φ must be shown to be bijective and $\varphi(L_i) \cong L_i$.

For $\varphi(L_i) \cong L_i$, note that $L \otimes_K K_v \cong \prod_i L_i$ as an isomorphism of topological rings as both sides are finite dimensional vector spaces over the complete field K_v and thus have a unique topology induced by the sup-norm (which by uniqueness of norm-topologies on f.d. vector spaces, is the same as the topology on$\prod_i L_i$). Next, the image of the canonical embedding $L \hookrightarrow L \otimes_K K_v$ defined by $\ell \mapsto \ell \otimes 1$ is dense because $K \subseteq L$ is dense in K_v: given any $\ell \otimes x$ in $L \otimes_K K_v$ with $\ell \in L$ and $x \in K_v$, choose $y \in K^\times$ arbitrarily close to x so that $\ell y \otimes 1 = \ell \otimes y$ is an element of the image of L arbitrarily close to $\ell \otimes x$ (and similarly for sums of pure tensors). The image of L is therefore also dense in $\prod_i L_i$ and has dense image under the projections $\prod_i L_i \twoheadrightarrow L_i$ and $\prod_i L_i \to L_i \times L_j$ ($i \neq j$). Now, if $\varphi(L_i) = L_w$ then $L_i \cong L_w$ since L is dense in the complete field L_i, and L_w is the completion of L with respect to the restriction of the absolute value on L_i to L, by the universal property of completions ([4]).

For injectivity of φ, note that if $\varphi(L_i) = \varphi(L_j) = L_w$ for some $i \neq j$ then image of the diagonal embedding $L \to L_w \times L_w$ is not dense in $L_w \times L_w$ (its closure is isomorphic to L_w!), but the image of L is dense in $L_i \times L_j$; a contradiction.

For surjectivity of φ, for each $w|v$ define a continuous homomorphism of finite étale K_v-algebras and topological rings:

$$\varphi_w : L \otimes_K K_v \to L_w \qquad \ell \otimes x \mapsto \ell x.$$

The map φ_w is surjective because its image contains L and is complete, and L_w is the completion of L. It follows that φ_w factors through the projection of $L \otimes_K K_v \cong \prod_i L_i$ on to one of its factors L_i and induces a homeomorphism from L_i to L_w. But then $L_i \cong L_w$ as topological fields, so $\varphi(L_i) = L_w$ and φ is surjective.

Corollary 5.1.6: Places and Irreducible Factors

Let L/K be a finite separable extension of global fields, ν a place of K, and $f \in K[x]$ an irreducible polynomial such that $L \cong K[x]/(f(x))$. Then there is a bijective correspondence between the irreducible factors of f in $K_\nu[x]$ and the places of L lying above ν. In particular, if $f = f_i \cdots f_r$ are the factors of f in $K_\nu[x]$, then:

$$L_{w_i} \cong \frac{K_\nu[x]}{(f_i(x))} \qquad 1 \leq i \leq r$$

for appropriate w_i's.

***Proof*:**
exercise.

Corollary 5.1.7: Classifying Embeddings

Let L/K be a finite separable extension of global fields and ν a place of K. Then given a fixed algebraic closure $\overline{K}_\nu$, there is a bijection

$$\frac{\mathrm{Hom}_K(L, \overline{K}_\nu)}{\mathrm{Gal}_{K_\nu}(\overline{K}_\nu)} \leftrightarrow \{w|\nu\}$$

where the left hand side is the $\mathrm{Gal}_{K_\nu}(\overline{K}_\nu)$-orbits of K-embeddings of L into $\overline{K}_\nu$, and the right hand side is the places of L above ν.

***Proof*:**
Let $L \cong K(\alpha) = k[x]/(f)$ for some $\alpha \in L$ with minimal polynomial $f \in K[x]$. Then from field theory, there is a bijection between $\mathrm{Hom}_K(L, \overline{K}_\nu)$ and the roots of α_i of f in $\overline{K}_\nu$ that is compatible with the action of $\mathrm{Gal}_{K_\nu}(\overline{K}_\nu)$ on both sets.

Now, if $f = f_1 \cdots f_r$ is the factorization of f in $K_\nu[x]$, each f_i corresponds to an orbit of the galois action on the roots of f, which by corollary corollary 5.1.6 we get a bijective correspondence with the places of L above ν, as we sought to show.

In the special case of $K = \mathbb{Q}$ and $v = \infty$, we have that

$$\frac{\mathrm{Hom}_{\mathbb{Q}}(L, \mathbb{C})}{\mathrm{Gal}_{\mathbb{R}}(\mathbb{C})} \leftrightarrow \{w|\infty\}$$

which gives the Archimedean embedding theorem from [10, chapter 22]

Definition 5.1.8: Real and Complex Place

Let K be a number field. Then the elements of $\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{R})$ are the *real embeddings* and the elements of $\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ whose image does not lie strictly in $\mathbb{R}$ are called the *complex embeddings*

Corollary 5.1.9: Index Given Real and Complex Embeddings

Let K be a number field with r real places and s complex places. Then:

$$[K:\mathbb{Q}] = r + 2s$$

***Proof*:**
This is computation: take $K \cong \mathbb{Q}[x]/(f)$ for some irreducible separable $f \in \mathbb{Q}[x]$ so that:

$$[K:\mathbb{Q}] = \deg f = \#\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C})$$

Then the action of $\mathrm{Gal}_{\mathbb{R}}(\mathbb{C})$ on $\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ as r orbits of size 1 and 2 orbits of size 2, and the rest follows.

Example 5.1.10: Real and Complex Places in Non-trivial Field
Take $K = \mathbb{Q}[x]/(x^3 - 2)$. Then there are three embeddigns $K \hookrightarrow \mathbb{C}$ for each root given by:

$$x \mapsto \sqrt[3]{2} \qquad x \mapsto \zeta_3 \sqrt[3]{2} \qquad x \mapsto \zeta_3^2 \sqrt[3]{2}$$

The first is a real embedding, the other two complex and conjugate to each other. Thus, K has a real place and a complex place, and we get that $[K:\mathbb{Q}] = 1 \cdot 1 + 2 \cdot 1 = 3$.

5.2 Extending Prime Factorization to Places

In this section, we shall continue the analogy between primes and places by showing that they respect many of the similar identities and properties that prime ideals do. We want to show an analogy for residue field, norm, inertia degree and ramification index, the fundamental identity, picard group, and divisors. In this section, we shall use the words prime and place interchangeably, distinguishing them as finite and infinite primes.

Let us first define the residue field for infinite primes. We shall see that

$$\kappa(\mathfrak{p}) = K_{\mathfrak{p}} \qquad \mathfrak{p}\big|\infty$$

will be the right notion. Now, for each prime $\mathfrak{p}$ we define the (canonical) homomorphism:

$$\nu_{\mathfrak{p}} : K^{\times} \to \mathbb{R}$$

where if $\mathfrak{p}$ is finite, it is the usual $\mathfrak{p}$-adic exponential valuation and $\nu_{\mathfrak{p}}(K^{\times}) = \mathbb{Z}$, and if $\mathfrak{p}$ is infinite then:

$$\nu_{\mathfrak{p}}(a) = -\log(|\tau a|)$$

where $\tau : K \to \mathbb{C}$ is the embedding which defined $\mathfrak{p}$. Now, for any $\mathfrak{p}/p$ (whether p is finite or infinite, define:

$$f_{\mathfrak{p}} = [\kappa(\mathfrak{p}) : \kappa(p)]$$

Where if $\mathfrak{p}\big|\infty$ we have $f_{\mathfrak{p}} = [K_{\mathfrak{p}} : \mathbb{R}]$ and

$$\mathfrak{N}(\mathfrak{p}) = \begin{cases} p^{f_{\mathfrak{p}}} & \mathfrak{p} \nmid \infty \\ e^{f_{\mathfrak{p}}} & \mathfrak{p}\big|\infty \end{cases}$$

where e here is Euler's number, not the ramification. In fact, we may consider e as an infinite prime number, and the extension $\mathbb{C}/\mathbb{R}$ can be thought of as an unramified extension with inertia degree 2. Next, define the absolute value $|-|_{\mathfrak{p}} : K \to \mathbb{R}$ via:

$$|a|_{\mathfrak{p}} = \mathfrak{N}(\mathfrak{p})^{-\nu_{\mathfrak{p}}(a)}$$

and $|0|_{\mathfrak{p}} = 0$. If $\mathfrak{p}$ is an infinite prime with associated embedding $\tau : K \to \mathbb{C}$, notice that:

$$|a|_{\mathfrak{p}} = |\tau a| \qquad |a|_{\mathfrak{p}} = |\tau a|^2$$

where $\mathfrak{p}$ is real and complex respectively.

Next, if L/K is a finite extension of K, denote the primes $\mathfrak{P}/\mathfrak{p}$ of L the valuations in the class $\mathfrak{P}$ which, when restricted to K, give $\mathfrak{p}$. In the case of an infinite prime $\mathfrak{P}$, we get the inertia degree and ramification index to be:

$$f_{\mathfrak{P}/\mathfrak{p}} = [L_{\mathfrak{P}} : K_{\mathfrak{p}}] \qquad e_{\mathfrak{P}/\mathfrak{p}} = 1$$

Thus, we may general the fundamental identity and many other results

Proposition 5.2.1: Fundamental Identity For Places

Let $\mathfrak{P}/\mathfrak{p}$ be an extension of primes (finite or infinite). Then:

1. $\sum_{\mathfrak{P}/\mathfrak{p}} e_{\mathfrak{P}/\mathfrak{p}} f_{\mathfrak{P}/\mathfrak{p}} = \sum_{\mathfrak{P}/\mathfrak{p}} [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = [L : K]$
2. $\mathfrak{N}(\mathfrak{P}) = \mathfrak{N}(\mathfrak{p})^{f_{\mathfrak{P}/\mathfrak{p}}}$
3. $\nu_{\mathfrak{P}}(a) = e_{\mathfrak{P}/\mathfrak{p}} \nu_{\mathfrak{p}}(a)$ (for $a \in K^\times$)
4. $\nu_{\mathfrak{p}}(N_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(a)) = f_{\mathfrak{P}/\mathfrak{p}} \nu_{\mathfrak{P}}(a)$ (for $a \in L^\times$)
5. $|a|_{\mathfrak{P}} = |N_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}|_{\mathfrak{p}}$ (for $a \in L$)

***Proof*:**
exercise

5.2.1 Galois Theory of Global Fields

Next, let us take a moment to extend the theory of places when L/K is a galois extension of global field. Then we have a galois group $\mathrm{Gal}_K(L)$ acting on the set of places ω of L via $\omega \mapsto \sigma(w)$ where $\sigma(w)$ is the equivalence classes of absolute valued given by

$$\|\alpha\|_{\sigma(w)} = \|\sigma(\alpha)\|_w$$

This generalizes the case from the finite prime case. We can generalize the decomposition group:

Definition 5.2.2: Decomposition Group For Places

Let L/K be a galois extension of global fields and let w be a place of L. Then the *decomposition group* of w is the stabilizer:

$$D_w = \{\sigma \in \mathrm{Gal}_K(L) \mid \sigma(w) = w\}$$

We shall see that in this case we can get infinite primes that ramify. By letting $L \cong \mathbb{Q}[x]/(f)$, we get a one-to-one correspondence between the embeddings of L into $\mathbb{C}$ with the root of $f \in \mathbb{C}$. Then each embedding of L into $\mathbb{C}$ restricts to an embedding of K into $\mathbb{C}$. This map induces a map that sends each infinite place w of L to an infinite place v of K that extends to w. Now, this map may send a complex place to a real place, which happens when a pair of distinct complex conjugate embeddings of L restrict to the same embedding of K (implying it must be a real embedding). In this case, we say that v (and w) is *ramified* in the extension L/K, and the ramification index is $e_\nu = 2$ (and $e_\nu = 1$ otherwise). In this case, we shall also have $f_\nu = 1$ for $\nu\big|\infty$, and $g_\nu = \#\{w|\nu\}$. With all of this, we may generalize the fundamental identity to:

$$e_\nu f_\nu g_\nu = [L : K]$$

and more generally:

Definition 5.2.3: Ramification Index For Global Infinite Primes

Let L/K be a galois extension of number fields L/K. Then:

$$e_0(L/K) = \prod_{\nu\nmid\infty} e_\nu \qquad e_\infty(L/K) = \prod_{\nu\big|\infty} e_\nu$$

$$e(L/K) = e_0(L/K)e_\infty(L/K)$$

Now as L/K is galois, it induces a transitive action on the embeddings and their corresponding places. As $L \cong k[x]/(g)$, each embedding of K into $\mathbb{C}$ gives rise to $[L : K]$ distinct embeddings of L into $\mathbb{C}$ that extend it, one for each root of g in $\mathbb{C}$. Thus ,for each infinite place ν of K, the galois group acts transitively on $\{w\big|\nu\}$, and so either all of the places $w|\nu$ are ramified or none are, which is determined by the decomposition group for places (namely, whether the decomposition group is order 1 or 2).

Let us conclude by generalizing Dirichlet's unit theorem. It is named after *Herbrand* for its link to the Herbrand quotient developed in the sequel *Class Field Theory*:

Theorem 5.2.4: Herband Unit Theorem

Let L/K be a galois extension of number field. Let $w_1, ..., w_r$ be the real places of L and $w_{r+1}, ..., w_{r+s}$ be the complex places of L. Then there exists $\epsilon_1, ..., \epsilon_{r+s} \in \mathcal{O}_L^\times$ such that

1. $\sigma(\epsilon_i) = \epsilon_j$ if and only if $\sigma(w_i) = w_j$ for all $\sigma \in \mathrm{Gal}_K(L)$
2. $\epsilon_1, ..., \epsilon_{r+s}$ generate a finite index subgroup of $\mathcal{O}_L^\times$
3. $\epsilon_1\epsilon_2 \cdots \epsilon_{r+1} = 1$ and every relation among the ϵ_i is generated by this one.

***Proof*:**
This proof is straight from Sutherland's notes. I need some more time to internalize it before typing it out myself. It is here for ease of reference

1. Pick $\epsilon_1, \ldots, \epsilon_{r+s} \in \mathcal{O}_L^\times$ such that $\|\epsilon_i\|_{w_j} < 1$ for $i \neq j$; the existence of such ϵ_i follows from the strong approximation theorem that we will prove in the next lecture; the

product formula then implies $\|\epsilon_i\|_{w_i} > 1$. Now let $\alpha_i := \prod_{\sigma\in D_{w_i}} \sigma(\epsilon_i) \in \mathcal{O}_L^\times$. We have $\|\alpha_i\|_{w_i} = \prod_{\sigma\in D_{w_i}} \|\epsilon_i\|_{w_i} > 1$ and $\|\alpha_i\|_{w_j} = \prod_{\sigma\in D_{w_i}} \|\epsilon_i\|_{\sigma(w_j)} < 1$, since $\sigma \in D_{w_i}$ fixes w_i and permutes the w_j with $j \neq i$. Each α_i is fixed by D_{w_i}.

Let $G := \mathrm{Gal}(L/K)$. For $i = 1, \dots, r+s$, let $r(i) := \min\{j : \sigma(w_j) = w_j \text{ for some } \sigma \in G\}$, so that $w_{r(i)}$ is a distinguished representative of the G-orbit of w_i. For $i = 1, \dots, r+s$ let $\beta_i := \sigma(\alpha_{r(i)})$, where σ is any element of G such that $\sigma(w_{r(i)}) = w_i$. The value of $\sigma(\alpha_{r(i)})$ does not depend on the choice of σ because $\sigma_1(w_{r(i)}) = \sigma_2(w_{r(i)})$ if and only if $\sigma_2^{-1}\sigma_1 \in D_{w_{r(i)}}$ and $\alpha_{r(i)}$ is fixed by $D_{w_{r(i)}}$. The β_i then satisfy (i).

2. The β_i also satisfy (2): a product $\gamma_j := \prod_{i\neq j} \beta_i^{n_i}$ cannot be trivial because $\|\gamma_j\|_{w_j} < 1$; in particular, $\beta_1, \dots, \beta_{r+s-1}$ generate a subgroup of $\mathcal{O}_L^\times$ isomorphic to $\mathbb{Z}^{r+s-1}$ which necessarily has finite index in $\mathcal{O}_L^\times \simeq \mathbb{Z}^{r+s-1} \times \mu_L$ (see Theorem 15.12). But we must have $\prod_i \beta_i^{n_i} = 1$ for some tuple $(n_1, \dots, n_{r+s}) \in \mathbb{Z}^{r+s}$ (with $n_i = n_j$ whenever w_i and w_j lie in the same G-orbit, since every $\sigma \in G$ fixes 1). The set of such tuples spans a rank-1 submodule of $\mathbb{Z}^{r+s}$ from which we choose a generator $(n_1, \dots, n_{r+s})$ (by inverting some β_i if necessary, we can make all the n_i positive if we wish). Then $\varepsilon_i := \beta_i^{n_i}$ satisfy (1), (2), (3) as desired.

5.2.2 Picard Group for Places

Let us extend fractional ideals of K by taking into account infinite primes

Definition 5.2.5: Replete Ideal

Let K be a field. Then a *replete ideal* of K is an element of the group:

$$J(\overline{\mathcal{O}}) = J(\mathcal{O}) \times \prod_{\mathfrak{p}\infty} \mathbb{R}_+^\times$$

For notational consistency, for any infinite prime $\mathfrak{p}$ and any real number $\nu \in \mathbb{R}$, let:

$$\mathfrak{p}^\nu := e^\nu \in \mathbb{R}_+^\times$$

Next, given a ysstem of real numbers $\nu_\mathfrak{p}, \mathfrak{p}\big|\infty$, let $\prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_\mathfrak{p}}$ denote the vector:

$$\prod_{\mathfrak{p}\big|\infty} \mathfrak{p}^{\nu_\mathfrak{p}} = (\dots, e^{\nu_\mathfrak{p}}, \dots) \in \prod_{\mathfrak{p}\big|\infty} \mathbb{R}_+^\infty$$

and *not* the product of the quantities $e^{\nu_\mathfrak{p}} \in \mathbb{R}$. Then every replete ideal $\mathfrak{a} \in J(\overline{\mathcal{O}})$ admits a unique factorization:

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_\mathfrak{p}} = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{\nu_\mathfrak{p}} \times \prod_{\mathfrak{p}\big|\infty} \mathfrak{p}^{\nu_\mathfrak{p}}$$

we shall denote the finite portion $\mathfrak{a}_f$ and the infinite portion $\mathfrak{a}_\infty$ so that

$$\mathfrak{a} = \mathfrak{a}_f \times \mathfrak{a}_\infty$$

Then $\mathfrak{a}_f$ is a fractional ideal and $\mathfrak{a}_\infty$ is an element of $\prod_{\mathfrak{p}|\infty} \mathbb{R}_+^\times$. Viewing $\mathfrak{a}_f$ and $\mathfrak{a}_\infty$ as elementst of

$$\mathfrak{a}_f \times \prod_{\mathfrak{p}\big|\infty} 1 \qquad (1) \times \mathfrak{a}_\infty$$

we can think of all the elements of $J(\overline{\mathcal{O}})$ as

$$\mathfrak{a} = \mathfrak{a}_f \cdot \mathfrak{a}_\infty$$

Now, to each $a \in K^\times$, we can associated to it a *replete principal ideal*:

$$|a| = \prod_{\mathfrak{p}} \mathfrak{p}^{n u_{\mathfrak{p}}(a)}$$

This forms a subgroup $P(\overline{\mathcal{O}}) \subseteq J(\overline{\mathcal{O}})$ and hence we can define the *replete picard group* or *replete class group*:

Definition 5.2.6: Replete Picard Group

$$\text{Pic}\,(\overline{\mathcal{O}}) = \frac{J(\overline{\mathcal{O}})}{P(\overline{\mathcal{O}})}$$

Next,

Definition 5.2.7: Absolute Norm For Replete Ideals

The *absolute norm* of a replete ideal $\mathfrak{a}$ is defined to be the positive real number:

$$\mathfrak{N}(\mathfrak{a}) = \prod_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}}$$

The absolute norm is multiplicative and induces a surjective homomorphism:

$$\mathfrak{N} : J(\overline{\mathcal{O}}) \to \mathbb{R}_+^\times$$

By using a generalization of proposition 1.1.10, the replete norm of a principal ideal a is equal to 1

$$\mathfrak{N}(a) = 1$$

Hence, we in fact get a surjective homomorphism on the replete Picard group

$$\mathfrak{N} : \text{Pic}\,(\overline{\mathcal{O}}) \to \mathbb{R}_+^\times$$

There is still a bit more in Neukirch, p.187. I gotta leave it alone for now to finish other work, but must come back

5.3 *Haar Measure and the Product Formula

A natural property that the real and complex numbers have is the existence of a *translation invariant Lebesgue measure* (unique up to scaling; or unique if the unit interval/square must have measure 1).

With the places introduced, it is natural to ask whether a measure can be defined on $\overline{K}_\nu$ and if these measure have some similarities to the real and euclidean one. As $\overline{K}_\nu$ are locally compact abelian groups, there is always a unique *Haar measure* defined on them. The build-up for this measure would require a course in analysis (see [11] for the full details), so the following is a summary of the important concepts

5.3.1 Sutherland vs. these notes n Sutherland, this is the beginning of what is the "lattice" section in [10, chapter 22]. I am covering this here because the Haar measure of local fields cam up in my research, but it is not strictly necessary for the rest of these notes.

Definition 5.3.2: Radon Measure and Haar Measure

Let X be a locally compact Hausdorff space with Borel measure μ. Then μ is a *Radon measure* if for every Borel set S:

1. $\mu(S) < \infty$ is S is compact
2. $\mu(S) = \inf\{\mu(U) \mid S \subseteq U, U \text{ is open}\}$
3. $\mu(S) = \sup\{\mu(C) \mid C \subseteq S, C \text{ is compact}\}$

If X is furthermore a group, a Radon measure is a (left/right) *Haar* measure if it is *(left/right) translation invariant*:

$$\mu(S) = \mu(x+S) \qquad [\text{for right } \mu(S+x)]$$

If X is abelian, the left/right measures are identical and the measure is simply called a *Haar measure*.

Theorem 5.3.3: Existence and Uniqueness of Haar Measure

Let G be a locally compact group. Then there exists a Haar measure μ on G. If μ' is another Haar measure, then there exists a $\lambda \in \mathbb{R}_{>0}$ such that $\mu'(S) = \lambda\mu(S)$.

***Proof*:**
see [11, Existence of Haar Measure, Uniqueness of Haar Measure]

Example 5.3.4: Haar Measure On Real Space
The standard Haar measure on $\mathbb{R}^n$ is the familiar:

$$\mu\left(\prod_i (a-i, b_i)\right) = \prod_i |b_i - a_i|$$

which is also the unique Haar measure where the cube has measure 1.

Let us apply this to local fields. By proposition 1.3.3, every local field K is locally compact, and the corresponding valuation ring A is compact, in particular $A = B_1(0)$. Given a Haar measure μ on K, it shall always be normalized so that $\mu(A) = 1$; with this condition this makes the μ the unique Haar measure on K.

Proposition 5.3.5: Haar Measure On Local Fields

Let K be a local field with discrete valuation v, residue field k, and absolute value

$$|-|_v := (\#k)^{-v(-)},$$

and let μ be a Haar measure on K. For every $x \in K$ and measurable set $S \subseteq K$ we have

$$\mu(xS) = |x|_v \mu(S).$$

Moreover, the absolute value $|-|_v$ is the unique absolute value compatible with the topology on K for which this is true.

***Proof*:**
This is straight from Sutherland!

Let A be the valuation ring of K with maximal ideal $\mathfrak{p}$. The proposition clearly holds for $x = 0$, so let $x \neq 0$. The map $\varphi_x : y \mapsto xy$ is an automorphism of the additive group of K, and it follows that the composition $\mu_x = \mu \circ \varphi_x$ is a Haar measure on K, hence a multiple of μ, say $\mu_x = \lambda_x \mu$, for some $\lambda_x \in \mathbb{R}_{>0}$. Define the function $\chi : K^\times \to \mathbb{R}_{>0}$ by $\chi(x) := \lambda_x = \mu_x(A)/\mu(A)$. Then $\mu_x = \chi(x)\mu$, and for all $x, y \in K^\times$ we have

$$\chi(xy) = \frac{\mu_{xy}(A)}{\mu(A)} = \frac{\mu_x(yA)}{\mu(A)} = \frac{\chi(x)\mu_y(A)}{\mu(A)} = \frac{\chi(x)\chi(y)\mu(A)}{\mu(A)} = \chi(x)\chi(y).$$

Thus χ is multiplicative, and we claim that in fact $\chi(x) = |x|_v$ for all $x \in K^\times$. Since both χ and $|\cdot|_v$ are multiplicative, it suffices to consider $x \in A - \{0\}$. For any such x, the ideal xA is equal to $\mathfrak{p}^{v(x)}$, since A is a DVR. The residue field $k := A/\mathfrak{p}$ is finite, hence A/xA is also finite; indeed it is a k-vector space of dimension $v(x)$ and has cardinality $[A : xA] = (\#k)^{v(x)}$. Writing A as a finite disjoint union of cosets of xA, we have

$$\mu(A) = [A : xA]\mu(xA) = (\#k)^{v(x)}\chi(x)\mu(A),$$

and therefore $\chi(x) = (\#k)^{-v(x)} = |x|_v$ as claimed. It follows that

$$\mu(xS) = \mu_x(S) = \chi(x)\mu(S) = |x|_v \mu(S),$$

for all $x \in K$ and measurable $S \subseteq K$. To prove uniqueness, if $|\cdot|$ is an absolute value on K that induces the same topology as $|\cdot|_v$ then for some $0 < c < 1$ we have $|x| = |x|_v^c$ for all $x \in K^\times$. Let us fix $x \in K^\times$ with $|x|_v \neq 1$ (take any x with $v(x) \neq 0$). If $|\cdot|$ also satisfies $\mu(xS) = |x|\mu(S)$ then

$$\frac{\mu(xA)}{\mu(A)} = |x| = |x|_v^c = \left(\frac{\mu(xA)}{\mu(A)}\right)^c,$$

which implies $c = 1$, meaning that $|\cdot|$ and $|\cdot|_v$ are the same absolute value.

Exercise 5.3.1

1. Let K be a number field with s complex places. Show that the sign of the absolute discriminant Δ_K is $(-1)^s$.

5.4 *Arakelov Compactification

As we have seen, the infinite places behave analytically like boundary points to the finite primes. In standard algebraic geometry, the affine scheme $\operatorname{Spec}(\mathcal{O}_K)$ is not proper (compact). To geometrically compactify this space, we must formally append the infinite places to create a space often denoted $\overline{\operatorname{Spec}(\mathcal{O}_K)}$.

However, as the "local ring" at an infinite place is a closed bounded interval (which is not closed under addition), we cannot simply glue a new affine scheme using prime ideals. Instead, Arakelov geometry compactifies the space by altering the category of divisors and line bundles, gluing analytic metrics onto the algebraic data.

Definition 5.4.1: Arakelov Divisor

Let K be a number field. An *Arakelov divisor* is a formal sum:

$$\widehat{D} = \sum_{\mathfrak{p} \nmid \infty} n_{\mathfrak{p}}(\mathfrak{p}) + \sum_{\nu \mid \infty} x_\nu(\nu)$$

where $n_{\mathfrak{p}} \in \mathbb{Z}$ and $x_\nu \in \mathbb{R}$. The group of Arakelov divisors is denoted $\widehat{\operatorname{Div}}(K)$.

Notice that the coefficients for the infinite places are allowed to be real numbers, reflecting their continuous, Archimedean nature. This is the additive counterpart to the replete ideals defined in section 5.2.

Definition 5.4.2: Principal Arakelov Divisor

For any $a \in K^\times$, its *principal Arakelov divisor* is defined as:

$$\widehat{(a)} = \sum_{\mathfrak{p} \nmid \infty} \nu_{\mathfrak{p}}(a)(\mathfrak{p}) - \sum_{\nu \mid \infty} \log |a|_\nu(\nu)$$

The negative sign on the infinite places is a standard convention so that the product formula manifests as a degree-zero condition, exactly as it does for meromorphic functions on a compact Riemann surface (where the number of zeros equals the number of poles).

Definition 5.4.3: Arakelov Degree

The *degree* of an Arakelov divisor $\widehat{D}$ is the homomorphism $\deg : \widehat{\operatorname{Div}}(K) \to \mathbb{R}$ given by:

$$\deg(\widehat{D}) = \sum_{\mathfrak{p} \nmid \infty} n_{\mathfrak{p}} \log \mathfrak{N}(\mathfrak{p}) + \sum_{\nu \mid \infty} x_\nu$$

Theorem 5.4.4: Arakelov Degree of Principal Divisors

Let $a \in K^\times$. Then $\deg(\widehat{(a)}) = 0$.

***Proof*:**
This is an immediate consequence of the Product Formula. We have:

$$\deg(\widehat{(a)}) = \sum_{\mathfrak{p}\nmid\infty} \nu_{\mathfrak{p}}(a)\log \mathfrak{N}(\mathfrak{p}) - \sum_{\nu|\infty} \log|a|_\nu$$

By the definition of the normalized absolute values from section 5.2, exponentiating this sum yields $\prod_\nu |a|_\nu = 1$. Taking the logarithm of both sides proves the sum is 0.

This theorem shows that by adding the infinite places, $\overline{\mathrm{Spec}(\mathcal{O}_K)}$ behaves exactly like a projective curve. Finally, at the level of sheaves, this compactification is realized by equipping algebraic line bundles with analytic data at the boundary.

Definition 5.4.5: Hermitian (Arakelov) Line Bundle

An *Arakelov line bundle* $\overline{\mathcal{L}}$ on $\overline{\mathrm{Spec}(\mathcal{O}_K)}$ is a pair $(\mathcal{L}, (\|\cdot\|_\nu)_{\nu|\infty})$ where $\mathcal{L}$ is an invertible sheaf on $\mathrm{Spec}(\mathcal{O}_K)$, and for each infinite place ν, $\|\cdot\|_\nu$ is a Hermitian metric on the complex vector space $\mathcal{L}\otimes_\nu \mathbb{C}$.

Thus, rather than constructing a topological stalk at ∞, we compactify the line bundle by providing a notion of "size" for its sections at the infinite boundaries.

The machinery of Arakelov divisors and Hermitian line bundles is not just a formal trick; it allows us to import the powerful global theorems of algebraic curves into number theory. For a projective curve X, the group of divisors modulo principal divisors forms the Picard group $\mathrm{Pic}(X)$.

Definition 5.4.6: Arakelov Class Group

The *Arakelov class group* (or arithmetic Picard group) $\widehat{\mathrm{Pic}}(\mathcal{O}_K)$ is defined as the quotient group:

$$\widehat{\mathrm{Pic}}(\mathcal{O}_K) = \widehat{\mathrm{Div}}(K)/\{\widehat{(a)} | a \in K^\times\}$$

Because $\overline{\mathrm{Spec}(\mathcal{O}_K)}$ is now properly compactified, $\widehat{\mathrm{Pic}}(\mathcal{O}_K)$ behaves like the Picard group of a compact Riemann surface. Its structure elegantly combines the ideal class group of K (from the finite primes) with the unit group $\mathcal{O}_K^\times$ (regulated by the infinite metrics). Furthermore, this framework provides the exact geometric setting required to state and prove the Arithmetic Riemann-Roch Theorem, which computes the "dimension" (measured via the volume of Euclidean lattices) of the space of global sections of $\overline{\mathcal{L}}$.

A Remark on $\mathbb{F}_1$ and Purely Algebraic Compactification

While Arakelov geometry provides a rigorous and beautiful hybrid of algebra and complex analysis, the necessity of gluing analytic metrics leaves a lingering geometric dissatisfaction. As noted at the beginning of this section, the "local ring" at an infinite place (such as the closed interval $[-1, 1]$ for the standard Euclidean valuation) is a multiplicatively closed monoid, but it fails to be a commutative ring because it is not closed under addition (e.g., $1 + 1 = 2 \notin [-1, 1]$).

Because Grothendieck's scheme theory—and specifically the Proj functor used to algebraically compactify spaces—strictly requires commutative rings, the infinite places are fundamentally locked out of classical algebraic geometry.

This strict requirement is the primary motivation behind the highly active research program of $\mathbb{F}_1$-*geometry* (geometry over the field with one element). In this speculative framework, the foundational algebraic objects are relaxed from rings to monoids. If $\mathbb{Z}$ can be successfully recast as an algebra over the deeper base of $\mathbb{F}_1$, the monoid $[-1, 1]$ would become a perfectly valid local algebra. Under such a theory, the analytic workarounds of Arakelov geometry would no longer be necessary, and $\overline{\mathrm{Spec}(\mathcal{O}_K)}$ would emerge naturally as a literal, purely algebraic compactification.

A

The Hasse-Arf Theorem

This appendix proves the Hasse-Arf theorem stated in theorem 3.5.9: for a finite *abelian* extension L/K of local fields, the jumps of the upper-numbering ramification filtration are integers. The result is separated out because its proof, while elementary, is longer than the development in section 3.5 and would interrupt the thread there. It also cannot be routed through class field theory, which lies downstream of this book; the argument is the self-contained one, resting on the ramification theory of sections 3.4 and 3.5 and the different of section 3.6.

The proof has two clean reductions and one arithmetic core. The reductions bring the general abelian case down to a single congruence between consecutive ramification breaks of a cyclic extension; the core establishes that congruence.

A.1 Reduction to Cyclic Extensions

The upper numbering was built to pass to quotients, and that is exactly what turns the abelian case into the cyclic one.

Proposition A.1.1: Reduction to the Cyclic Case

If every finite cyclic extension of local fields has integral upper-numbering jumps, then so does every finite abelian extension.

***Proof*:**
Let L/K be abelian with group G, and let v be an upper break, so $G^v \neq G^{v+\epsilon}$ for all $\epsilon > 0$. The quotient $G^v/G^{v+\epsilon}$ is a nontrivial finite abelian group, so it carries a character not identically 1; extend it to a character $\chi\colon G \to \mathbb{C}^\times$ with $\chi(G^v) \neq 1$ and $\chi(G^{v+\epsilon}) = 1$. Set $H = \ker \chi$, so that

$G/H \cong \operatorname{im}\chi$ is cyclic.

By Herbrand's theorem in the upper numbering (corollary 3.5.8), $(G/H)^w = G^w H/H$ for all w. If $G^v H/H = G^{v+\epsilon}H/H$, then any $g \in G^v$ would equal $g'h$ with $g' \in G^{v+\epsilon}$ and $h \in H$; but then $\chi(g) = \chi(g')\chi(h) = 1$, contradicting the choice of a $g \in G^v$ with $\chi(g) \neq 1$. Hence $(G/H)^v \neq (G/H)^{v+\epsilon}$, so v is also an upper break of the cyclic quotient G/H. By hypothesis $v \in \mathbb{Z}$. As v was an arbitrary upper break of G, the abelian case follows, completing the proof.

A.2 The Break Congruence

It remains to prove Hasse-Arf for a cyclic extension. Passing to the inertia field removes the tame part (whose only break, at $v = 0$, is already integral), so assume L/K is cyclic and totally ramified of degree p^n. Let its lower ramification breaks be $\lambda_1 < \cdots < \lambda_n$, where G_i drops from A_{k-1} to A_k at $i = \lambda_k$, writing $A_k = p^k G$ for the subgroup of index p^k. The following computation converts integrality of the jumps into a congruence among the λ_k.

Proposition A.2.1: Cyclic Hasse-Arf as a Congruence

With the notation above, the upper-numbering jumps of L/K are the numbers $\varphi_{L/K}(\lambda_k)$, and

$$\varphi_{L/K}(\lambda_k) = \lambda_1 + \frac{\lambda_2 - \lambda_1}{p} + \cdots + \frac{\lambda_k - \lambda_{k-1}}{p^{k-1}}$$

Consequently all upper jumps are integers if and only if $\lambda_k \equiv \lambda_{k-1} \pmod{p^{k-1}}$ for $2 \leq k \leq n$.

***Proof*:**
On the interval $(\lambda_{k-1}, \lambda_k]$ the ramification group is $G_i = A_{k-1}$, of index $[G_0 : A_{k-1}] = p^{k-1}$, so $\varphi_{L/K}$ has slope $p^{-(k-1)}$ there (proposition 3.5.3). Integrating from 0, and using $\varphi_{L/K}(\lambda_1) = \lambda_1$ (slope 1 up to the first break), gives the displayed formula. The upper jumps are the images $\varphi_{L/K}(\lambda_k)$ of the lower breaks by definition of the upper numbering. Finally, arguing by induction on k: given $\varphi_{L/K}(\lambda_{k-1}) \in \mathbb{Z}$, the value $\varphi_{L/K}(\lambda_k) = \varphi_{L/K}(\lambda_{k-1}) + (\lambda_k - \lambda_{k-1})/p^{k-1}$ is an integer if and only if $p^{k-1} \big| \lambda_k - \lambda_{k-1}$, which is the stated congruence, completing the proof.

Everything is now reduced to the congruence $\lambda_k \equiv \lambda_{k-1} \pmod{p^{k-1}}$. By considering the degree-p^2 subquotients of the cyclic tower it suffices to treat $n = 2$: a cyclic totally ramified extension L/K of degree p^2 with breaks $\lambda_1 < \lambda_2$, for which the claim is $\lambda_2 \equiv \lambda_1 \pmod{p}$. Let $M = L^{A_1}$ be the intermediate field of degree p, so L/M and M/K are each cyclic of degree p. Different-transitivity pins down the intermediate break.

Lemma A.2.2: The Intermediate Break

In the degree-p^2 tower $L/M/K$ above, the break of M/K equals λ_1.

***Proof*:**
The ramification groups give $\nu_L(\mathfrak{d}_{L/K}) = (\lambda_1+1)(p^2-1)+(\lambda_2-\lambda_1)(p-1)$, $\nu_L(\mathfrak{d}_{L/M}) = (\lambda_2+1)(p-1)$, and $\nu_M(\mathfrak{d}_{M/K}) = (\mu+1)(p-1)$ where μ is the break of M/K (theorem 3.6.3). Transitivity of the different (theorem 3.6.7) with $e_{L/M} = p$ reads $\nu_L(\mathfrak{d}_{L/K}) = \nu_L(\mathfrak{d}_{L/M}) + p\,\nu_M(\mathfrak{d}_{M/K})$; dividing by $p-1$ and simplifying collapses to $\lambda_1 p = p\mu$, so $\mu = \lambda_1$, as we sought to show.

By proposition A.2.1 applied to M/K (degree p, trivially integral) together with transitivity $\varphi_{L/K} = \varphi_{M/K} \circ \varphi_{L/M}$, the top upper jump of L/K is $\varphi_{M/K}(\lambda_2) = \lambda_1 + (\lambda_2 - \lambda_1)/p$, so integrality is again the congruence $\lambda_2 \equiv \lambda_1 \pmod{p}$. This last congruence is the arithmetic heart of the theorem, and is where the norm map enters.

Lemma A.2.3: The Break Congruence

Let L/K be cyclic and totally ramified of degree p^2 with lower breaks $\lambda_1 < \lambda_2$. Then $\lambda_2 \equiv \lambda_1 \pmod{p}$.

The problem reduces to a valuation computation. Let π be a uniformizer of L and $u = \sigma(\pi)/\pi$ with σ a generator of G; then $\nu(u-1) = i_G(\sigma) - 1 = \lambda_1$, so $u \in U_L^{(\lambda_1)} \setminus U_L^{(\lambda_1+1)}$. The telescoping identity $\sigma^k(\pi) = \pi \prod_{j=0}^{k-1} \sigma^j(u)$ gives

$$\frac{\sigma^p(\pi)}{\pi} = \prod_{j=0}^{p-1} \sigma^j(u), \qquad \lambda_2 = i_G(\sigma^p) - 1 = \nu\Big(\prod_{j=0}^{p-1} \sigma^j(u) - 1\Big)$$

Writing $u = 1 + \epsilon$ with $\nu(\epsilon) = \lambda_1$ and expanding, $\prod_j (1 + \sigma^j \epsilon) - 1 = \sum_{k=1}^{p} e_k$, where e_k is the k-th elementary symmetric function of $\sigma^0\epsilon, \dots, \sigma^{p-1}\epsilon$; the congruence is thus a statement about the least valuation occurring in this sum.

A.2.4 Warning: valuation analysis flagged for verification It remains to show $\nu\big(\sum_k e_k\big) \equiv \lambda_1 \pmod{p}$. Since G acts trivially on $\mathfrak{P}_L^{\lambda_1}/\mathfrak{P}_L^{\lambda_1+1}$, each $\sigma^j\epsilon \equiv \epsilon$ there, but the valuations $\nu(e_k) \geq k\lambda_1$ carry no automatic divisibility by p, and which e_k dominates depends on λ_1 against $\nu_L(p)/(p-1)$, with cancellations among the leading terms. Resolving this correctly is the norm-filtration computation of Serre, *Local Fields*, Ch. V; it is a full development rather than a short step, and is left flagged rather than reconstructed from memory, since a shortcut is wrong in at least one regime. TODO: complete the valuation analysis of $\prod_{j=0}^{p-1} \sigma^j(u) - 1$ establishing $\lambda_2 \equiv \lambda_1 \pmod{p}$ (Serre, Local Fields, Ch. V).

As a check that the congruence is the right statement, the cyclotomic tower realizes it exactly: for $L = \mathbb{Q}_p(\zeta_{p^n})$ the lower breaks lie at $\lambda_k = p^k - 1$ (example 3.4.6), and $\lambda_k - \lambda_{k-1} = p^{k-1}(p-1) \equiv 0 \pmod{p^{k-1}}$, so the upper jumps $\varphi(\lambda_k) = k$ are integers, consistent with example 3.5.10.

Theorem A.2.5: Hasse-Arf, restated

Modulo lemma A.2.3, every finite abelian extension of local fields has integral upper-numbering jumps.

***Proof*:**
By proposition A.1.1 it suffices to treat cyclic L/K, and after removing the tame part, cyclic totally ramified of degree p^n. By proposition A.2.1 integrality of the jumps is the family of congruences $\lambda_k \equiv \lambda_{k-1} \pmod{p^{k-1}}$, which reduce to the degree-p^2 congruence of lemma A.2.3 applied to the successive subquotients, completing the proof.

What Next?

This book has built the local fields and the ramification theory of their extensions, stopping just before the theorem that organizes all abelian extensions of a local field. Several roads lead onward.

Local class field theory. The immediate sequel classifies the finite abelian extensions of a local field K by the finite-index subgroups of $K^\times$, through the local reciprocity map $K^\times \to \mathrm{Gal}_K(K^{\mathrm{ab}})$; the Hilbert symbol of section 3.7 and the norm groups met throughout are its raw material, and Lubin-Tate formal groups give it an explicit shape. This is developed in *Class Field Theory* [9]. The classical accounts are Serre's *Local Fields* and the Cassels-Fröhlich *Algebraic Number Theory.*

Galois cohomology and local duality. The Brauer group of a local field, the local duality theorems, and the cohomological reformulation of class field theory are the subject of *Group and Galois Cohomology* [7], following Serre's *Galois Cohomology.*

The local Langlands program. Beyond abelian extensions, the smooth representation theory of GL_n over a local field encodes the whole Galois group; the case of GL_2 is *Langlands for GL2(F)* [8]. Bushnell and Henniart's *The Local Langlands Conjecture for GL(2)* is the definitive treatment.

p-adic Hodge theory. The Witt vectors of section 1.5 are the entry point to the study of p-adic Galois representations and the comparison theorems relating p-adic étale and de Rham cohomology. Fontaine's period rings and the CMI notes of Brinon and Conrad are where to begin; this lies beyond the present series.

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